1. Excellence

1.1 Objectives

Most plastic packaging has deficient end of life (EoL) options leading to a large burden on the environment and depletion of often used non-renewable resources. Only 31% of plastic is currently recycled while 41% is incinerated and 27% landfilled[1]. Indeed, on the one hand, many food packaging use multilayer structures to meet required performances but those are currently very difficult to recycle. On the other hand, polylactic acid (PLA) is currently the most affordable bio-based and biodegradable plastic but its biodegradability, and that of many other biodegradable biopolymers, is not granted in every environment: PLA is currently only biodegradable under thermophilic conditions in industrial composting or digestion plants at limited thickness. Several companies, such as NOVAMONT, BASF, etc., have developed polymers that can meet home compostability requirements. Nevertheless, further improvements are needed to provide cost effective solutions with high bio-based contents and suitable performances for demanding packaging applications while still achieving compostability in mild conditions.

Using sustainable sourced comonomers, additives and fillers to formulate novel **PLA copolymers and compounds**, BIOnTOP project will deliver recyclable-by-design **cost competitive** packaging solutions that can be mechanically **recycled** or **industrially/ domestically** composted and are even suitable for anaerobic digestion (AD). We will tailor the bioplastic biodegradation through **copolymerisation** with selected diacids, **reactive extrusion** (REX) with specific additives that enhance the biodegradation potential, as well as formulation of **blends with natural fillers** from agro-wastes to speed up disintegration.

Furthermore, high permeability often limits the application of bioplastics for packing sensitive food products. In order to solve this limitation, the barrier properties of the **bioplastics** will be enhanced using previously developed **protein-based coatings, as well as novel fatty acid (FA) grafting** (nanometric layers), which, in spite of being applied in multilayer films, can be later removed and **do not jeopardize their posterior recyclability or their biodegradability.** The hydrophobic grafting will further contribute to reducing product waste at end of packaging use through a release/easy emptying effect.

Likewise, in the field of **textiles,** most used coatings (polyamide, polyurethane...) are not bio-based and of different nature than the fibres they are applied on, limiting the properties of the recycled materials. Here, **PLA** (or BIOnTOP copolymers) based **coatings and FA grafting will be developed**, allowing **reprocessing of the coated fibres** without significant loss of properties.

Tuning the BIOnTOP materials/ packaging structures[2], we will respond to end users' needs for tailored EoL options. To demonstrate the broad potential of our bio-based packaging, we have selected 7 synergistic disposable packaging materials types with very broad markets where current waste management situations have to be improved:

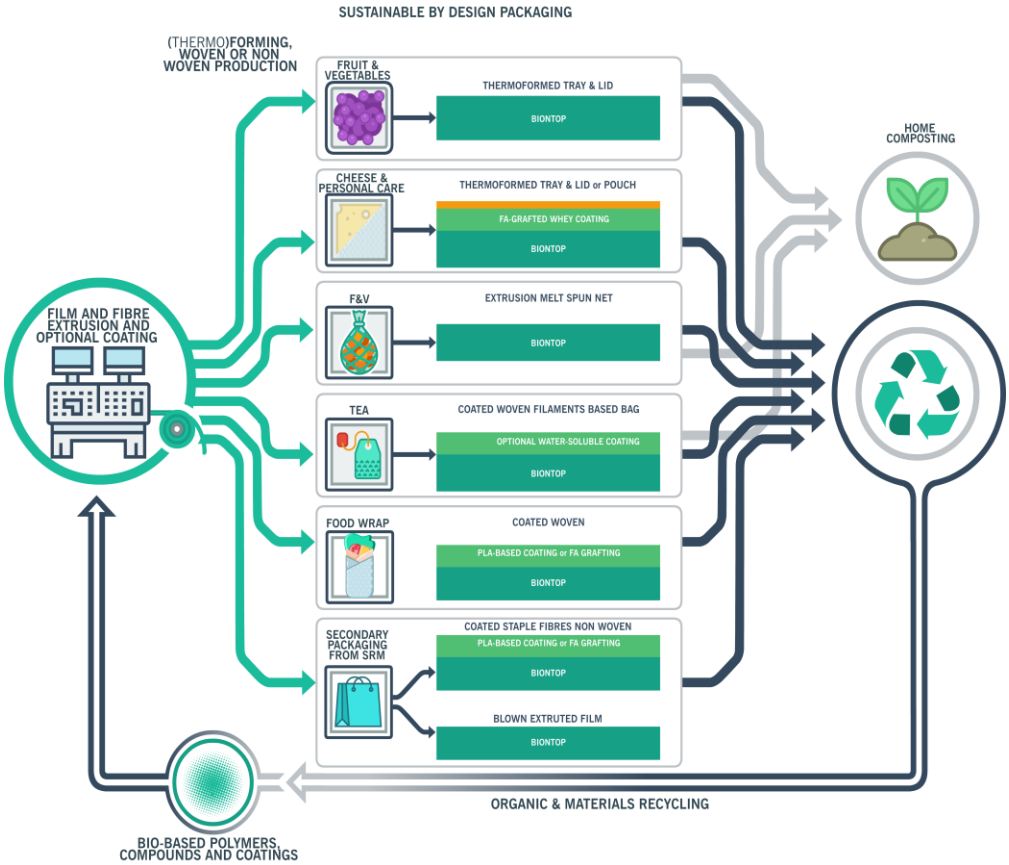


Fig. 1: BIOnTOP innovations in the target supply and value chains

1. Recyclable (material & organically), home-compostable monomaterial **trays & films for** fruits and vegetables (F&V).
2. Recyclable (material & organically) **multilayer trays & films compatible with MAP** (modified atmosphere packaging) for e.g. dairy and personal care products.
3. Home compostable & organically recyclable **nets for F&V.**
4. Home compostable & organically recyclable **coated textiles** e.g. monofilament woven fabric tea bags.
5. Recyclable (material & organically), reusable **coated woven fabrics** e.g. food wraps. Recyclable (material & organically), reusable **secondary packaging** from secondary raw materials (**SRM**):
6. Extruded **blown bags**.
7. **Non-woven bags**.

1 Plastics – the Facts 2017 An analysis of European plastics production, demand and waste data, Plastic Europe report
2 The new PLA copolymers and, blends thereof or with PLA from the project will be respectively referred to as BiontopBiontopBIOnTOP copolymers, compounds or derived materials in the proposal

The **best end of life scenario needs to be selected on a case-to-case basis** (material, product, availability of collection/sorting logistics and infrastructure, amounts, value, etc.) as key eco-design criteria. While considering industrial recycling as a favourable option, wherever possible and to improve acceptability in the current waste management context, near infrared (NIR) and hyperspectral imaging (HSI) solutions will be delivered to assess the correct sorting of the new bioplastics towards either organic or material recycling streams.

The **possibility to reprocess** the BIOnTOP formulations into new valuable products will be also carried out: the recycled SRM will be tested for similar applications, but also as secondary packaging bags for food and non-food applications (given the current regulatory limitations). This will expand the impact of the project to a number of sectors & consumers e.g. multiple use bags above 50 microns thickness complying with 94/62/EC Directive on packaging & packaging waste (amendment 720/2015).

Whereas reprocessing can mainly be applied to non contaminated rigid monomaterial and new multilayer packaging solutions, **compostability** can be more suited for specific applications. Indeed, on the one hand, according to SAVEFOOD statistics, around 22% of F&V in the EU get wasted yearly at the retailer and consumer level[3] (often while still packed). Likewise, for a range of food packaging textiles (e.g. coffee pads or tea bags), recycling or industrial composting is extremely difficult or even impossible as collected materials contain huge amounts of residues. On the other hand, lightweight is a trend to maximise resource efficiency, however films or nets made out of conventional materials are very difficult to recycle mechanically and in fact even disturb the recycling of the rigid packaging stream. The compatibility of the new solutions with existing waste management facilities and the need for additional or improved standards will be studied. Moreover, predictive models for BIOnTOP materials' biodegradation kinetics will be developed to support the eco-design of products with suitable durability as a function of their conditions of use and target waste treatment.

Thus, a **new circular bio-based value chain** will be created: linking from biomass suppliers (agri-food processors whose waste will be used in our formulations) to the relevant waste management bodies, recyclers, encompassing biopolymer producers, end users of the new bio-based products, and consumer representatives, etc. The developed materials will mainly cover the food, cosmetic, pharma and secondary broad scope packaging of early adopters present in the consortium, but their relevance goes far beyond this, with potential impact in e.g. pharma packaging, tableware and agricultural (mulching films), biomedical or sanitary sectors (covered by our advisory board, AB).

T**he main goal of BIOnTOP is to deliver novel bio-based biodegradable packaging based on versatile copolymers and coatings that optimally preserve the packed products but also our resources** (packaging based on significantly >85% renewable resources, partly produced from by-product biomass and recyclable). The different conversion processes as well as post-consumers waste sorting and recycling will be upscaled following advanced industry 4.0 approaches. Based on tailored PLA formulations and removable barrier or monomaterial coatings, BIOnTOP packaging will be compatible with **a broad range of packaging applications' requirements** but also **multiple EoL options** (materials & organic recycling, home & industrial composting and biogas production by AD)**.** Besides enabling recyclability, our coatings will also, where needed, bring UV and barrier properties as well as innovative functionalities to the consumer, like non-stick/product-release/easy emptying packaging as well as waterproofed fabrics. They will also be **safer for the environment** (reduced environmental footprint and improved value from recycling to incentive a limited leak leading to pollution of vulnerable natural habitats). **Complying with the industry needs** (cost/performance competitive vs. fossil-based non-biodegradable counterparts and legislative compliance e.g. for safe food contact), BIOnTOP will have a **great impact** on EU bioplastics & end users' sectors.

To realise this over a **48-month** work plan, the following **specific objectives**, linked to their respective WPs and key performance indicators (KPIs), later used as milestones and check points for the smooth project implementation, will be fulfilled:

Objective 1: To define the specifications for the bio-based packaging with tailored biodegradability and properties *(WP1)*

Based on an updated state of the art review, and the consumer & market requirements for the new biopolymer-based solutions for each of the target applications, a survey of available bioplastics will be defined by the end users, aiming at a substitution of the currently used fossil-based materials. This survey will be conducted in synergy with the regulatory/waste management landscape analysis performed in WP7. The process specifications will be defined by the plastic converters (steps, throughput, scale, etc.) as well as the strategy to be implemented to maximise the overall economic/environmental viability of the BIOnTOP products. ***Performance Indicator:*** By M6, the target properties, analytical methods to prove them, benchmarks for the bio-based products with tailored biodegradability and performance as well as the process specifications to produce them, will be defined.

Objective 2: To copolymerise selected food contact approved monomers from biomass with lactide, lactic acid and PLA prepolymers to obtain novel copolymers that can biodegrade in mild conditions *(WP2, 4)*

Respective batch and reactive extrusion-based copolymerisation routes will be developed with commercially available cost competitive bio-based diacids (succinic and malic acid, derived from corn and fruit waste, respectively). The resulting biodegradation and material properties will be tested. ***KPI:*** By M13, first copolymers will be available to start biodegradation tests. By M18, at least one copolymer for which biodegradation can be triggered in home composting conditions will be identified and upscaled for further conversion.

3**https://www.save-food.org/**

Objective 3: To produce PLA or copolymer blends[4] and resulting films & fibres with tailored biodegradability *(WP2)*

After selecting affordable additives that can be extracted from biomass (waxes, citrates produced by citric acid, oligomers of lactic acid, cardanol derivates, plasticizers/organic fillers from wheat), optionally reactive blends will be prepared based on PLA and the copolymers from the previous WP. Films and filaments will be processed to select the most suitable formulations. ***KPI:*** By M22, initial PLA based compounds will be characterised in view of their convertibility into the selected applications and films and fibres will be available for coating trials.

Objective 4: To develop bio-based grafting and coating to apply on BIOnTOP substrates (WP3)

PLA based (biodegradable) coatings will be developed for delivering hydrophobic properties and improved barrier & tackiness to textiles. A FA based humidity barrier nano-scale coating for films (resulting from solvent free chemical grafting) will be developed to enable deposition of monomolecular layers that drastically improve the barrier properties. Oxygen barrier whey protein coating developed in previous projects will be applied on the BIOnTOP plastic films before FA grafting. The FA grafting as well as polysaccharides will also be tested on PLA textiles as aroma barrier. ***KPI*:** By M28, the multifunctional coatings with a bio-based content above 95% according to ASTM D6866, respectively enhancing PLA's oxygen and humidity barrier by a factor greater than 10, suitable for sensitive food and personal care applications, and coated PLA textiles with better barrier, tackiness and hydrophobicity (water contact angle > 120°) properties will be available.

Objective 5: To evaluate the compounds' biodegradation in different environments and model it *(WP4)*

The biodegradation of the new polymers, compounds and coatings from WP2 and 3 will be studied via EN 13432 for industrial compostability, and the new French standard NF T 51-800 for home compostability. Extensive data will be generated to develop predictive models regarding the biodegradation & disintegration rate. Biogas potential tests will be made too. ***KPI:*** By M34, it will be shown that BIOnTOP packaging can respectively meet conditions for industrial and home composting (even at greatest thickness for thermoformed trays) and by M36 biodegradation models will be available to support the eco-design of the new products by selecting the right compound formulation for each target.

Objective 6: To set up an automated in-line identification system for the developed compounds & coatings and to study their recyclability *(WP4)*

Although our focus is on the development of new biodegradable bioplastics, recycling is a (more) valuable solution whenever possible. Therefore, prior sorting is needed. The suitable technology for identifying each developed material and evaluating their recyclability by reprocessing to packaging and textile fibres or as minor fractions in more mainstream materials, will be studied. ***KPI:*** By M36, the ability to sort and recycle BIOnTOP materials will be established.

Objective 7: To demonstrate the environmental & economic sustainability of developed compounds & products *(WP4, 7)*

LCA (life cycle analysis), s-LCA (social LCA) and LCC (life cycle cost) analysis will be performed. It will include the study of all involved process steps and developed products. The results of the suitability for all targeted end of life scenarios will be also taken into account. ***KPI:*** By M24, the interim LCA/LCC results will be used, among other criteria, to select the materials for the demonstrators, and by M48 it will show their reduced environmental impact vs. selected benchmarks feeding into post project industry, stakeholders and policy waste management recommendations to be delivered in WP7.

Objective 8: To produce and characterize bio-based monomaterial trays & films at pilot scale and to use SRM to obtain bags *(WP5)*

Compounding of the relevant formulations will be upscaled to pilot scale. The compounds will be used for the flat film extrusion of monolayer thin films (lids) and thick films to be thermoformed as trays in a pilot plant in an industry-relevant environment. The F&V packaging demonstrators will be tested by microbiological, sensory and chemical-physical analysis as well as in terms of suitability for food contact according to EC 10/2011. They will be compared with standard counterparts. SRM will be used in the blown extrusion of bags. ***KPI:*** By M42, bio-based trays & films for F&V with the potential to biodegrade in home composting and recyclable bags from SRM will be available and by M48, their potential to substitute fossil benchmarks will be demonstrated.

Objective 9: To produce and characterize bio-based multilayer packaging at pilot scale *(WP5)*

Films similar to those used for monomaterial packaging will be used for application of whey and FA coatings to produce multilayer trays for cheese and related applications as well as flow packs for personal care uses in an industry-relevant environment. They will be compared to conventional counterparts. ***KPI:*** By M42, multilayer package which material have the potential to be organically recyclable or reprocessed after coating removal will be available, and by M48, their potential to substitute selected benchmarks will be demonstrated.

Objective 10: To produce and characterize bio-based nets and fabrics at pilot scale *(WP6)*

The selected compounds developed in WP2 will be upscaled for the extrusion of nets and filaments that will be further processed into textile fabrics and will be coated with the coatings developed in WP3 in an industry-relevant environment. The resulting solutions will be compared to conventional counterparts. Moreover, the application accessible for the new textile materials and

4 new PLA and copolymers blends from the projects will be referred as BiontopBiontopBIOnTOP compounds and substrates in the proposal

coatings, as well as its counterpart from SRM, will be preliminarily prospected (e.g. tea bag, reusable food wrap, carrier bags, etc.). ***KPI:*** By M42, optionally functionalised (grafted, coated) nets and fabrics with the potential to biodegrade together with their organic content in home composting, as well as reusable food wraps and textile bags from SRM with suitable properties to substitute set benchmarks will be available (M48).

Objective 11: **To study the EoL options for the new bio-based packaging and validate the prediction from the biodegradation models** *(WP4)*

The final products (including materials alone but also together with packed F&V) will be studied in terms of organic recycling as in objective 6. In terms of acting as contingency for accidental littering and specify further possible applications, preliminary marine biodegradation and soil tests will also be performed. The predictions will be compared with tests results which will be used to improve the models. ***KPI:*** By M48, the new products' biodegradability in application relevant conditions and biodegradation models will be validated (target prediction +/-20% accuracy on compounds' time to reach full biodegradation).

Objective 12: **To define sustainable circular value chains & cross-sectorial connections and survey of consumers perception** *(WP7)*

A value chain sustainability analysis will be performed based on existing best practices in the field of bio-economy, markets for the products, as well as the readiness of key actors to shift to such bioplastic solutions. Furthermore, the consumer perception of such products and the policy and regulatory barriers will be studied. The value chain study will also address how the results could tap into the much broader potential of the technologies developed. ***KPI***: By M24, an interim report providing an insight on the value chains where BIOnTOP results could be applied as well as robust economically, environmentally and socially beneficial business models will be provided. Its final version will be delivered at M48 and will also include a thorough market analysis across the involved sectors.

Objective 13: **To maximise the innovation impacts by contributing to the post-project development and uptake of the project results for growth and jobs** *(WP7)*

The project knowledge and results will be managed and protected. Business and exploitation plans will be delivered for all Key Exploitable Results (KERs) to map the post-project phase and business models will be set up. A professional communication campaign to outreach, engage with and mobilise our key stakeholders will be designed and executed. In addition, significant efforts will be dedicated to encourage policy makers to support beneficial (or at least non-discriminatory) legislative frameworks for BIOnTOP solutions and contribute to the development of new or improved industry standards. In particular, our project will help to give initial feedback to regulators, with potential for contributing to standards regarding specific biodegradation frameworks, which will increase the impact of BIOnTOP. Of course, REACH compliance will be considered at all development stages. Regular knowledge transfer activities will be organised among the partners to secure the post-project stage. Besides, specific trainings will be targeted to scientists, end user industry and policy makers to guarantee upskilling and knowledge exchange. ***KPI:*** By M6, a robust communication plan to support the recognition of BIOnTOP as a reference in the bioplastics market place will be available, as well as to secure stakeholder engagement with the project. From M24 to M48, at least 3 training sessions will be held to ensure knowledge exchanged. By M48, final plans will be in place for the transition to the post-project phase to advance the TRLs and reach the market, sources of further funding will be analysed focusing on applications where a return on investment is feasible based on the assessment performed in WP7, as well as policy and regulatory recommendations.

1.2 Relation to the BBI-JU Annual Work Plan

The BIOnTOP proposal, addressing the topic BBI.2018.SO3.R10 ***- Develop bio-based packaging products that are biodegradable/ compostable and/or recyclable***, will contribute significantly to the KPIs set out in the Strategic Innovation and Research Agenda (SIRA)[5] and is fully aligned with the BBI roadmap as will be further discussed in section 2. Our overriding goal aligns with its 4 strategic orientations (SO) (fig. 2): SO1) we will use biomass derived feedstocks (significantly >85%, although minor non bio-based compounds, e.g. processing aids or catalysts, may be used if no alternative exists); favouring local resources of second or third generation wherever possible, e.g. waste from agro-industrial value chains: plasticizers, malic acid as well as lactic acid from fruit processing, FA from oil by-products, and wheat fillers as milling residue; SO2) the core of our activities will deal with the eco-design of packaging based on novel copolymers and compounds from selected bio-based building blocks; SO3) we will tackle the requirements of several mass products together worth 192 bn €/year for which the materials will be validated (SO4) awareness raising campaigns will allow us to meet the needs of our EU end user industries as well as the consumers' demand for greener products. Besides creating socio-

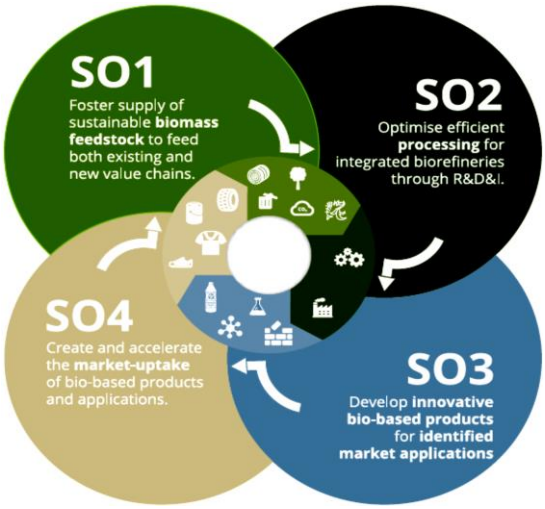


Fig. 2: The 4 SO in the BBI SIRA 2017

5 Strategic Innovation and Research Agenda (SIRA) BIO-BASED INDUSTRIES for Development & Growth in Europe – May 2017

economic value, the project will also help the EU in implementing better waste management strategies as well as reducing the major societal challenge reflected in the European Plastics strategy. Further extending the impact of the BBI-JU, we also have beneficiaries from countries where no BBI participant in awarded funded projects are based like Czech Republic, as well as for personal care & pharma packaging which are, to our knowledge, not yet covered in the running projects. We further cover cross cutting advanced materials and processing KETs. The detailed relation of the proposal to the specific challenge and scope addressed by the topic are outlined in table 1 below:

Table 1: How BIOnTOP relates to the scope and challenges of the BBI 2018.SO3.R10 topic

Topic text	How BIOnTOP addresses these challenges/scope
To make the packaging EoL significantly more sustainable. *Design new processing systems for functional bio-based packaging products that are reusable, recyclable, and/or compostable and biodegradable in various environments to reduce their overall environmental footprint., as an alternative to the current identified benchmark products.*	The packaging formats that we focus on are generally both neither renewably sourced, nor recyclable at EoL independently of being based on monomaterials (light packaging) or multimaterials (multilayer or coated products). As seen, even biodegradable polymers have weaknesses in terms of use (properties) or even EoL (limited biodegradability in mild conditions or at high thickness). Innovations in the area of bio-based packaging give unique opportunities to make their production and use more circular. As will be tested in WP4, our new bio-packaging will be eco-designed to be compatible with the most suitable **end of life options on a case to case basis according to best practice studies from ISWA (member of our AB) on a general and regional basis**. Applying Circular Economy (CE) principle, reusable packaging (bags & wraps) will be delivered to maintain the resources in use at their outmost initial value for longer. For single use materials and at the end of the multiple use packaging where applicable, final material recycling will be considered. Since the upstream sorting of EoL waste is not always optimal, we will check that each material can be properly identified as a pre-requisite to be handled in the right waste management stream and, if applicable, recycled into new materials for packaging or textile applications (we can note that there are increasing efforts for recycling PLA as already done by ROMEI). For applications contaminated with organic substances after use, different composting routes will be available. Our solutions will be suitable for industrial composting even at high thickness where this EoL option is available, for home composting and for AD. **Soil and marine biodegradability** will also be tested. Where applicable, eco-toxicity tests will be performed on the developed materials along with the biodegradation trials. The **eco-design of our new materials** will be further enabled by the **novel biodegradation predictive models** developed in WP4. All these materials and processes along the product life cycle will allow meeting society's and users' demands to improve the EoL options to avoid landfilling and support manufacturers, who are responsible for the suitability of the materials they commercialise, to enforce relevant **waste legislations**. Incrementing our circularity, we will use biomass by-products as raw materials fibres from wheat bran, monomers such as lactic and malic acids from fruit waste, coatings from cheese by-products and nano-scale surface modifications from ubiquitous fatty acids from plant oils (most sustainable will be selected). Although outside of our scope that is rather on later packaging conversion, research into biorefining additional building blocks such as LA from second (non edible) and third generation (waste) feedstock at suitable purity is currently pursued intensely. As soon as on the market, those could be substituted by the relevant monomers to further increase BIOnTOP sustainability. In case of multilayer materials, we will either resort to removable coatings (whey protein +FA grafting), so separated monomaterial can be obtained in the pre-washing step of the recycling process, or to coated textiles using similar polymeric matrix as the textile it is applied on.
In case of multi-layer products, consider the feasibility of producing multi-layer/single-polymer solutions, provided they meet all required functionalities and outperform state-of-the-art alternatives for sustainability. *Other innovations in functionality and production are assets.*	In terms of multifunctionality brought by our coatings, we will bring innovations in terms of providing packaging films and textiles with: i) easy emptying behaviour (relevant to minimise residual product at EoL of e.g. cosmetic sachets), ii) non-stick properties (relevant to improve consumer experience e.g. for cheese release lids, iii) waterproof properties (e.g. for carrier/ shopping bags). Furthermore, our protein barrier coatings also ensure UV barrier while keeping coated film transparency, again a preferable feature vs. e.g. non-transparent metallised barrier. Finally, alginates will be used as aroma barrier for tea bags providing an added asset vs. fossil counterparts or emerging PLA solutions. Further maximising resource efficiency (optimising properties and use of materials by barrier layers that can reduce packaging thickness), the materials to be delivered will have a number of unique selling points vs. conventional counterparts: **i) close to 100% bio-sourced, ii) superior food preservation in case of barrier packaging**, **iii) meeting converters' needs in terms of processability** (standard converting techniques used also for novel materials) and **iv) EoL options adjustable to the respective applications**. Other properties like mechanical, thermal resistance, optical and barrier properties, etc., will be tuned to match the requirements of each of the selected applications employing or combining several approaches, e.g. modular copolymers structure, fillers, REX, etc. (see section 1.4). Benchmarking will be done vs. currently broadly used fossil and emerging bioplastic solutions along with the validation of the new materials for each application and economic as well as environmental sustainability studies of both delivered products and processes to demonstrate such comparison on a cradle to cradle basis.
Address the production process, including the necessary improvements to coating and chemical FA grafting steps to obtain the target end-products and their specifications. TRL3 to 5.	We will develop different approaches for obtaining **biopolymer-based packaging for multiple food and personal care products as well as multisector bags (secondary packaging) through innovative processes**. We will also evaluate the theoretical applicability of our materials to other applications (further textiles products, pharma packaging, among others). Therefore, we will initially select different monomers, fillers and additives that will be compounded with PLA, as available bio-based polymer, but also with new BIOnTOP copolymers. We will upscale the coating process for both plastic films and textiles. This encompasses new coatings processes based on whey protein and their further top coating using fully bio-based fatty acid anhydrides nanolayer. If needed, in specific packaging formats, lamination will be performed but it is not expected to be modified while using our new materials. The start, progress and end TRL of each technical component is justified in table 3. In each WP, from the bioplastics & coatings development in WP2-3, to the production and characterisation of the derived bio-based packaging for each target application (WP5-6), we will work subsequently **at lab and pilot scale** obtaining significant amounts of material for an industrially relevant further characterisation and validation of the demonstrators. Different techno-economical aspects of scalability will be considered in WP5-6 to give recommendations for successful transfer and sustainable production scale up, so the results TRL is further matured to reach the market.
Actively industry	The whole **circular bio-based value chain** is covered by BIOnTOP partners. All WPs involve industrial partners in

participation to demonstrate integration potential for the developed concepts into current industrial landscapes or existing plants towards quicker deployment and scaled up to apply industrial-wide.	the **supply of the raw materials as well as in the upscaling and testing of the new packaging materials** based on the initial industrial requirements defined in WP1. The consortium has a strong industry component including the participation of 9 SMEs, 6 Large enterprises and many more associated via membership in the participating industry association. It is noteworthy that 4 partners are associated and 2 full BIC members and therefore committed to making the BBI roadmap successful. In terms of end users**,** the **food & personal care** sector are covered by 3 industrial partners. This dense ecosystem will allow researchers and early adopters alike to cooperate, understand the specific market demands, and set the basis for the long-term success of the new value chains. They will also drive the **knowledge transfer** across all involved sectors and the **upscaling** of the results obtained by the RTDs with a view to the smooth post-project replication.
Demonstrate the benefits versus the state-of-the-art and existing technologies. This could be done by providing evidence of new processing solutions and new products obtained. Involve consumer organisations, together with recyclers and composting plant representatives.	The validation of the demonstrators obtained for each sector (nets, trays, flowpacks/sachets, bags of different types…) will be performed in WP5 and 6. This activity will encompass the testing of the processability of the new compounds and coatings by the converters and testing by final users according to their established quality control approaches but also the **comparison of the BIOnTOP solutions with existing counterparts on the market**. Any additional standards that the new products need to comply with will be considered (e.g. the packaging will be tested by migration to establish **food contact safety** in WP5-6). A list of selected substances and food contact approval number in compliance with existing legislation for use in food contact materials is given in section 2.1. We are confident that our developments can convert into a sustainable supply of usable material with **functionalities that outperform existing biopolymers and fossil-materials but employing processes compatible with the existing packaging lines**. Our consortium, optimally constituted as a preliminary seed for the future value chain, involves a number of partners that are linked to end of life management: OWS(terminated)/OWS RF (composting & biogas plant design), SILON (reprocessing packaging into textile), ROMEI (material recycling) and BIO-MI (production of materials and final products and recycling). Furthermore, we mobilise a consumer organisation (MC) who will support understand driving factors in consumer acceptance in our requirements in WP1 and in their final perception in WP7.
Assess the environmental impacts of the developed processes or products using LCA methodologies based on available standards, certification, accepted and validated approaches (section 2.2.5 BBI JU AWP 2018). *Allow for pre- and co-normative research necessary for developing the needed product quality standards.* *Any potential hazards associated with the developed processes and products should be analysed to ensure that the products comply fully with REACH legislation and other toxicity, safety requirements and any relevant EU legislation.*	In WP4, a thorough **LCA** will be performed resulting in an identification of the hot spots and recommendations for the selection of products and processes in the project to improve sustainability of the products and EoL options. In terms of **pre-normative analysis**, we will contribute to the discussion on **emerging standards** for biodegradation in specific environments that are still under construction as a **harmonized EU framework** (e.g. home compostability and AD) and, to that end, provide a number of biodegradation test results, especially in mild conditions and anaerobic conditions whereby reliable systematic studies lack (see section 1.4). This task will be embedded in our comprehensive set of activities focused on **standardisation and certification**, adherence to and compliance with agreed standards and best practices is essential for the market introduction of new materials and products. Therefore, the relevant and applicable standardisation and certification landscape for BIOnTOP will be thoroughly assessed covering: i) applicable standards for biodegradation such as EN 13432 (industrial composting of packaging), or NF T 51-800 (home compostable plastics); ii) applicable standards for determining bio-based content (e.g. EN 16640, EN 16785) and relevant certification schemes; iii) current developments, e.g. in the CEN working groups CEN/TC 411 (Bio-based products) and CEN/TC 249 (Plastics); iv) relevant sustainability certification schemes, e.g. for sustainable sourcing of bio-based materials. The lack of standards regarding biodegradable plastics/products in AD systems will also be tackled since it is also a valuable organic recycling procedure (to some extent even more than recognized industrial composting as also biogas is produced. We will identify areas in which BIOnTOP can contribute to the improvement and development of standards. In order to bring any relevant findings of the BIOnTOP action to bear on the field of standardisation and certification, the consortium will engage in an in-depth exchange with the relevant standardisation bodies, in which EUBP and ISWA (AB) are involved, and with the most important independent certifiers in the field of bioplastics, namely TÜV AUSTRIA Belgium and DIN CERTCO, who are also EUBP members. In terms of policy and regulation, the current political and policy framework for bio-based and biodegradable plastics will be surveyed, detailing any beneficial and detrimental legislation and initiatives, and the key actors on the EU level, and in some exemplary Member States that exhibit either very progressive or very inhibitive legislation for bioplastics, as well as depending on their waste management land scape. In order to positively influence the political framework for BIOnTOP exploitation, specific fact-based, scientifically backed information will be used in policy recommendations in WP7, for example in the areas of waste management and increased resource efficiency, to help utilise the potential of bioplastics in implementing circular bioeconomy principles. BIOnTOP will reach out to EU decision-makers to disseminate this information and thus contribute to progressive policy initiatives supporting the bio-based economy. In terms of safety, our safe by design packaging will consider food contact legislation from the project outset and we will also perform migration tests on the primary as well as secondary materials from the project. Furthermore, a task in WP7 will be dedicated to the analysis of REACH compliance of our innovation.
Include an economic viability performance check (value chain and market analysis) of the developed products and processes, analysis of social impacts. Social science as cross cutting priority	The proposal focuses on improved copolymers derived from PLA that is the **most affordable biodegradable biopolymer nowadays** available at industrial scale with a high maturity. It is therefore suited for mass short-term use applications where EoL has a major socioeconomic impact. To further improve its **cost competitiveness**, it will be blended with low cost waste **biomass derived additives like plasticizers and organic fillers.** Likewise, the very thin (nano-scale) fatty acid (FA) grafted layers are very material efficient and therefore cost effective. Our target for the new packaging is to be cost competitive (iso-cost vs. PLA or 40% maximum increase vs. fossil-based benchmarks, see estimates table 5). An LCC will be performed in WP4 and economic aspects will be included in the business plan in WP7 together with a value chain analysis. Social aspects will be covered both in the consumer perspective study in WP7 and by the s-LCA in WP4.

In keeping with the **expected scope of proposal under BBI 2018.SO3.R10 topic,** the BIOnTOP proposal has a budget of ca. €5.5 million and an EC contribution of ca. €4.2 million, therefore being within the range guided by the call text (including 31% in kind contributions by partners as described in section 2). The project will generate excellent value for money for the bio-based industries value chains while delivering a number of bio-based solutions as well as new applications valorising agro-food wastes.

1.3 Concept and methodology

BIOnTOP is a 4-year RIA project with 21 partners from 8 countries across Europe (including a pan-European association as well as global players). The consortium is very strong with mature suppliers pioneering bioplastic & packaging converting companies, innovative technology companies and RTDs, as well as large end users. The proposal aims to develop and validate the technical and economic feasibility of new eco-designed bio-based packaging films, trays and textile solutions responding to actual market needs with sustainable end-of life stages, that will preserve our resources and environment.

While modern economies highly depend on plastics in Europe, in fact consumes in the range of 50 Million tonnes per year, only 25-31% of the plastic waste in Europe is currently recycled and the larger share is incinerated or even landfilled[1] (high variability between countries). In the light of the EU's policy objective of achieving a resource efficient recycling society it is hard to accept that in Europe we still landfill nearly 40-50% of plastic waste[6]. This all contributes to the depletion of limited resources. Proving renewably sourced solutions in our target sectors is of greatest relevance as they consume more than 40% of all plastic produced each year. On the other hand, while growing in market share, bioplastics still have a very limited share among plastics (approximately 1%). They can, however, provide advantages in terms of switching to renewable non-fossil feedstocks and EoL modularity, especially for those short terms single use applications which, by nature (e.g. lightweight nets) or due to contamination (fouling or leftover products), are not recyclable (mechanically or organically). To realise this potential, more research and development has to be invested in further improving bioplastics' specific functionalities, and more cost-competitive solutions have to be found to ease the eventual large-scale superior performance and market-driven uptake. Besides those often non-optimal disposal routes, **plastics**, in particular packaging (bags, bottles, lids, etc.), **account for 89-95%** or roughly 8 million tonnes (Mtons) of **marine litter** and 17 Mtons of plastic packaging is irreversibly dispersed on land every year worldwide[7]. Thus, the very slow rate of degradation of plastics leads to an accumulating pollution persisting hundreds of years.

The BIOnTOP project will:

- Develop **a range of complementary bio-based material formulations with tailored biodegradability and other relevant applicative properties** which, if possible, are derived from available agro-food by-products.
- **Validate the use of the resulting bioplastics, coatings and biocomposites** in a number of **food & personal care packaging** as well as of **their environmental impacts and economic feasibility** in comparison with existing benchmarks.
- **Provide recommendations for suitable business approaches** for the new materials from an integrated bio-based **value chain perspective** including, among others, consumers & end users requesting more sustainable solutions and waste management bodies through a series of outreach activities exchanging knowledge between the relevant actors.

The considered technologies are summarised in Table 2, together with their purpose and materials used. Resulting solutions will be industrially compostable or mechanically recyclable in case appropriate collection is feasible, while providing bio-sourcing, among other technical assets vs. market benchmarks.

Table 2: BIOnTOP bio-based materials innovations to achieve project goals in terms of tailored biodegradability & properties

Technology used	Effect	Trays & films		F&V nets	Tea bags	Coated textiles	Multiuse bags
		Monolayer	Multilayer				
Copolymerisation with bio-based diacids	Modulated biodegradation rate and temperature triggering	x	x	x	x	x	x
Addition of organic fillers, plasticizers, compatibilizers from agro-food waste	Impact on: processability & processing, biodegradation, cost reduction	x	x	x	x	x	
Protein coating +FA grafting on films	Oxygen and water vapour barrier for MAP packaging and easy emptying properties		x				
PLA coating or FA grafting on textile	PLA textile with PLA based coating: bio-based mono-material					x	x

In BIOnTOP, our trans-disciplinary team combines all the relevant skills and organisations needed to address the full extent of the challenge of developing and validating innovative sustainable bio-based coatings and new bio-copolymers in plastic packaging solutions. The participation of a number of leading industry players proves the actual soundness of the concept to generate large impact and will maximize the consortium capability in **transferring this know-how** to enable post project further demonstration and qualification into broadly available market ready solutions. Furthermore, the participating association/NGO, covering not only the provision of required biopolymers-based solutions but also the consumers, will also help raising **environmental awareness** and policy outreach **that, besides their great performance and competitive cost, could further encourage the adoption of our new solutions**.

To complement our consortium, further stakeholders will be involved in an Advisory Board (AB). The AB will consist of Packaging Cluster, ISWA, REVERDIA, BIOTREM, FERRER, EU Textile Platform (ETP), Ginegar, Flandersfood, Roll & Eat, Colruyt Group, etc. (details and support letters annexed to section 5), number of them from the BIC. They will be consulted as necessary for both technical, policy and business-related insights and further AB members may be appointed if needed.

BIOnTOP will enable know-how transfer through its **value chain of actors**: from technology institutes & academia (4 RTDs:

6 http://ec.europa.eu/environment/waste/plastic_waste.htm
7 THE NEW PLASTICS ECONOMY RETHINKING THE FUTURE OF PLASTICS, Ellen MacArthur foundation, 2016

AIMPLAS, INSTM, CTB and ASU) to industry participants (9 SMEs, an industry association and 6 Large enterprises leaders in their field): monomers and other bio-based raw materials to be used in the new compounds will be supplied by partners (WEAREBIO, TOTAL-CORBION) and AB members (REVERDIA), polymer producers (TOTAL-CORBION, and other blend partners producers through EUBP members), plastic/packaging converters (PLANET, EMSUR/INNOTECH, SILON, BIO-MI, CMSA, SIOEN), end users packers (WEAREBIO, ENTREPINARES, UBESOL and further ones in AB covering even retail stage), recycler, technological solutions H&S and environment assessment providers (ROMEI, IRIS, ARCHA and OWS(terminated)/OWS RF) and an innovation specialist (ENCO).

Positioning of the project

Most activities performed in BIOnTOP project start **at TRL3**– experimental proof of concept or technology validation in lab (TRL4**)**. To maximise our chances of success for the overall interdependent results, we have selected a number of complementary approaches to tailor biodegradation and properties. Most technologies are based on prior research from involved partners for the same or other base biopolymers. As certain elements actually start with a more advanced TRL (e.g. whey protein based oxygen barrier, converting processes), that will help us focusing the efforts where required, it is deemed realistic to state that **all project technologies will arrive in at least a consolidated TRL5 by the end of the project** when our technologies will be combined in the final project demonstrators to be produced first at RTDs pilot scale (extensive validation at TRL5 and production trials at MRL5) and some demonstrators will undergo preliminary demonstration tests in semi-industrial environments (TRL6). Table 3 provides a summary of the TRL advances that will be achieved for the key technological components of the project.

Table 3: Summary of the TRL advancements that will bring about as result of the BIOnTOP project

Technology	Initial TRL	Final TRL	Comment
Copolymers from bio-based (malic, succinic) diacids and lactide/ lactid acid or PLA prepolymers	3	5	New applications of bio-based diacids for biodegradable polymers have been the subject of extensive literature (see 1.4) but generally not for modulating their biodegradation profiles. On the other hand, several LA copolymers are commercially available but not meeting home compostability conditions. In BIOnTOP, after initial lab trials to identify best formulations, the production of the copolymers will reach a TRL5 in an industrially relevant environment pilot scale (25 kg of copolymers by batch process and 75 kg by REX). The systematic understanding of the effect of the copolymer structure on the biodegradation together with the derived biodegradation predictive model promises ground breaking innovations.
PLA reactive extrusion **resp. for copolymers based on lactic acid or for PLA*	3-4*	5-6*	INSTM has a patent licensed to PLANET for production of PLA based transparent films with the use of reactive plasticizers. The formula is suitable to obtain films with properties comparable to PP or HDPE, and research is ongoing to achieve properties such as LDPE. PLANET has experience with large scale compounding of PLA-based formulations, but the process needs to be adapted to the copolymers developed in the project, considering its specific processability and using reactive renewable additives to match the suitable performances requirements. In BIOnTOP, the production will reach initial trials at TRL6 in PLANET (200 kg of selected compounds).
Biocomposites with renewable agricultural filler	4	5-6	Prototypes (pots, cream jars, sheet for thermoforming) were produced by INSTM on pilot scale with PLA with natural fillers such as wood, algae, potato, bran, legumes. In BIOnTOP, wheat fibres will be compounded with the new copolymers, also considering REX approaches, to modulate EoL behaviour and cost of the materials for the different applications
Fatty acid grafting	4	5-6	FA grafting was performed on paper at TRL up to 7. Whey protein coatings were already applied at TRL 7 and above but only combined with FA grafting in preliminary trials at TRL3. Tests in pilot printing lines will allow reaching TRL 5.
Melt spinning of BIOnTOP compounds for production of nets, filaments for textiles and nonwovens	4	5-6	Although PLA is already industrially processed to certain textile materials, the processability of the new biodegradable formulations needs to be investigated and demonstrated at pilot level. The processing of nets, nonwovens and filaments via the melt spinning process will need adjustments (e.g. processing temperature, drawability). In BIOnTOP, nets will be produced up to TRL6 in CMSA's pilot based on their know-how on commercial PLA nets. Likewise, after tests at CTB, SIOEN and SILON will respectively test at TRL5-6 textile conversion to filaments and nonwovens.
PLA Coating for textile	3-4	4-5	PLA coatings in general are poorly developed. The main application where PLA is used in coating processes is cardboard, where PLA provide waterproofness to e.g. paper cups. These coatings are however not suited for textiles (too hard and incompatible extrusion coating process). There are several initiatives to develop PLA coatings for different textile items in laboratory prototypes but they are not mature enough for industrialization (TRL3-4). Moreover, coating of the newly developed PLA materials, with improved biodegradability, will need further finetuning. Tests at CTB will prove the feasibility at lab scale after which the coatings will be upscaled to pilot scale by SIOEN (TRL4-5).
Blown extrusion of recycled and virgin PLA copolymers	3	5	PLA copolymers as prime materials and PLA copolymers as SRM recycled in the frame of project will be introduced as a base for the bag applications at BioMi (currently produced at TRL 9 using PP, PE, PET, Ecoflex and starch based bio-plastics) in subsequent blowing extrusion trials at TRL 3-5

National or international research and innovation activities linked to the project

BIOnTOP will build on prior state-of-the-art as well as IP belonging to several project partners. A selection of relevant projects that the BIOnTOP partners are/have participated in are listed in Table 4, and derived expertise, methodologies, lab or pilot processes that will feed into BIOnTOP (projects ordered along the relevant individual technology components/product life cycle).

Table 4: Relevant research and innovation activities (additional ones can be found in the respective partner profiles)

Project (and reference), Description	Partners involved	Opportunity for carryover to BIOnTOP
Bread4PLA Demonstration Plan Project to produce Poly-lactic acid (PLA) Biopolymer from waste products of bakery (LIFE+) (2011-2014)	AIMPLAS	Proofs of feasibility to obtain LA and derived PLA from 3rd generation LA (processed from bread waste).
BIOREFINE-2G Development of 2nd Generation Biorefineries (FP7-KBBE-613771) (2013-2017)	AIMPLAS	Production of Dicarboxylic Acids and Bio-based Polymers Derived Thereof.
WHEYLAYER2 Barrier biopolymers for sustainable packaging. (FP7-315743 FP7) and WHEYLAYER- Whey protein-coated plastic films to replace expensive	IRIS, INSTM (via UNIPI)	Developed whey protein based oxygen barrier coatings for plastic films (feeds in WP3).

polymers and increase recyclability (FP7-218340-2)		
PLASTICISED Innovation in biopolymer applications by the use of optimised plasticisers and novel processing techniques (Flanders traject) (2015-2019)	CTB	Plasticisers for PLA to allow application of PLA coatings.
AGRIMAX Agri & food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications (H2020-BBI-PPP-720719) (2016-2020)	IRIS, INSTM, OWS (terminated), AIMPLAS, ARCHA	Bio-based blends and composites with PLA, PHAs, bio-based additives and natural fibres.
CITRUSPACK - Revalorization strategies within the circular economy for the use of citrus waste in green packaging and cosmetics (LIFE16 ENV/ES/000171)	OWS (terminated)	PLA reinforced with fibers to produce compostable packaging
EFFECTIVE (H2020 BBI) - Advanced Eco-designed Fibres and Films for large consumer products from bio-based polyamides and polyesters in a Circular Economy perspective	BIO-MI	Developing new monolayer packaging solutions via several transformation technologies.
BIOBOTTLE Biodegradable solutions for packaging of liquid dairy products (FP7-SME-606350) (2014-2016)	AIMPLAS, OWS (terminated)	New biopolymer blends development and processing for dairy packaging applications (pouches and bags)
DIBBIOPACK Development of injection and blow extrusion moulded biodegradable and multifunctional packages by nanotechnologies: improvement of structural and barrier properties, smart features, and sustainability (FP7-NMP-280676) (2012-2016)	INSTM, ARCHA	Production of PLA based packaging by injection moulding and by calendering (sheets).
PLA4FOOD Active multilayer packaging based on improved performance PLA formulations for 4th range food products (FP7-SME-262557) (2010-2013), BIO4MAP Transparent and high barrier biodegradable film and sheet for Modified Atmosphere food packaging (FP7-SME-606144) (2013-2016)	AIMPLAS	Multilayer PLA processing for film and sheet extrusion and thermoforming for food barrier packaging applications.
ECOBIONET -Industrial implementation of biodegradable and compostable packaging nets for agricultural and shellfish products (Eco-innovation)	AIMPLAS	Development of biodegradable material for packaging nets.
BB100 Pure Nature: 100% bio-based (INTERREG, 2018-2020)	CTB	Use of bio-based additives for the textile sector.
BIOAGROTEX Development of new agro-textiles from renewable resources and with a tailored biodegradability (FP7-NMP-213501) (2008-2012)	CTB	Development of PLA based agrotextiles (at present on the market) – Not biodegradable under mild conditions.
BIO4SELF Bio-based self-functionalised self-reinforced composite materials based on high performance nanofibrillar PLA fibres (H2020-NMP-685614) (2016-2019)	CTB	Development of PLA based self-reinforced composites with high mechanical properties which can be used as basis for BIONTOP textiles.
POLYBIOSKIN High performance functional bio-based polymers for skin-contact products in biomedical, cosmetic and sanitary industry (H2020-BBI-PPP-745839) (2017-2020)	INSTM, EUBP, IRIS	Showcase the usability of biopolymers in other sanitary applications like diapers as well as biomedical and cosmetic sector.
SUPERTEX (ECO/10/277225/SI2.596871) Sustainable Flame Retardant Technical Textile from Recycled Polyester	CTB, ROMEI, INSTM (UNIPI)	Recycling of multilayer scraps from packaging industry to obtain materials for technical textiles
BIO-EOL End-of-life possibilities of biopolymer based consumer products (ERANET-Cornet) (2014-2015)	CTB	Developing EoL recycling routes for PLA consumer products; mechanical and chemical recycling.
MULTICYCLE (H2020 SPIRE) Advanced & sustainable recycling processes and value chains for plastic-based multi-materials, in grant preparation	IRIS, ISWA, CTB, AIMPLAS, SILON	Use of packaging SRM in textile, optical sorting, etc.). Dissemination strategy around a Circular Plastic Helix that can help creating synergies.
OPEN-BIO Opening bio-based markets via standards, labelling and procurement (FP7-KBBE-613677) (2013-2016)	OWS (terminated)	Pre-standardisation work on biodegradation in marine conditions, soil and organic recycling (industrial and home composting, AD).

In addition to these projects, during Task 7.5, we will build links with numerous other synergistic projects and initiatives from H2020 (e.g. BBI, SPIRE, SC2, SFS, SC5, and BIOTEC calls), as well as national and regional schemes (RIS3) for exchange of information and best practices. In addition, BIOnTOP will engage with the **relevant policy and standardization body** to act as a catalyst for innovation-related actions but also with selected associations to foster the creations of cross sectorial connections with BIOnTOP project results, supporting the transition to a post-petroleum society by use of sustainable bio-based materials across the whole bio-based value chain. In addition, we will also link to other activities of BBI-JU through project partners (number of which are already involved in awarded BBI projects linked to biopolymers applications in packaging and beyond), as well as to the SPIRE PPP (that has included in its Roadmap the use of sustainable new raw materials for the European Process Industries, etc.), SUSCHEM, CEFIC, Plastics Europe, and other associations from the plastic, packaging, textile sectors e.g. personal care and food end users with a primary focus on organic segment.

Overall approach and methodology

Our approach and methodology involve two main components:

1) **R&D and validation:** Research activities with the main focus to deliver packaging materials with modulated biodegradability and compatibility within a number of EoL scenarios. New copolymers will be developed in batch reactor and by REX at lab scale and upscaled to a small pilot scale to produce sufficient amount for the validation of 7 demonstrators case studies covering the needs of different target applications in terms of properties, format, available waste management and converting processes. Besides, biodegradation studies will be performed along the development and the validation work will focus not only on the technical performance of the different materials and demonstrators, as well as scalability of their production process, but also on their environmental, social and economic sustainability using e-LCA, s-LCA and LCCA methodologies among others. The safety and compliance with other applicable standards of the developed products will be studied.

2) **Innovation:** This work will accompany and cross-cut the R&D work with the aim of preparing for post-project further development leading to the future replication of the project results. It consists of the following activities:
   a) **guidelines and recommendations for future full-scale implementation of the proposed technologies** via analysis of the **new value chains**, of the regulatory, social, consumer perception, safety and relevant ethical aspects.
   b) **dissemination** of the project results, including a series of activities to exchange knowledge between the stakeholders to validate their interest in the results of the project.
   c) **exploitation** and business planning to further increase the TRL of the developed processes and products beyond an expected final TRL5 to carry them over the notorious 'valley of death' towards commercial adoption.

How sex and/or gender analysis is taken into account in the project's content

In relation to gender analysis, this concept is relevant to the content of the project in several aspects. To this end, innovation will consider the gender dimension and cater for the needs, motivations and differences of females where possible. In the development state, taking gender into account will help yielding new research results and spark creativity, opening new avenues for future research as recognized in the past for those who give it serious attention[8]. Gender analyses will be highly relevant in consumer perceptions study (task 7.2) which outcome will be used in the communication and dissemination context to ensure our message is appropriate to reach both males and females. For example, the consideration of environmental factors on material selection is likely to vary depending on the gender, as well as the concern on their EoL. Women are predominantly involved both in the purchase and waste sorting of packaging at household level. In addition, female inputs to new business models will also be considered. Indeed, while the presence of females in the plastic converting sectors is traditionally lower than of males, there is a strong presence of females along the value chain of the bioeconomy industries where they play an increasingly prevalent role in consumer decision-making.

The gender dimension of our consortium and our measures to align with the article 16 of the H2020 programme are presented at the end of section 3.3. This is important in ensuring the education and training of the next generation of scientists, technologist and others needed in the EU to foster the future of the circular bioeconomy and maximise its creativity and innovation potential.

1.4 Ambition

Advance beyond the state-of-the-art and the extent to which the proposed work is ambitious

The use of bio-based materials is a very interesting way to decrease the environmental impact of end users and packaging industries. However improvements in the functionality and properties of bio-based materials are required to fill performance and price gap with fossil materials that generally hampers their replacement. In this sense, **BIOnTOP project has the ambition to develop novel packaging products in different formats (trays, films, flowpacks and textiles) using new copolymers based on PLA and other bio-based diacids as well as innovative barrier coatings** that will lead to significant knowledge advancements of the state-of-the-art in packaging materials with tailored end-of-life phase as follows:

State of the art in the field of home and industrially compostability and degradation under anaerobic conditions

Industrial compostability was the first process of organic recycling for which a test strategy including specific requirements was developed. This resulted in the harmonized EU standard EN 13432 *Requirements for packaging recoverable through composting and biodegradation - Test scheme and evaluation criteria for the final acceptance of packaging* (2000) according to which a broad range of products are certified. However, during home composting, the high temperatures (> 50°C) required by some polymers such as PLA for starting the degradation and obtained during industrial composting processes, are mostly not reached. At lower temperatures, degradation rate is reduced or even limited. Furthermore, in order to be home compostable, a material must demonstrate sufficient biodegradation and disintegration at ambient temperature. This principle was first applied by the OK compost HOME Conformity Mark of certification institute TÜV AUSTRIA Belgium. Currently, also the German Certification Institute has developed a logo: DIN-Geprüft Home Compostable. On a European level, no standard has been established yet for home compostable products, but recently home composting has become an important factor in France, with the ban of conventional lightweight plastic shopping bags, the encouragement of bioplastics compostable at domestic level and the introduction of a French national standard on home compostable materials NF T 51-800 *Plastics – Specifications for plastic suitable for home composting* (2015). Moreover, several European companies, such as BASF, BIOTEC, LIMAGRAIN, NOVAMONT, FUTAMURA, etc. have developed polymer resins that are home compostable, but they have limited thermomechanical behaviour and processability. Most of these products are starch, polyhydroxy butyrate (PHB) or polyhydroxy succinate (PBS) based, although the Ecovio compounds of BASF contain *PLA* (in low amount when meeting home compostability requirements- rest being non-bio-based PBAT*)*. On the market, there is still a lack of products meeting the needs of the packaging applications (in terms of price, controlled biodegradability and product properties). Moreover, it should be noted that at present, PLA can be in the material formula (although it is not degrading at ambient temperature), as the current test requires that 90% absolute or relative to cellulose biodegradation is reached. However, in line with the International and American specifications for industrial compostable products and DIN-Geprüft Industrial Compostable label, in which it is requested that organic components between 1% and 10% must be tested separately, it is likely that the European standard in discussion follows a similar approach. In that case, materials including

8 Londa Schiebinger (2008). Gendered Innovations in Science and Engineering. ISBN: 978080475814

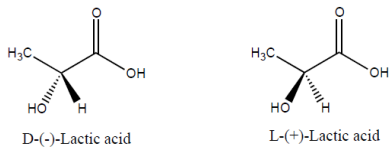
even minor PLA amounts and currently claiming home compostability would fail. In this case, the use of BIOnTOP copolymers would become mandatory.

Besides, it is well known that compostability rate is thickness dependent. While it may be possible to compost a specific bioplastic sample used in packaging film, a consumer product such as thick injected cosmetic jar, made from this bioplastic may not compost at a rate sufficient to meet the compostability standard, simply because it is too thick. This is true even for starch derived or PLA based materials at high thickness used for rigid food packaging (trays). For example, PLA for industrial composting conditions is certified up to 2000 µm thickness (Total-Corbion), but polybutylene succinate (PBS) up to 68 µm (Xinjian Blue Ridge Tunhe Polyester Co., LTD; China) and polybutylene adipate terephthalate (PBAT) up to 120 µm (BASF, Germany)[9.

As biodegradation is also determined by the characteristics of the environment (temperature, moisture content, pH, oxygen presence, microbial population…), specific test procedures were developed to determine biodegradation of plastics under anaerobic conditions (ISO 15985). However, although AD in biogasification plants is considered in the EU Waste Directive as a valuable treatment method for organic recycling, there is currently no standard existing that defines the specific requirements on biodegradation, disintegration and effect on the digestate that should be fulfilled (similar to EN 13432 for industrial composting). AD has so far only been poorly investigated as an EoL option for bioplastics and packaging. Still, it is an interesting technology due to additional biogas formation, suitable for electricity or heat production. In fact, the bioplastics can have a positive effect on biogas production due to their high C/N ratio. The results of this project will help to understand the behaviour of plastics and packaging in these anaerobic systems and can be a basis for development of such a standard.

Beyond the state of the art, BIOnTOP will combine several technologies discussed below to reach modular EoL that are currently not achievable with the available materials under control waste management scenario, while still ensuring the material meets the properties required for packaging productions. BIOnTOP will improve the biodegradation of PLA based blends and copolymers in home composting conditions, allowing to apply a higher PLA content and more cost-effective products. Moreover, the behaviour of PLA products in AD systems will be evaluated opening new EoL options, in both cases contributing with tests results towards new harmonised standards.

State of the art in the field of PLA, lactide (LD) and lactic acid (LA) based copolymers



Among the biodegradable polymers, polylactic acid (PLA)-based polymers are one of the most extensively studied biomass-derived synthetic polymers, used for a wide range of applications covering disposable household items, food packaging, agricultural films, drug delivery systems, and implantable biomedical devices. However, some PLA properties are still insufficient for their breakthrough in more potential applications, mainly due to its lack of toughness and impact resistance.[10]

Generally commercial grade of PLA are co-polymers of *L*-LA and *D*-LA or the corresponding lactide (LD) forms.[11]. The ratio of *L* to *D* enantiomers is known to affect the properties of PLA. Thus, the content of *D*-LA affects the glass transition of PLA, which decreases progressively, as well as the melting temperature and the crystallization rate, which decrease with the increase in the *D* units content. In general, in industrial packaging applications, the use of PLA with a content of *D* units of a 2–6 wt. % randomly distributed is very common as then, the crystallization of PLA is low, favouring the production of amorphous transparent films (in contrast to textiles needing high crystallinity). The crystallinity of PLA interferes with controlled degradation and is an obstacle to the application of PLA as a biodegradable plastic. Therefore, research aims to **equip PLA-based polymers with controllable biodegradation profiles**, tailored mechanical properties, reactivity for chemical modification, and other functionalities while keeping the favourable characteristics of PLA. In that sense, a **promising approach is the introduction of hydrophilic (reactive) segments, other bio-based monomers and/or branched structures into PLA**. Currently, **the lack of consistency in biodegradability studies**, with varying conditions and protocols being applied to different systems, **makes it very hard and controversial to compare the biodegradability of plastics and in particular of PLA and derived materials developed in the academic literature**.[12]

Biodegradation of PLA

As seen previously, the biodegradation of plastics is a complex process involving multiple steps and pathways and is highly dependent on the conditions. The initial degradation step for PLA is to be broken down into monomers or low molecular weight oligomers, where the ester bonds are cleaved hydrolytically to facilitate uptake into microorganisms.[13] PLA becomes water soluble when the molecular weight (Mn) is below 20,000 g/mol.[14] The initial degradation into smaller fragments is a rate limiting step in biodegradation processes.[15] After uptake into the microorganisms, subsequent metabolic activity breaks the polymers down into metabolic end products such as carbon dioxide and water, while a portion of the carbon is converted into biomass.[16] Composting of PLA proceeds efficiently under adequate conditions due to the combined effects of hydrolysis and microbial activity.[17]

9 http://www.okcompost.be/data/pdf-document/okc-mate.pdfdincertco.tuv.com/search/categories?category_id=442&locale=en
10 Mehta, et al..,Synthesis of poly (lactic acid): a review. J. Macromol. Sci. Polym. Rev., 2005; 45(4): 325–349 (2005)
11 http://www.natureworksllc.com/What-is-Ingeo/Where-Ingeo-Comes-From
12 Garrison et al.. Review Bio-Based Polymers with Potential for Biodegradability. Polymers 2016, 8, 262; doi:10.3390/polym8070262
13 Stlouk et al. Kinetics and mechanism of the biodegradation of PLA/clay nanocomposites during thermophilic phase of composting process. Waste Manag. 2015, 42.
14 Gorrasi et al., Effect of PLA grades and morphologies on hydrolytic degradation at composting temperature: Assessment of structural modification and kinetic parameters. Polym. Degrad. Stab. 2013, 98, 1006–1014
15 Husárová, et al. Identification of important abiotic & biotic factors in the biodegradation of PLA. Int. J. Biol. Macromol. 2014, 71, 155–162
16 Mueller, R.J. Biological degradation of synthetic polyesters-Enzymes as potential catalysts for polyester recycling. Process. Biochem. 2006, 41, 2124–8.
17 Ghorpade, et al., Laboratory composting of extruded poly(lactic acid) sheets. Bioresour. Technol. 2001, 76, 57–61.

Furthermore, elevated temperatures encountered during composting accelerate the hydrolysis process of PLA, especially when temperatures exceed 50ºC.[18] Extracellular enzymes released by specific microorganisms can cause cleavage of PLA chains, which may contribute to the degradation process. However, some studies have shown that extracellular enzymes do not significantly accelerate depolymerization. The reaction mechanism for the hydrolysis of polyesters is a bimolecular nucleophilic substitution reaction (SN2), which may be catalysed with either acids or bases.[19] Furthermore, the reaction rates for hydrolytic degradation are dependent on a number of factors: temperature, size and shape of sample, molecular weight (Mw), crystallinity, and relative humidity. Moreover, polymers that crosslink during irradiation tend to show reduced biodegradation because the higher Mw decreases the ability for uptake by microorganisms.[20]

PLA synthesis

The synthesis routines of PLA can be divided into polycondensation reaction from LA and ring opening polymerization (ROP) from LD via a coordination-insertion mechanism.[21,22]

-***Direct polycondensation (DP)***: DP of LA is usually performed in bulk by distillation of condensation water with or without a catalyst, while vacuum and temperature are progressively increased. A polymer of low Mw and uncontrolled stereoregularity is sometimes obtained being the main disadvantage of DP and restricts its use. The polymer thus possesses inferior mechanical properties. So, this method is normally employed only to obtain polymers of low Mw (e.g. prepolymers for WP2 or coatings).

-***Ring-Opening Polymerization (ROP)***: While DP of LA is cheaper, ROP of LD is the method used commercially. Many researchers have studied this ROP and some patents[23] have been filed (mostly already expired patents not focusing on copolymers as in the present project), however a huge research is still needed. As an example, NatureWorks LLC has developed a solvent-free, low-cost continuous process for the production of PLA from com-derived dextrose. The resulting PLA (Ingeo™) is the first synthetic polymer to be produced from annually renewable resources. The ROP of LD can be initiated by metal complexes, organic compounds or enzymes to yield high Mw PLA in excellent conversion and purity. The benefits of ROP in conjunction with a "living" method have enabled the controlled synthesis of block, graft, and star polymers. It is very important to control the stereochemistry of PLA because it plays a very important role in PLAs' physico-chemical, mechanical properties and biodegradability.

-***Reactive Extrusion (REX)***: REX has emerged as a powerful tool with potential to replace chemical batch reactor in some cases. REX, performed in a well-designed twin-screw co-rotating extruder, is an attractive route for bioplastic processing in order to carry out polymerization. The obvious advantages of an extrusion polymerization are that it is a **solvent free** melt and a **low-cost continuous** process, starting from monomers or prepolymers and resulting in polymers, while **controlling the chemistry** throughout the process. In this regard, REX processing of PLA-based materials has been investigated mainly at small scale in order to modify their properties. REX can therefore enhance the commercial viability and cost-competitiveness of these materials as already found for biodegradable polymers like PLA [24].

Beyond state-of-the-art copolymerization routes of diacids with lactic acid/lactide, throughout the BIOnTOP project we will develop two cost effective environmentally friendly approaches (direct polycondensation and REX synthesis; this latter is currently not well established in the Industry due to insufficient knowledge of its advantages) that will supply tailored EoL such as biodegradability in home composting conditions and increase the sustainability of the bio-based polymer synthesis process.

Chemical Structure Influence on Biodegradation

Several studies in the literature suggest relationships between the chemical structure of substrates and their degradation rates. Likewise, the polymers exhibiting the lowest mechanical properties are also generally the ones with the highest degradation rates and it is also why commercially available solutions for home compostable shopping bags or so are not suitable for higher demanding applications as tackled in the project. High biodegradability rates depend primarily on (1) the presence of terminal hydroxyl or acyl functional groups to allow for metabolic processes and (2) lower molecular weights. Such findings[25,26,27] suggest that in many cases biodegradability may be directed by physico-chemical phenomena, such as the mobility of substrate, availability, mixing, mechanical properties impacting breaking down of the macrostructure, and/or hydrophilicity of chemical species. Bio-based or bio-renewable monomers can lead to materials that may not necessarily be fully or significantly biodegradable and the biodegradability of a large number of novel bio-based systems has not been investigated yet. Changes in the functional groups of bio-based monomers, crosslink density and co-polymerization with non-biodegradable comonomers can lead to materials that exhibit various degrees of biodegradability, with varying environmental impact.

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19 Lyu, S.P.; et al.. Kinetics and time-temperature equivalence of polymer degradation. Biomacromolecules 2007, 8, 2301–10.
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21 Garlotta, D. A.,literature review of poly (lactic acid).J. Polym. Environ., 2001; 9(2): 63-84.
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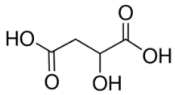
24 Processing, Characterization and Physical Properties of PLA Chapter 4 2014 on http://pubs.rsc.org | doi:10.1039/9781782624806-00099
25 Lu et al. Isomer-specific biodegradation of nonylphenol in an activated sludge bioreactor andstructure-biodegradability relationship. Water Res. 2015, 68
26 Genovese. New eco-friendly random copolyesters based on poly(propylene cyclohexanedicarboxylate): Structure-properties relationships. Express Polym. Lett. 2015, 9, 972
27 Beran, E.; Hull, S.; Steininger, M. The relationship between the chemical structure of poly(alkylene glycol)s and their aerobic biodegradability in an aqueous environment. J. Polym. Environ. 2013, 21, 172–180

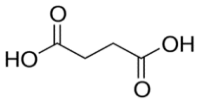
PLA Copolymers

The introduction of different monomers (dicarboxylic acids: DA) in the PLA chain **reduces PLA crystallization and the resulting copolymers will display enhanced biodegradability**. Furthermore, this modification allows obtaining different lengths and Mw distributions providing a good approach to **tune the biodegradation of the final compounds**. While the presence of a methylene in position *alpha* to the carboxylic acid of LA hinders its biodegradation, the selection of polymers without this steric hindrance will favour their biodegradability because the access of the microorganism to the carbonyl group is easier. The choice of initiator system, co-initiator as chain control agent, catalyst concentration, monomer-to-initiator ratio and polymerization temperature & time, significantly affects the polymer properties, such as the Mw, degree of crystallinity, and residual monomer content, which in turn affect the physico-mechanical properties and processing temperature range of the polylactide and its copolymers as well as their biodegradation behaviour.[28]

The US Department of Energy identified 12 strategic building block chemicals that can be produced from carbohydrates or sugars via biological or chemical conversions from biomass.[29] Most of them are bifunctional dicarboxylic acids among which BIOnTOP considered itaconic acid, adipic acid, tartaric acid, malic acid, succinic acid, glycolic acid and aspartic acid as potential candidates that can be copolymerized with LA to obtain biodegradable copolymers in suitable conditions. All of them, except glycolic acid, are bio-produced at low cost. Taking into account their price, bio-based availability in Europe, chemical structure, bibliographic references and patents to prepare copolymers with LA or lactide and their biodegradability, malic and succinic acid have been selected to be copolymerized with lactic acid in WP2.



Malic acid *(MA):* Companies such as Novozymes and Myriant have developed proprietary technologies based on aforementioned fermentation process. Currently, 80% of global malic acid is produced synthetically and the remaining is produced by renewable sources among other from **fruit waste**, as done by OENO (collaborator of INSTM who will act as supplier for WP2). Commercialization of bio-based malic acid at competitive pricing as compared to its synthetic counterpart is expected to create immense opportunities to be used as platform chemical in the future. The estimated bio-based MA price is at around 2400€/tonne.[30] At the market price of 1.9€/kg, MA production from soy molasses via PMA fermentation offers an economically competitive process. MA is used to produce bio-based polyesters which find application in pharmaceuticals used to treat cancer cells. It has been demonstrated that poly(malic acid-co-*L*-lactide) acts as a superb degradation accelerator for PLA in physiological conditions due to the superiority of poly(malic acid) vs. PLA to biodegrade.[31] Their findings also proved the usefulness to tune the degradation rate.



Succinic acid *(SA):* Nowadays, the bulk of SA is produced at a rate of 25,000 tons per year from maleic anhydride obtained by oxidation of the C4 fraction (n-butane or butadiene) of crude oil. However, the biotechnological production of SA by fermentation of carbohydrates has attracted recent attention.[32] At larger scale (typically 50 kTonnes), and due to lower costs of raw materials, bio-SA has the potential to be cheaper than its fossil-derived product and is expected to open up new markets and applications. Reverdia and Myriant's production cost for bio-SA is estimated to be lower than 1,040€/MT[33] whereas market prices for petroleum-derived SA are reported between $6,000 and $9,000/MT[34]. Building on such advantage, the most conservative market studies estimates 250 ktonnes of bio-based SA will be produced by 2020 with high opportunities for growth in EU. Some of the major companies dominating this market are Reverdia (AB), BioAmber and Succinity. As an example of copolymerization synthesis, LA, ethylene glycol and SA were copolymerised with different monomer feed ratios by direct polycondensation.[35] The lack of methyl groups in the succinic molecule should favour the biodegradation and the biodegradation of its derivatives such as poly(butylene succinate) (PBS) are well demonstrated.[36] In fact, PBS is home compostable according to Vinçotte certificate. For that reason, it is an interesting blend partner for PLA. However, most PBS on the market is not yet bio-based and its market is less established than that of PLA.[37] Possibility of terpolymers with adipic acid are interesting solutions to be addressed too.[38]

Beyond state-of-the-art copolymerization strategies, throughout the BIOnTOP project, novel tailor made copolymers of malic and succinic acids with lactic acid/lactide will be developed through different synthetic routines. Moreover, copolymers with different ratios MA or SA/LA, molecular weights, lengths of each segment and crystallinity will be designed which will allow breakthrough in their biodegradation behaviour in mild conditions, e.g. for home composting. In addition, the comparison of the degradability behaviour of different bio-based systems in a consistent manner will allow a systematic understanding of the effect

28 Duda, S. et al, "Polymerizations of ε-caprolactone and L,L-dilactide initiated with stannous octoate and stannous butoxide—a comparison," Macromolecular Symposia, vol. 153, no. 1, pp. 41–53, 2000. D. W. Grijpma et al., "Polymerization temperature effects on the properties of l-lactide and ε-caprolactone copolymers," Polymer Bulletin, vol. 25, no. 3, pp. 335–341, 1991
29 Werpy T, Petersen G (2004) Top value added chemicals from biomass, vol 1. U.S. Department of Energy, Oak Ridge
30 https://greenchemicalsblog.com/2012/09/05/novozymes-expands-bio-based-activities/
31 Poly(malic acid-co-L-lactide) as a superb degradation accelerator for Poly(L-lactic acid) at physiological conditions. DOI: 10.1016/j.polymdegradstab.2016.10.016
32 www.dsm.com/en_US/downloads/media/12e_09_dsm_and_roquette_commercialize_bio_based_succinic_acid.pdf, www.bio-amber.com/press_releases.php
33Paradigms in Green Chemistry and Technology, Angelo Albini,Stefano Protti, Springer
34 The National Non-Food Crops Centre 2013
35 Biodegradation behaviour of poly(lactic acid) and (lactic acid-ethyleneglycol-malonic or succinic acid) copolymers under controlled composting conditions in a laboratory test system, Oana Cadar et al. Polymer Degradation and Stability 97 (2012) 354e357
36 Polymer, Vol. 50, 5, 23, 2009, 1178-1186, A novel biodegradable multiblock poly(ester urethane) containing poly(l-lactic acid) and poly(butylene succinate) blocks; Biotechnol. J. 2010, 5, 1149–1163; G.-Q. Chen (ed.), Plastics from Bacteria: Natural Functions and Applications, Microbiology Monographs, Vol. 14
37 http://www.okcompost.be/data/pdf-document/okh-mate.pdf
38 X. Zheng, B. Wu, L. Wu, J. Hu, Z. Bu, B.-G. Li; "Poly(l-lactic acid)-block-poly(butylene succinate-co-butylene adipate) Multiblock Copolymers: From Synthesis to Thermo-Mechanical Properties", Ind. Eng. Chem. Res., 2014, 53 (9), pp 3550–3558, DOI: 10.1021/ie403623f

of the structure of the end of life behaviour that is currently lacking in the literature and a contribution to biodegradation standards.

Use of biomass derivatives and blends for tailoring of biopolymers biodegradability in composting environments

As seen, polymer biodegradability is connected with polymer crystallinity, crosslinking and molecular weight. In addition, the selection of proper additives, in particular plasticizers and fillers can significantly speed up the biodegradation of polymer based materials[39]. Plasticizers lower the glass transition temperature of polymers and promote the rubber state that is easier to biodegrade, as experienced by INSTM and OWS (terminated) on cellulose di-acetate blends plasticized with triacetin[40], a GRAS compound according to FDA. In PLA-based blends[41] the selection of proper plasticizer type and amount allows promoting the biodegradability. In order to avoid migration of excessive plasticizers with consequent changes in mechanical properties with ageing and possible contamination in case of packed items, the use of reactive plasticizers and their reaction with the polymeric matrix, possibly in the extrusion step, is an interesting approach. As such, REX will be also used in the preparation of blends besides the copolymerisation step. It has been proven by INSTM for the production of PLA based transparent films with epoxy functionalised plasticizers[42], recently updated by INSTM with bio-based and for food contact approved epoxydised agro-based derivates plasticizers. In BIOnTOP, INSTM plans to use mainly bio-based plasticizers such as acetyl tributyl citrate and functionalised (epoxy, maleic) oligomers of LA as well as epoxy derived natural oils such as soybean oil derivates. INSTM has also already experienced positively the use of LA oligomers as plasticizers in PLA achieving valuable mechanical properties and compostability. The effect of nucleating agents such as PDLA[43], talc, calcium carbonate or even montmorillonites[44,45,46,] etc. will be also studied in order to modulate the biodegradation properties, which are connected with the distribution of the PLA amorphous and different crystalline phases. Such inorganic fillers in small amounts also improve processing avoiding sticking on the mould or of blow moulded films.

Likewise, biocomposites using natural fibres or fillers (NF) have been researched intensively over the last years yielding to new applications due to their processing advantages, biodegradability, low cost, low relative density, high specific strength and renewable nature. Several examples of biodegradable composites are reported in literature using different types of matrixes, e.g. starch based blends and PLA, and natural reinforcements such as fillers from pine, cellulose, potato pulp, rice husk[47]. NF considerably reduces the price of the final material since generally natural fillers are cheaper than bio-based polymeric matrices such as PLA. Moreover, if proper compatibility is achieved, fillers can strengthen the material and increase the Young's modulus, or impact resistance. Besides, composites can promote biodegradability, as the presence of fillers increase the polymer surface available for microbial attack especially when the fillers are based on easily degradable natural components[48], such as cellulose or by-products with high content in hemicellulose. The swelling and degradation of the fillers highly promote the disintegration of composite material[49]. The amount of fillers loaded in a composite is limited by weak adhesion at the fibre-polymeric matrix interphase, poor mechanical properties and difficulties in processing[50]. Therefore, the use of compatibilizers such as maleic or itaconic anhydride may allow to obtain a higher load of fillers in PLA polymeric matrix. As well as fillers pre-treatment, the amount and properties of the coupling agent are also crucial for the properties of the composites[51].

In terms of alternative routes, while the polymer blending with home compostable polymers like PCL or PBS is limited by low properties/high costs thereof, and, as discussed earlier, the presence of even low amount of PLA can prevent meeting related standards, there is also a significant amount of academic literature regarding the use of enzymes mixed in different polymers showing variable results depending on the matrix. The BIOCLEAN project[52] among other studies[53] concluded that none of the tested enzymes showed degradation capability toward tested conventional polymers. In the case of PLA, Carbios, a French company, has developed a pilot technology for embedding an enzyme in plastic at the time of production.[54] In contrast, to avoid issue with the further thermal sensitivity of the enzyme that may lead to a loss of enzyme activity and therefore a slower biodegradation, BIOnTOP will prefer additivation and copolymerisation routes to tune the polymer behaviour.

In BIOnTOP, an accurate study of polymer crystallinity will be performed as function of additives and fillers content and processing

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43 L. Aliotta et al., European Polymer Journal 2017, (Effect of nucleating agents on crystallinity phases, and properties of Poly (lactic acid) (PLA)
44 Ray et al.., 2003. New polylactide-layered silicate nanocomposites. 2. Concurrent improvements of material properties, biodegradability and melt rheology. Polymer 44, 857–866. Stloukal et al. Kinetics and mechanism of the biodegradation of PLA/clay nanocomposites during thermophilic phase of composting process Waste Mngt 42 (2015) 31–40
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49 GUIDANCE DOCUMENT INDUSTRIAL COMPOSTABILITY CLAIMS CHECKLIST, JUNE 2016, The Plastics Industry Trade Association (SPI) Bioplastics Division. PLA and PHA Biodegradation in the Marine Environment, California Department of Resources Recycling and Recovery March 5, 2012
50 E.Chiellini, et. al. Biomacromolecules, 2001, 2, 806-811; E. Chiellini et. al., Macromolecular Bioscience, 2004, 4, 218-231.
51 M Seggiani et. al., Chemical Engineering Transaction, 2015, 43, 1813-1815
52 New Biotechnological approaches for biodegrading and promoting the environmental biotransformation of synthetic polymeric materials
53 http://www.ows.be/lab-consulting-services/publications/
54 https://carbios.fr/

conditions[55] to address the connection between polymers morphology, mechanical properties and biodegradability. The formula patented[42] by INSTM of flexible films based on PLA and reactive plasticizer targeting the production of films for shopping bags will be optimized by REX between PLA (or the new copolymers thereof) and food contact, biodegradable, bio-based plasticizers such as FA based derivatives and modified oligomers of LA or adipic acid, and acetyl tributyl citrate. To tune crystallinity, a proper selection of nucleating and strengthening agents, or low molecular weight oligomers of LA, will be made. For selected packaging formats such as trays, food contact approved natural cellulose based fillers such as wheat milling by-products will be used to tune disintegration and lower material cost. Furthermore, the NF/matrix interface will be improved with wax coating and coupling agents such maleic anhydride. The combination of NF, copolymers and suitable additives will be optimized to achieve our biodegradability targets.

State of the art and progresses in the field of net packaging for fresh produce (F&V)

Nets are a widely used packaging format for fruits and vegetables. ECOBIONET project faced up the EoL treatment of plastic nets if mixed with the household waste. The management of this waste causes problems to household waste treatment due to their low apparent density and high strength (i.e. material is difficult to be cut and the nets may get entangled and collapse treatment machines). The main objective of the project was to industrialize the process and technologies related to the fabrication of different types of biodegradable compostable nets, obtained by **extrusion melt spinning process (one-step process)**, for agriculture and shellfish packaging products. ECOBIONET goal was to manufacture completely biodegradable and compostable packaging nets while being both cost-competitive and technically comparable to current polyolefin-based nets. For this, a new biodegradable compound was obtained, derived from the synergic combination of different immiscible biodegradable polyester materials (including an important percentage of PLA) with plasticizers and compatibilizers. However, ECOBIONET nets are only completely biodegradable in the conditions of a composting plant[56].

Building on the knowledge regarding the production of industrially compostable PLA-based nets developed by AIMPLAS and CMSA in ECOBIONET project, BIOnTOP will develop suitable formulation for home composting which is an asset due to difficulties in collecting such light packaging, but also for countries with limited composting plants as well as for domestic valorisation as a fair share of F&V are spoiled in pack.

State of the art and progresses in the field of biopolymers and PLA for textile fibres and filaments.

The use of biopolymers in textile applications is nowadays only via extrusion and is estimated at 100 kTon per year[57], which has the potential to increase strongly in the coming years if broader applicability and better price with adapted polymer formulations are achieved. Mostly used (partially) bio-based polymers are often “drop-in” variants of the oil-based counterparts such as bio-PE, bio-PET, bio-PTT, etc. Current bio-based and biodegradable textiles consist mainly of PLA[58]. TPS (thermoplastic starch) is rarely used since its mechanical properties are insufficient for most textile applications. Other (partially) bio-based materials are in research for textile applications as well (e.g. PHA, PBS, PEF and bio-PA). **PLA is** currently and will be in the near future **the only bio-based and biodegradable polymer resulting in high-quality products** which is also available in **sufficient amounts and at acceptable price**. However, although the feedstock is bio-based, for fully environmentally sustainable solutions, improvements are still required for EoL not to be limited to high temperature and humidity composting conditions. As other bottlenecks, PLA is more expensive than fossil-based textiles, PLA processibility is more sensitive (mainly due to hydrolysis) and a better understanding and control of the biodegradation potential are needed.

PLA fibres have a number of characteristics that are similar to many other thermoplastic fibres, such as controlled crimp, smooth surface and low moisture regain. The physical properties and structure have been studied by several researchers[59], and suitability as textile fibre is already acknowledged. Its mechanical properties are considered to be broadly similar to those of conventional PET[60] and, probably due to their lower melting and softening temperatures, comparisons to polypropylene are also appropriated[61]. PLA filaments are available on an international market, but volumes are still limited. Examples of their producers include Monotex[62] (Biolene - PLA monofilament), Teijn Fibers[63] (Biofront - scPLA), Polisilk[64] (Polisilk HT PLA), Diolen[65] (Swicofil HT PLA). Also, PLA based textiles for clothing application are being developed[66].

PLA-based textiles are commercially available: some examples are groundcovers for weed control, textiles for crop protection,

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57 http://www.european-bioplastics.org/market/applications-sectors/
58 L. Ruys, Raf Van Olmen - New developments in agrotextiles based on the results of the FP7 project: “BIOAGROTEX” – Unitex, nr 5, p 4-7, 2012.
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60 Lunt, J. and Bone, J., AATCC Rev., 2001, 1 (9), 20
61 Palade, L.I., Lehermeier, H.J. and Dorgan, J.R., Macromolecules, 2001, 34 (5), 1384
62 http://monotex.ro/bio.php
63 https://www.teijnfiber.com/english/products/specifics/biofront.html
64 http://www.polisilk.com/
65 http://www.swicofil.com/diolen.html
66 H2020 project ‘Fibfab -Industrialization Of Biobased Textile Fabrics For Clothing Applications’ , see fibfab-project.eu

etc. PLA-based materials are also gaining interest in textile domains like apparel and interior textiles where biodegradability is not such an issue as recycling is a more valuable EoL solution. Often these products are mainly to demonstrate the feasibility of a sustainable/bio-based product and they have so far no or only a very limited market share.

An example of a more successful introduction of a PLA bio-based textile is being realized by our project partner **SIOEN who applies a woven fabric from very fine PLA monofilaments** (30dtex) in **teabags.** The main PLA disadvantage for this application is however the lack of home compostability needed due to disposal with content and difficult collection.

PLA coatings for textiles. Coatings are used on textiles for a number of aesthetic and functional purposes and are generally based on non bio-based materials such as PU and PA. There are multiple applications in which PLA coatings could substitute existing coatings. Indeed, PLA has an excellent UV-stability and decent flame retardance (FR) properties which makes it applicable for interior and exterior applications for a broad range of items. So, there are several initiatives to develop PLA coatings for textiles, but so far, no suited PLA coating formulations are commercially available. One of the main reasons why PLA coatings have not yet reached the market is their brittleness. This is due to the high molecular weight of commercial PLA (ca 100.000 g/mol) vs. what is needed (ca 10.000 g/mol) for coatings. As a result, the coatings need the addition of plasticizers as can for example be seen on fig 3. from partner CTB. Therefore, like for compounding, developers of PLA coatings need to look into most suited environmentally friendly (preferably bio-based) molecules available to plasticize PLA. Plasticizers are not always compatible with PLA and issues with migration of the plasticizer arise. This causes not only increased brittleness over time but also an unpleasant feeling of the coating. Migration of the plasticizer can be limited using highly compatible molecules and/or grafting them on PLA.

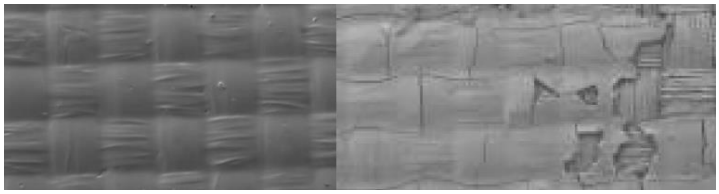

Fig. 3: Plasticized (left) vs. unplasticized PLA coating on woven textile

Beyond the state of the art, the development of new tailored copolymers and blends with controlled biodegradability from the BIOnTOP project will be applied to extrude PLA filaments that will be further processed into woven fabrics. As applications, we want to develop **the first fully home compostable woven teabag** as BIOnTOP new formulations have the potential to enhance the biodegradability in mild conditions. We will also look into developing a PLA coating or alternative FA grafting strategies (discussed below) for PLA fabric to be used in a reusable food wrap that at the end of its life is recyclable or compostable. Those will impart durable hydrophobicity enabling industry **to replace fluorocarbons which are now currently used in hydrophobic coatings but are toxic, persistent and bioaccumulative.** In addition, the secondary use of the recycled PLA materials will be shown by **processing the recycled PLA material into nonwovens, that will be functionalised by our waterproofing coatings** and used in a demonstrator carrier bag.

State of the art and progresses in the field of bio-based oxygen and aroma barrier coatings and films

Protection against oxygen is a key factor for sensitive/fresh food, cosmetic and pharmaceutical products to achieve a sufficient shelf life. Otherwise colour or taste deviation, oxidation of grease, formation of micro-organisms or degrading nutrients will take place. To achieve low oxygen permeability co-extruded or laminated multilayer plastic films are widely used in the packaging industry whereby mineral oil derived EVOH, or PVDC, are the mostly used oxygen barrier materials. However, theses polymers are neither biodegradable nor bio-based. Therefore, research into sustainable packaging materials that barrier maintains the performance has been recently intensified. In recent European[67,68] projects with participation of INSTM and IRIS, oxygen barrier coatings based on whey protein have been developed. **Whey continues to be a widely available and underutilized (only 40% is valorised and a much lower amount would in fact allow substituting all EVOH for packaging worldwide) by-product from cheese making** containing around 13% protein in dry matter. The whey protein-based coatings are formulated using Whey Protein Isolate (WPI) as well as bio-based plasticizers and/or reactive additives such as urea to enhance thermoformability. The whey formulation is subsequently coated on a plastic substrate and dried. In this process step the protein is denatured in order to form a crosslinked network. This patented step is necessary to achieve an oxygen permeability as low as 1.5 $cm^3/(m^2*d*bar)$ (Q_{100}). Finally, the coated film is laminated into the final structure and eventually additional conversion processes such as thermoforming, sealing etc. can be carried out to obtain the packaging items. Application of both standard and thermoformable whey coating formulations have been proven in pilot scale in roll-to-roll lacquering and lamination lines. Resulting films were reported to have excellent mechanical as well as gas and UV barrier properties, beyond most existing biopolymers (fig. 3). Whey coatings are thus capable of replacing synthetic oxygen-barrier layers such as EVOH in multilayer packaging. Furthermore, the whey coating is both bio-based and biodegradable. On PLA, based on preliminary tests, barrier improvement factors (BIF) 6 were obtained (although further ones planed here are needed to reach the full potential of the whey coating on this substrate as its intrinsic barrier calculated on PET should have led to a 60 fold improvement vs. PLA). The use of such nitrogen rich materials

67 Wheylayer (Grant Agreement FP7- 218340-2) and Wheylayer 2 (Grant Agreement: FP7- 315743)
68 ThermoWhey (EraNet-ManuNet 02PN2020)

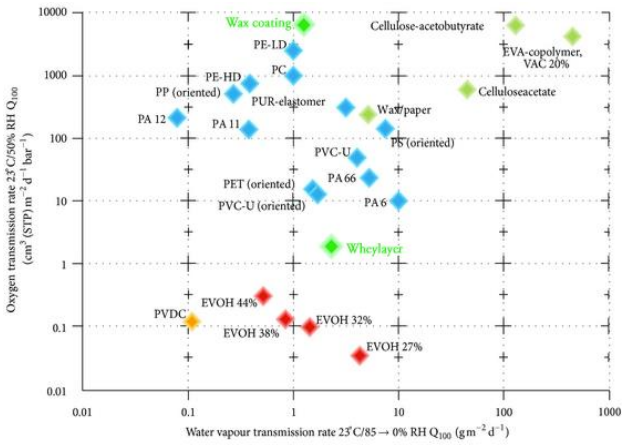

Fig. 4: Barrier properties of whey-based coating vs. other plastics commonly used in food packaging industry (normalized to 100 µm)

protein coatings on biodegradable based films will not compromise biodegradability but in fact even accelerate it[69]. Thus, whey protein-coating, which are in market introduction phase, represents a new application for an agro-food industry by-product while safeguarding the performance and enabling the recyclability of multilayer films which are generally landfilled. Additionally, the whey protein-based coatings used in multilayer laminates fulfil the requirements of European Commission Regulation (EU) No 10/2011.

In addition, bio-based coatings can be used as aroma barrier. Alginate is widely used in various industries such as food, beverage, textile, printing and pharmaceutical as thickening agent, stabilizer, emulsifier, chelating agent, encapsulation, swelling, suspending agent or used to form gel, film and membrane. Sodium alginate is the most common salt of alginate. It is commercially available and can be applied as coating to serve as aroma and oxygen barrier for products with low relative humidity[70].

State of the art on nanoscale surface modification by chemical grafting for improved hydrophobicity and to optimize barrier properties

As seen, films which are based on protein isolates have excellent oxygen barrier properties due to their polar nature and crosslinked polymer network[71], however, the water vapour barrier needs to be improved in order to make this novel bio-based coating suitable for many applications sensitive as well to humidity uptake or loss (e.g. dry pharma or food products, or wet products such as wet wipes). To overcome these drawbacks, polar substrates are generally combined with hydrophobic layers, such as polyethylene layers, via wet lamination (with adhesives), extrusion lamination (without adhesives), wet and extrusion coating. In the "wet" processes, water borne options are favoured for environmental reasons. Such solutions and dispersions of waxes, starches or poly(vinyl alcohol) are available. Additionally, physical (vacuum deposition) and chemical vapor deposition (plasma coating) are possible on a planarization coating. It can consist in bio-ORMOCER, an organic-inorganic hybrid polymer which is only partially coming from renewable resources and not providing any intrinsic barrier without inorganic coatings. However, the inorganic coatings can be scratch sensitive so protected by another top coat.

An innovative approach will improve also the water vapour barrier properties of bio-based films and coatings by green chemical grafting with FA from renewable resources (using palmitoyl chloride as benchmark and the fully bio based fatty acid anhydrides) to monomolecular graft a hydrophobic layer on the protein films[72,73]. The modified protein films are expected to provide significantly improved barrier properties towards water vapour (unpublished first results showed that water vapour BIF of 6 have been reached on protein based cast films without process and material optimization). The chemical grafting will be applied on protein based films and coatings by the so-called transfer method which was developed for the surface grafting of PVOH-based films[74]. The main attractions of this solvent free grafting process are its low cost, its use of existing industrial bio-based reagents and its potential to be applied on a printing machine at high speed and high efficiency. It is also a clean chemistry process that does not give rise to by-products or to polluting effluents. In addition, the FA green grafting is expected to provide easy emptying properties due to significantly reduced surface energy of the grafted surfaces. These rebalance properties are comparable to emerging non bio-based market available solutions such as LIQUIGLIDE™ developed by the Massachusetts Institute of Technology or Toyal Lotus® developed by Toyo Aluminium K.K. or AsahiGuard E-Series developed AGC Chemicals.

Beyond the current state of the art, BIOnTOP will develop the novel **combination of the polar whey protein-based oxygen barrier with a non-polar bio-based FA water vapour barrier coating to offer a barrier both against humidity and oxygen**. It is expected that grafting process optimization on tailored protein film can lead to similar BIF values as already achieved on PVOH-films (approx. BIF 20). In addition, this will provide easy emptying functionality and easy product (e.g. packed cheese) release functions which are beyond the current state of the art as non-migrating alternative to existing solutions. The developed formulations and processing parameters will be upscaled to obtain flexible and thermoformable laminates tailored to the target applications. Furthermore, the project will consider alginate coatings and FA-grafting as a novel route for functionalising textiles respectively for aroma barrier in tea bags and for waterproofing reusable food wraps and bags.

State of the art in the field of packaging recycling

Plastic packaging sorting

The sorting of polymers is of vital importance since even very low amounts of cross contamination can ruin the entire batch. Material recycling from mixed waste plastics is challenging and based on sophisticated process chains combining different sorting

69 P.Cinelli, M. Schmid, E. Bugnicourt, J. Wildner, A. Bazzichi, I.Anguillesi, A. Lazzeri; "Whey protein layer applied on biodegradable packaging film to improve barrier properties while maintaining biodegradability" Polymer Degradation and Stability, 2014, 108, 151-157,
70 Jost et al (2014). "Influence of plasticiser on the barrier, mechanical and grease resistance properties of alginate cast films." Carbohydrate Polymers 110: 309-319.
71 schmid et al. Properties of Whey-Protein-Coated Films and Laminates as Novel Recyclable Food Packaging Materials with Excellent Barrier Properties. Int Journ Polymer Sc
72 SAMAIN, D. 2002. Method for treating a solid material to make it hydrophobic, material obtained and uses. Google Patents.
73 BERLIOZ, S. 2007. Etude de l'estérification de la cellulose par une synthèse sans solvant. Application aux matériaux nanocomposites. Université Joseph-Fourier-Grenoble I.
74 SCHMID et al. 2012a. Fundamental investigations regarding barrier properties of grafted PVOH layers. International Journal of Polymer Science, 2012.

steps like screening, wind sifting, NIR sorting, electro-magnetic removal of different metals and others. Equipment to automatically sort plastics are already widely commercially available (e.g. via spectroscopic identification systems, including near-infrared, X-ray, colour, or laser, and combinations thereof), but currently many recyclers still sort plastics by hand. Such an approach is unreliable, expensive and not suitable for all transparent packaging.

Nevertheless, all the used identification techniques require a pre-calibration which has to be developed for innovative materials as BIOnTOP, for which the right identification approach has to be proposed both to allow its subsequent recycling but also allow industry acceptance for new biopolymers in the main stream of currently recycled materials. According to a study from Natureworks contradicting the initial reluctance from PET recycling facilities, PLA cannot be considered a threat for the current recycling plants as it is easily automatically sorted out from other polymers included PET based on conclusions from Titech (>97% NIR accuracy, consistent with sorting efficiencies for other materials) but also Unisensor (laser efficiency on flakes>96-99%) and MSS which confirmed PLA unique polymeric NIR "signature" to identify it from PET, PE, and other plastics. Furthermore, a study from Tomra sorting solution conclude that automatic sorting of PHB, Starch, TPS, etc. or blends is also possible by NIR[75].

In the case of multilayer, although plastic/plastic composite films are predominately incinerated, they sometimes end up in material flows of different target fractions e.g. in case there is no automated sorting or if the outer layer of a composite is thick enough to let the sorting systems identify the outer layer only. If the density of the composite material is similar to the density of the target fraction, further floatation system often in place at recyclers will not work either. As such, efforts are needed to improve the available systems for the case of multilayer recycling. That will be made more easily when using our new bio-based barrier coatings, which have very specific spectroscopic signatures vs. other plastics. Indeed, in the previous Wheylayer project, the capacity to detect the protein presence by IR-TF (fig. 7)[76] was shown. The FA layer is so thin that it will not need identification even when applied on flexible (e.g. 20 microns) films. IRIS has experience in developing tailored fast online VIS/NIR spectroscopic tools for plastic processing monitoring to lead to identify precisely their composition and traces of unwanted substances[77,78]. In case of multilayers, results from IRIS show the capacity using HSI to distinguish colored from natural HDPE and from LDPE from multilayer films including PE among its layers using chemometrics analysis selecting different principal components (fig. 6).

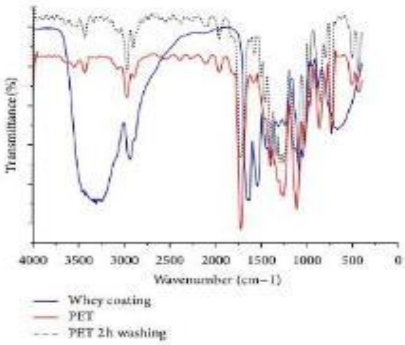

Fig 7: FT-IR spectra of PET, whey coating and coated PET after 2 h washing

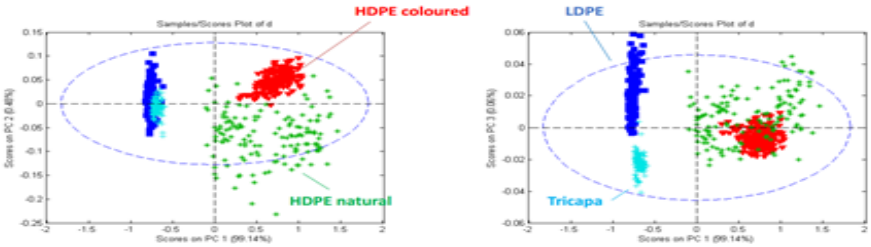

Fig 6: Example of chemometrics analysis on NIR HSI results on waste plastics

Recycling of the monomaterials of interest

Whereas 57% of the petrochemical based PET going to bottles or containers are recycled into bottles or textiles mainly, and polyolefins as well as mixed plastics can also be commonly found in recycled grades for downgraded applications at lower cost than the virgin grades they compete with, when it comes to more speciality materials used in packaging (or even less for textiles), there is generally no current industrial solution. Currently some companies dedicated to recycling of plastic materials, as ROMEI in Italy, collect and commercialize post-industrial thermoforming scraps coming from packaging companies after the necessary operations of grinding and eventually compounding. Such SRM were also considered in previous works for applications in the automotive field in the EVOLUTION project[79]. The use of recovered PLA in biodegradable blends or composites for durable applications thanks to reactive processing techniques is broadly considered.

The recycling of monomaterial PLA via grinding of sheets and trays, washing of the scraps, drying and the processing in extrusion machine to obtain pellets was also studied in PLA4FOOD and BIO4MAP projects[80]. The mechanical and rheological results revealed that flowability of the melt and mechanical properties of the materials produced are maintained during several processing cycles and this assures the possibility to use recycled PLA in blend with virgin PLA without loss of material characteristics. In fact, our SME partner ROMEI already performs such activity industrially, collecting PLA scraps from thermoforming companies and reprocesses them to technical sheets. Due to its low depolymerization temperature and possible presence of impurity, as an alternative to mechanical recycling, PLA is a good candidate for an environmentally sound chemical recycling as it can be readily hydrolysed with water to form LA, which is then purified and polymerised to remake prime polymer[81].

75 http://www.plasticseurope.org/documents/document/20120821135405-new_trends_within_polymer_sorting_to_tackle_for_tomra_sorting_solutions.pdf
76 "Processing and Validation of Whey-Protein-Coated Films and Laminates at Semi-Industrial Scale as Novel Recyclable Food Packaging Materials with Excellent Barrier Properties"; E. Bugnicourt et al.; Advances in Materials Science and Engineering, vol. 2013, Article ID 496207, 10 pages, 2013.
77 OPTIJECT project (Grant Agreement no. 262623), www.optiject.eu
78 B2B contracts for the detection of good batches of recyclates as well as for HSI-based monitoring solution for plastic identification in recycling line
79 Phuong, VT et al, Compatibilization and property enhancement of poly(lactic acid)/polycarbonate blends through triacetin-mediated interchange reactions in the melt, Polymer 55 (2014) 4498e4513
80 Signori, F, Coltelli, MB, et al., Thermal degradation of poly(lactic acid) (PLA) and poly(butylene adipate-co-terephthalate) (PBAT) and their blends upon melt processing, Polymer Degradation and Stability, 94, 1, 2009, 74-82
81 Natural fibers, biopolymers, and biocomposites, Taylor & Francis 2005, Chapter 16, Polylactic Acid Technology, David E. Henton, Patrick Gruber, Jim Lunt, and Jed Randall

Note that modification strategies such as the FA grafting (the thickness of a C16 FA grafted layer is approx. 2,5nm which would be less than 0.005% (v/v) vs. a 50 µm multilayer film) or whey protein coating when applied on rigid packaging, could still be behave as monomaterials. They are for example excluded by "Der Grüne Punkt– Duales System Deutschland GmbH (DSD)" which is nowadays the leading provider of take-back systems that considers packaging as monomaterials when other materials are used in content below 5% (w/w). Indeed, such "traces" can be removed in the cleaning stages during reprocessing and do not normally harm the property of the SRM.

Multilayer packaging recycling

PET/PE bilayer, widely used for packing fresh foods could only be recycled by grinding and extrusion. As PET and PE are immiscible, their compatibility must be improved in order to employ them in high-value applications. This is possible by using proper compatibilizers[82,83,84] but the resulting cost is **not industrially viable** although some example of PET/PE recycling in highly performant technical textile applications was achieved in previous works[85] thanks to the application of proper REX techniques[86,87]. Furthermore, it was observed that the degradation of the adhesive between the two layers, often consisting in poly(ethylene-co-vinyl acetate) (EVA), can result in yellowing of PET/PE blend granules[88], thus discouraging applications where a light colour is required. Another example is multilayer packaging containing polyolefins and a coextruded barrier layer of EVOH that cannot be separated and is incompatible with PE and PP[89] during recycling. While flexible films cannot be recycled at all, only final recycled products with low mechanical properties can be obtained from such rigid multilayer packages (low EVOH ratio). All in all, for multilayer packaging composites, especially flexible ones, there is no high-quality recycling established currently in Europe, and most material is used for energy recovery only, which is achieved by incineration plants with an average efficiency of 35%, or landfilled depending on the country, which in both cases is counteracting current efforts towards the CE[90].

In contrast to described direct reprocessing, there are several other approaches to achieve recycling of multilayer films. One of them is the solvent based CreaSolv® technology[91]. Specific solvents dissolve one polymer type out of the composite. A multi-step procedure separates other components of the multilayer and recovers the polymer in high purity by removal and recirculation of the solvent. Such approach is still in its industrial implementation phase. Another approach addresses the removal/dissolution of the adhesive layers of multilayer packaging. This results in separation of the components of a multilayered polymer composite but requires a subsequent separation of the segregated films by density, NIR or electrostatic. The technology is provided in technical scale by the German company Saperatec GmbH[92] and uses a micro emulsion to separate composite films. However, the application of this concept is not yet universal, as many different adhesives are used in the multilayer market. Likewise, the use of **removable barrier layers in multilayer structures is an emerging approach to significantly contribute to increasing the recycling potential for this post-consumer waste** (fig. 8)[93].This has already been extensively proven at TRL 4 for the **whey protein** coating, whereby the washing stage of the recycling process needs to be tuned to allow the hydrolysis of the whey coating using enzymes or a temperature of 30-40ºC depending on the needed process speed. This recycling approach for multilayer materials is specific to our bio-based barrier coatings but requires very little changes and investment in the recycling lines which can use their own sorting capabilities (when existing, they will only need to be tuned with the right chemometrics package to detect the new materials), shredders, washing baths, and optionally a density separation unit (e.g. in case of multimaterial structures used with the whey coating such as PET/barrier/PE) before each material can be reprocessed independently in usual compounding steps. In fact, in recent years other companies have been attempting to use this capacity to remove PVOH to claim recyclability[94]. Nevertheless, PVOH is not bio-based so it doesn't compete directly with our project but the acceptance of such new recycling approach by the recyclers could in fact be synergistic with our proposed process.

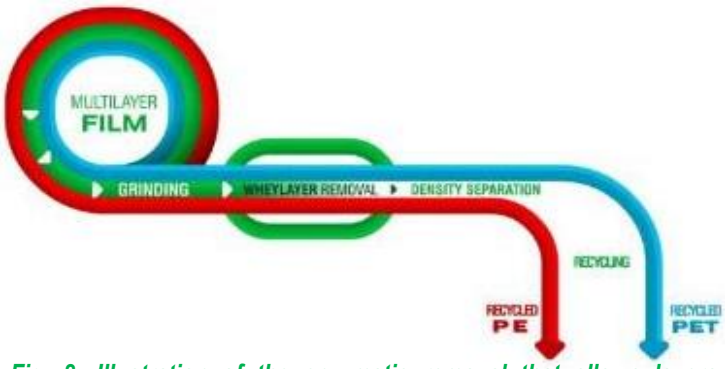

Fig. 8: Illustration of the enzymatic removal that allows layers combined with whey coating in multilayer packaging to be recyclable

82 Coltelli, M.B.; della Maggiore, I.; Savi, S.; Aglietto, M.; Ciardelli, F. Modified styrene-butadiene-styrene block copolymer as compatibiliser precursor in polyethylene/poly(ethylene terephthalate) blends. Polym. Degrad. Stab. 2005, 90, 211–223
83 Aglietto et al. Postconsumer polyethylene terephthalate (PET)/polyolefin blends through reactive processing. J. Mater. Cycles Waste Manag. 2004, 6, 13–19
84 Yousfi, M.; Soulestin, J.; Vergnes, B.; Lacrampe, M.F.; Krawczak, P. Morphology and Mechanical Properties of PET/PE Blends Compatibilized by Nanoclays: Effect of Thermal Stability of Nanofiller Organic Modifier. J. Appl. Polym. Sci. 2013, 128, 2766–2778
85 De Schrijver et al., Supertex-sustainable technical textile from recycled polyester, Chemical Fibers International, 1/2014, 44-46
86 Raffa, P., Coltelli, MB, et al., Chain extension and branching of poly(ethylene terephthalate) (PET) with di- and multifunctional epoxy or isocyanate additives: An experimental and modelling study, Reactive & Functional Polymers 72 (2012) 50–60
87 Raffa P., Coltelli MB, et al., Expanding the Application Field of Post-Consumer Poly(ethylene terephthalate) Through Structural Modification by Reactive Blending, J. Appl. Polym. Sci. 2014, 131, 40881.
88 Bartoli et al. Conversion of post-industrial PET-PE scraps into compatibilized plastic blends for new applications. AIP Conf. Proc. 2012, 1459, 160
89 Wang, Q.; Qi, R.; Shen, Y.; Liu, Q.; Zhou, C. Effect of high-density polyethylene-g-maleic anhydride on the morphology and properties of (high-density polyethylene)/(ethylene–vinyl alcohol) copolymer alloys. J. Appl. Polym. Sci. 2007, 106, 3220–3226
90 BUND *Wege zu einer nachhaltigen Abfallwirtschaft*; 2010.
91 Mäurer, A. et al. (2004): Good as new. Recycling plastics from WEEE and packaging wastes. Waste management world Vol 4, May/June, pp. 33-43.
92 9.AG, S. http://www.saperatec.de/technology/#technologie_top (accessed 19.07.2018).
93 P Cinelli et al., Recyclability of PET/WPI/PE Multilayer Films by Removal of Whey Protein Isolate-Based Coatings with Enzymatic Detergents, Materials 9(6):473, June 2016
94 http://www.g-polymer.com/eng/gasubariasei/

Furthermore, when applying the whey coating on bioplastic substrates (based on either cellophane or PLA), fully compostable solutions with improved barrier properties were obtained[95]. As such our multilayer structures containing a whey protein-based intermediate layer will benefit from improved both improved material recyclability when combined with standard substrates and organic recyclability when combined with biodegradable substrates.

Coated textile recycling

Many textile materials are treated with a coating or finishing for improving material functionality. These coated textiles currently cannot be recycled because the coating layer is difficult to impossible to separate from the textile substrate material, especially as the coating layer is typically cured and therefore not reprocessable. Some possible routes are mentioned in literature such as coatings that debond upon a trigger[96] or previously mentioned Creasolv® chemical recycling based on selective extraction. These solutions are however complex, require specific sorting and handling and are thus currently not yet economically feasible. A mono-material solution (textile and coating from same polymer class) could therefore be a valuable alternative.

BIOnTOP will **demonstrate the capacity for recycling all materials under study where such EoL is relevant**. To tackle the different challenges related with sorting of new materials, we will optimize the hardware, combining several spectroscopic technologies (possibly NIR and visible HSI), software thanks to advanced chemometrics analysis and possibly data fusion. The reprocessing of BIOnTOP monomaterials (or those with a very thin FA grafting) will be optimised. In addition, the use of a new PLA based coating on top of a PLA textile will allow the recycling of the coating textile material as a whole, without the need of a preceding separating step. Where needed for multilayer packaging, the removal of the whey protein coating will be adjusted to BIOnTOP substrates and overall recycling process upscaled in a pilot scale at ROMEI.

State of the art in the biodegradation modelling

As previously seen, polymer biodegradation is a poorly controlled and understood mechanism influenced by the crystallinity, crosslinking, molecular weight, additives, but also by the environment (moisture, temperature, pH, microbial activity…). While different models, such as BIOWIN, BESS, METABOL, etc. exist for the prediction of the biodegradation of organic compounds (chemicals), **no model is available for the prediction of the biodegradation of polymers**.

Due to the underlying complexity and number of influencing variables, it seems unrealistic to envisage a mechanistic model that would properly describe the actual behaviour of new bioplastic item in real life conditions. In contrast, data driven approaches can cope with such challenges. Data representing biodegradation profiles both in industrial and home composting conditions can be modelled by benchmarking different machine learning algorithms which use a fitness function as objective output to optimize the biodegradation curve, accounting for the effects of the known parameters influencing the disintegration and biodegradation. Therefore, at least two fitness functions can be considered: the first in terms of decomposition time and % decomposition achieved (absolute and/or relative to cellulose); and the second in terms of the time to reach a target decomposition, which is one of the final quality criteria of the applicable standards. Similar metrics can also be devised for the disintegration process. Two initial definitions of fitness functions for biodegradation are as follows:

$$Fitness_1 = \left(\left(1 - \frac{Tp}{Tmax}\right) \times \alpha\right) + \left(\left(\frac{Pp}{Pmax}\right) \times \beta\right) \quad (1)$$

$$\text{Fitness}_2 = 1 - \frac{To}{Tmax}, where\ P_o = 0.90 \quad (2)$$

where Tp is the time to reach the plateau, To is the time to reach 90% bio-degradation (set standard requirement), $Tmax$ is the maximum time of the standard test, Pp is the percentage biodegradation when reaches plateau, $Pmax$ is the maximum percentage of biodegradation achievable by any method, and α, β are two user defined weights. In Fig. 9, two example characteristic bio-degradation curves are shown with their corresponding values for Tp, $Tmax$, Pp, $Pmax$ and Po (where relevant). The number of experiments can be reduced by, for example, employing appropriate statistical techniques such as the Cholesky decomposition[97], to generate simulated data to cover a wider distribution range for key variables. Also, a "calibration" approach can be considered, using a Monte Carlo[98] method to generate candidate inputs (the parameter settings, e.g. composition) which give the result which is closest to the desired output (i.e., the optimum values for biodegradation in given condition). Data modelling algorithms will be used such as SVR (Support Vector Regression) and RI (Rule Induction): the SVR is a black-box technique adequate for numerical data, whereas RI generates a model as explicative rules which can also be used for data interpretation.

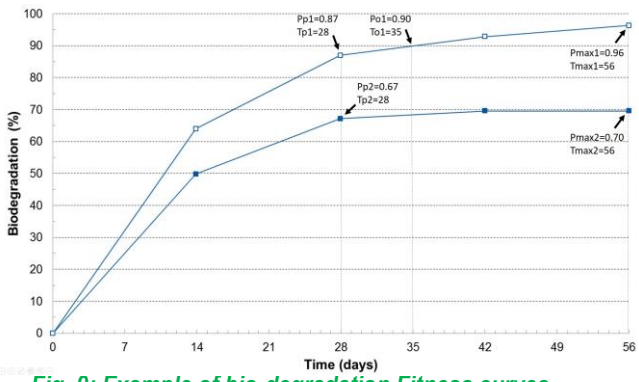

Fig. 9: Example of bio-degradation Fitness curves

"Deterministic" controllable test parameters and properties (crystallinity, crosslinking...) and the less controllable environment characteristics (exact compost composition in organic recycling or moisture, temperature, microbial activity in home composter) can be differentiated. The latter may be considered as having a "stochastic" aspect (that is, some random/chance element which cannot be defined in predefined rules). This supports the use of machine learning algorithms which can cope with a certain amount

95 Patrizia Cinelli, Markus Schmid, Elodie Bugnicourt, Jessica Wildner, Agostino Bazzichi, Irene Anuillesi, Andrea Lazzeri, Whey protein layer applied on biodegradable packaging film to improve barrier properties while maintaining biodegradability, Polymer Degradation and Stability. 07/2014
96 Patents EP2519596 (B1), EP2519565 (A1), EP2265681 (A2), EP1814935 (B1), EP1611217 (B1), WO2013/189956A1
97 Press et al. Cholesky Decomposition: The Art of Scientific Computing, 2nd ed. Cambridge, England: Cambridge University Press, pp. 89-91, 1992.
98 N. Metropolis et al., Equation of State Calculations by Fast Computing Machines, The Journal of Chemical Physics, 21, Vol. 6, pp. 1087-1092

of noise and stochastic behaviour. A CBR (Case Based Reasoning) approach can also be adopted in which successful configurations (i.e. environmental parameters and compositions) are stored in a historical database. CBR has been successfully used for relevant bio-industrial applications such as fermentation.[99].

Beyond the state of the art lacking any model for polymer biodegradation, based on the results of the biodegradation testing performed in different conditions along the formulation development during the first BIOnTOP project midterm, **data driven modelling that will account for parameters of composition, process, environmental use, thickness, etc. will be performed**. Machine learning algorithms, able to process complex datasets combining deterministic and stochastic behaviours, will be used to model both biodegradation and disintegration. The resulting predictive models will be a support tool to eco-design packaging that biodegrade according to the target EoL and standard (thickness, composition, etc.). They could also be relevant for the waste management plants to predict if compostability will be met for given packaging streams using the data from the previously discussed composition monitoring system used for sorting.

Innovation potential

While bioplastics[100] currently only represent about 1% of the about 300 million tonnes of plastic produced annually, experts predict a fourfold increase in production capacity within the next years[101]. According to EUBP data, in 2016 the split between bio-based/non-biodegradable and biodegradable (both bio-based, such as PLA, or fossil-based, such as PBAT) was 76.8% vs. 23.2% respectively with a predicted decline for the latter category. The EU currently has a very low presence in the overall bioplastics manufacturing industry and most of expected novel production sites will be established outside Europe as well. It is anticipated that the current European share (15%) in production capacity will decrease[101]. Nevertheless, countries like Germany and Italy have a very good position in the smaller segment of bio-based biodegradable packaging. It is therefore very important to equip Europe with differentiating bioplastic innovations and validate in sectors of high relevance in the EU economy.

Bio-based packaging

Europe is one of the largest consumers of plastics in the world and around 40% of the material processed is used in packaging[102]. **The food packaging industry is showing an increased interest in the development of new bio-based materials** to ensure that the food contained is safe and healthy and have an optimal conservation. Big brand owners have introduced bioplastic packaging, including Danone (Actimel, Activia, Volvic), Coca-Cola (PlantBottle), PepsiCo, Heinz, Tetra Pak and L'Occitane. But currently there is no market available cost-effective home compostable polymer with high bio-content that also complies with the requirements of sensitive products packaging. In 2014, 67.5% of produced biopolymers were used for packaging. The majority of the current bioplastic-packaging (86% according to EUBP data) consists of bio-based but non-biodegradable plastics such as bio-based PE and bio-based PET. However, there are undoubtedly applications where this feature is desirable (e.g. when food leftovers or material structure make material recycling non-feasible, or when collection is not properly handled). **In terms of biodegradable plastic packaging**, food packaging controlled more than 70% of the global market which is growing at over 18% CAGR globally from 2013 to 2019 to reach a value of $8,415.20 million[103].

Mono and multilayer Films and trays. The market driver for growth are new packaging material developments with improved properties, greater availability and lower price, increasing awareness of environmental issues, and the adoption of new regulatory requirements. While being biodegradable, in keeping with the waste management hierarchy, our solutions would still allow material recycling in the cases where such end of life option would be preferable. Using barrier materials, BIOnTOP also supports one of the general trends in packaging which is the **decrease of packaging weight** (thickness) to reduce the environmental impact due to resources efficiency and energy savings in transport. How this trend acts against easy material handling during recycling and further justifies the need for organic recycling as preferred EoL. Furthermore, such higher barrier packaging solutions can extend food shelf-life and maintain the initial food nutritional for longer, serving consumer interest in healthy food. This is especially relevant for the target group of organic food producers (such as WEAREBIO) who are very limited in terms of use of conservatives therefore request selected bio-sourced packaging, consistent with their products, to be ensure even better food conservation without requiring additives.

While the 2010 annual EU consumption was 10 billion pieces of multiuse bags, in the recent years, the case of shopping bags has widely evolved with the banning of single use fossil based non-biodegradable options in most EU countries. Complying 94/62/EC Directive on packaging and packaging waste and its amendment 720/2015, our innovative materials will tackle both the needs for bags used as recyclable secondary packaging used in multiple sectors, where thicker films, made out of SRM, will be blown extruded but also thinner films made of virgin BIOnTOP compounds compatible with primary F&V packaging. Indeed, the challenge of ultralightweight bags used to pack F&V still remains and it is estimated that over 10% actually leaks in the environment at EoL due to difficulties in their collection justifying the need for local home composting where possible (current materials meeting such target have too high cost or poor properties).

Textile primary and secondary packaging. A number of textiles are used in the packaging sector, such as nets, sachets, pads, etc. Mainly due to consumers' preferences in terms of convenience, as opposed to tea in bulk, individual serve tea bag is the

99 Montague, G., Martin, E., O'Malley, C. (2008) Forecasting for fermentation operational decision making, Biotechnol Prog. 2008 Sep-Oct;24(5):1033-41
100 including polymers that are bio-based, biodegradable or both, Entwicklung des Verpackungsverbrauchs nach Branchen. Gesellschaft für Verpackungsmarktforschung, 2010, Mainz, 3rd Edition.
101 Bioplastics- Facts and Figures, European Bioplastics, 2015, Berlin
102 Plastics- The Facts 2015, PlasticsEurope, 2015, Brussels Belgium
103 "Biodegradable Packaging Market by Packaging Type, Applications, and Geography– Global Trends & Forecast to 2019, RnRMarketResearch.com

leading format in the EU hot tea market. Tea bags were traditionally made of cellulosic non wovens with PE bags as secondary packaging. Nevertheless, improved materials based e.g. on PET are also used in new formats like pyramids. These pyramid bags are still limited in absolute numbers, e.g. €13million in Germany compared to standard bags (>€550million) [104]. However, whereas the revenue from standard bags is decreasing, the **pyramid tea bag revenue increased by more than 20% from 2015 to 2016**. Project partner SIOEN already supplies woven tea (pyramid) bags from PET and is also offering a biobased alternative from PLA. However, as already mentioned, that one is *not* yet home compostable. **To drive the application of (the slightly more expensive) PLA tea bags, biodegradation is key**. Further, following the trend for **minimizing packaging**, by our home compostable tea bags we will remove the need for secondary packaging, that way creating a unique truly sustainable solution.

Another driver for packaging innovation is the need for more reusable options. BIOnTOP will contribute to this via a textile based **reusable food wrap** (application relevant to AB member RollEat, see figure on the right) which are emerging solutions consumers can tap upon as primary carrier for dry food. The **global market for food wrap films was valued at US$ 10,038.6 million in 2017**[105]. The increased demand for on-the-go food across the globe is the key driver for the global food wrap films market. The market is forecast to expand at a CAGR of 2.1% during the forecast period, 2018-2026. Currently, market is dominated by aluminium and plastic wraps, together >80% of market. But, **the market is under pressure because of environmental concerns associated with the harmful effects of conventional plastic materials which are not biodegradable**. Moreover, the main materials used nowadays pose potential health threats[105]: (i) the **toxic effect associated with plasticizers in PVC** based food wrap films is hazardous for an individual; (ii) **LDPE based food wrap films may contain harmful chemicals** such as diethylhexyl adipate (DEHA); (iii) the **direct contact of food with aluminium-based food wrap films might cause dementia** and other adverse effects on the consumer health.

Furthermore, BIONTOP will also evaluate the use of SRM from our primary packaging applications into **reusable carrier bags** among other textiles use. Indeed, food contact is limited by the regulatory framework when working with recycled materials hindering similar applications as for prime material.

Besides assisting Europe in the transition towards a green economy in economically extremely important sectors in EU like food and personal care packaging, our innovative new materials will also be enablers for innovation in other sectors such as packaging for other goods which leftover at end of life prevent efficient recycling, other agricultural devices, disposable biomedical, non-packaging textiles and sanitary applications among others.

Overall benchmarking and IP position

While PLA is currently the most affordable bio-based and biodegradable polymer, the key BIOnTOP innovation potential arises from **improving PLA (or its monomers/prepolymers derived copolymers) formulations to match modular controlled biodegradability requirements under mild conditions as well as from the combination with novel coatings (PLA, whey and FA-based)**. At the same time, **price level reduction** can be aimed at, by the introduction of low-cost & low density NF vs. PLA while comonomers are iso cost vs. LA. This can help to bridge the current gap for further uptake in the considered packaging market where the economic aspects are extremely important. Likewise, the EoL issue (further discussed in section 2) is of extreme significance due to the volume generate by those single use items. **In addition to be almost solely based on renewable resources** (>85% based on renewable resources, partly produced from by-product biomass), the **waste management scenarios** of our new materials will also be improved, contributing from both ends to a **future EU circular bioeconomy**. In addition, this will on the long-term lead to a significantly reduced landfill (from currently not recyclable packaging) or incineration and **environmental burden from plastic derived products** (reduced carbon footprint and environment pollution by microplastics)**.**

Table 5 shows how we expect **BIOnTOP results to comply with the industry & consumer needs** comparing expected cost and performance vs. fossil-based non-biodegradable benchmarks (set at 100% for normalisation when no absolute value is available) and emerging bio-based/biodegradable counterparts based on set project KPIs from section 1.1.

Table 5: Benchmarking of our planned innovations vs. other commercial solutions

Demo materials	Compared properties	Standard (fossil) solution	Emerging bioplastic solution	BIOnTOP solution
Monolayer tray & films (F&V)	Bio-content	0% (e.g. PP tray and PE lid)	Close to 100% (PLA)	Close to 100%
	EoL	Limited recycling for rigid ones, landfilling when polluted	Industrial composting	Domestic and industrial composting, recycling for trays
	Approx. price /pack	100%	200%	140%
Nets (F&V)	Bio-content	0% (e.g. PE)	Close to 100% (PLA solution developed at TRL6 in EcoBioNet)	Close to 100%
	Tensile properties	Resistance:19 MPa Elongation: 680 % with standard material (PE)	Resistance:20 MPa; Elongation: 400 % with materials developed in ECOBIONET	100% /Maintaining the mechanical properties
	EoL	Landfilling	Industrial composting	Domestic and industrial composting
	Approx. price /pack	100%	200%	130%[102]
Multilayer tray & film (dairy, personal care)	Bio-content	0% (e.g. EVOH based multilayer structures)	Non available yet	>95%

104 https://www.mordorintelligence.com:81/images_db/RD_Images/Germany%20Tea%20Revenue%20By%20Packaging%20Type.png
105 https://www.transparencymarketresearch.com/food-wrap-film-market.html

	Food protection	100%	-	>100% (as UV-barrier is included in the protein coating)
	EoL	Incineration or landfilling	-	Domestic and industrial composting, recycling for trays
	Approx. price /pack	100%	-	100% (as very material efficient water vapour barrier due to the nano-scale grafting).
Textiles tea bags	Bio-content	0% (e.g. PET, PP)	Close to 100% (PLA)	Close to 100%
	Performance	Perfect by melt spinning	Limitations due to sensitivity to humidity and limited mechanical performance - Loss of odours and flavours during shelf life	Improved processability with new raw materials, preservation of flavours and odours thanks to water soluble coating.
	EoL	Landfilling/ when polluted difficult to collect	Limited *industrial* composting due to contamination, no home composting	Domestic and industrial composting
	Approx. price /pack	100%	200%	90 – 120 %
Reusable textile food wrap	Bio-content	0% (e.g. recycled PET with EVA lining).	Close to 100% (cotton based)	100% (PLA based with bio-based coating)
	Processabilty	Good	Good, but cotton production has big environmental impact	Good thanks to the use of optimised PLA compounds and bio-based coatings
	EoL	Incineration or landfilling	Compostable	Home compostable
	Approx. price /pack	100%	150%	120%
Secondary packaging (textile bags)	Bio-content	0% (e.g. PE, PP)	Close to 100% (cotton based)	100% (PLA based with PLA coating)
	Processabilty	Perfect (e.g. by staple fibre spinning & nonwoven production)	Good, but cotton production has big environmental impact	Good thanks to the use of optimised PLA compounds and bio-based coatings.
	EoL	Incineration or landfilling	Compostable	Material & organic recycling
	Approx. price /pack	100%	150%	150%
Secondary packaging (extruded blown bags)	Bio-content	0%- PE	0% (but from r-PE)	100%
	Mechanical properties	Tensile strength >25Mpa Elongation per break >400% Youngs modulus >350MPa Energy per break >4000KJ/m2	50-80% vs. virgin	Same as SRM from PE
	EoL	Landfilling, mechanical recycling (only for thick films e.g. 50 microns)	Landfilling, mechanical recycling (only for thick films e.g. 50 microns)	Composting, mechanical recycling (only for thick films e.g. 50 microns)
	Approx. price /pack	2€/kg	1,75€/kg- partially recycled content	2,80€/kg- partial recycled content

To maximise our innovation's commercial potential, beyond the definition of the new value chains and outreach activities, a comprehensive series of measures including consumer perception, standardisation and regulatory aspects, as well as business planning and knowledge /IPR management will be undertaken. The knowledge and IPs generated by the project will be exploited by partner companies, in order to improve their competitiveness, differentiating their products in the market, thereby tapping into increasing market and consumer demand for environmentally friendly packaging and textiles with high performance compared to traditional petroleum-derived counterparts and improved EoL options. The expected derived growth potential in terms of jobs and revenues are discussed in section 2 along the supply and value chain.

Freedom to operate was already given a major insight as it is vital for the commercial exploitation of the results, and a patent search has been performed around each of the technologies on which our target materials are based by the respective technology holders and experts (further reflected in the state of the art). The prior art is rich for all target applications in the project, and while certain areas are covered by patents which will be taken into account in the project development and the consortium holds several of the required patents that will be used as a basis for the planned advances in the project (see background table 13 in section 2), there is space for further innovating beyond the state of the art. Whenever possible, the project will consider not only developing confidential know-how but also results that could be protected by patents after deepening the IP search made in proposal preparation and ascertaining the business opportunity following a methodology further explained in WP7 and section 2. In particular, we expect to derive IP from the specific compositions that meet our EoL target (which will act as enabler for all derived products) and to copyright the biodegradation models (although the patentability of the derived eco-design tool will also be considered). In terms of relevant patents in our main innovation areas, we have reflected some in the following table, which differences with BIOnTOP are also highlighted (when relevant they will be further monitored along our technology watch activities to see the evolution of their approval and enforcement status):

Table 6: Results of the patent search focused on the main innovation area of the novel IP considered to be generated in BIOnTOP

Patent (reference, title, *status*)	Relevance to BIOnTOP	Conclusion in terms of impact on the project
US 611033A, PCT/IE1997000031 Acidic polylactic polymers, *withdrawn in EU*	Biodegradable polyesters from lactic acid and other hydroxyacids or acids	Although this patent is withdrawn and therefore does not affect our freedom to operate (FTO), we will consider different diacids when possible to maximise our copolymer novelty and combine it with other approaches not contemplated in this patent
KR20040063236A, Process for producing biodegradable polylactic acid by direct bulk polycondensation, *published*	Process for producing a high Mws PLA, excellent in biodegradability, tensile property, by bulk-polycondensation of an organic divalent carboxylic acid selected from a SA, an adipic acid, or a mixture thereof, and a diol.	Different comonomers will be applied but in the case of SA our FTO is not affected because this patent is only filled in Korea. BIOnTOP's novelty will be ensured also by synthesis strategies (REX)
CN102786672B, Preparation method of polylactic acid segmented copolymer, *published, only filled in China*	SA and butanediol react to obtain poly(butylene succinate) in a molten condition; and poly(butylene succinate) and lactide react to obtain the PLA segmented copolymer.	BIOnTOP will apply direct polycondensation and reactive extrusion (REX) approaches
JPH0753685A, Polylactic acid	Biodegradable copolymer of lactic acid, dicarboxylic	This patent does not affect our FTO, nor our novelty because a

copolymer, *published, not renewed*	acid (succinic acid), glycol (ethylene glycol) and/or alpha-oxyacid and/or a cyclic lactone	different copolymerization approach is applied involving other comonomers (an additional alpha-oxyacid and/or a cyclic lactone), catalysts and conditions to obtain a low melting copolymer.
US6342268B1, Method for treating a solid material to make it hydrophobic, material obtained and uses	A solid material is treated, the chemical structure of which defines reactive protogenic hydrophillic functions accessible to gases, applying a gas stream onto a microdispersion of at least 1 grafting reagent.	BIOnTOP will partially use the described process which was filled in 1997. Thus, the term of the patent expired already and therefore will not affect our FTO.
JPH10101911, Emulsification-evaporation technique for PLA coating; only granted in Japan	This patent (among other related but none leading to commercial products) describes a possible method for coating PLA.	BIONTOP will develop both improved PLA copolymers and formulations with plasticizers allowing the application of a PLA coating on top of a PLA textile but use a totally different technology.

2. Impact

2.1 Expected impacts

In a world of finite resources, the transition to a knowledge-based post-petroleum economy is imperative. To support this transformation towards a low-carbon Europe, reducing its heavy dependency on fossil resources, the EC has presented a roadmap targeting the **establishment of sustainable and competitive bio-based industries and value chains**[106]**.** The Bioeconomy Strategy and its Action Plan pave the way to a more innovative, resource-efficient and competitive EU society. That reconciles food security with the sustainable use of renewable resources for industrial purposes, while generating new incomes and ensuring job creation. In this context, the BBI's SIRA aims to accelerate, among other things, the **development of bio-based packaging and plastic sector**, decoupling economic growth from resource depletion and environmental impact. Perfectly in line with this goal, BIOnTOP addresses the need to bridge technology gaps, generating the necessary enabling knowledge and technologies to build and boost the new value chain from bio-monomers to end markets. In this vision, the bio-based industries play a strategic role in complementing and replacing product streams from fossil oil and providing innovative bioplastic solutions for different applications. In the strategic area related to **bio-based materials,** the BBI contributes to the ambition that the market **supplied by bio-based polymers and composites** will be **5 times higher than today** in 2020 **and 10 times higher in 2030.** This goal implies also a comparable quality ratio to fossil alternatives. Moreover, replacing fossil-based materials have proven to have an equal or overall better sustainability, in terms of biodegradability recyclability and material efficiency. The bio-based raw used in BIOnTOP are **locally, renewably and sustainably sourced chemicals and bioplastics. The multifunctional coatings developed will further help differentiating the bio-packaging by unique performances complementing their multiple EoL compatibility.** In addition, increased consumer awareness and acceptance of bio-based products, concerted policy and labelling will make an important contribution to improve market penetration. As such, the BIOnTOP packaging constitutes an **enormous business opportunity** helping to nurture the realisation of the EU CE[107] that requires also the optimal management of bioplastic waste. Nowadays, plastics are used in various sectors with a demand of 48 million tons per year; packaging makes up for 40% of this use, of which food packaging accounts for around 90%. In this sector, **the EoL treatment of plastics is underperforming**, especially when compared to more homogenous materials such as metals or glass, traditionally used in food packaging. According to PlasticsEurope[1], in 2017, 31% of plastic waste was recycled in Europe; nevertheless, landfill and incineration are still dominant treatments for plastic post-consumer waste. The recently adopted EC's plastics strategy aims at decoupling plastics production from virgin materials, reducing their leakage into the environment, improving the economics and uptake of recycling to support the transition from a linear to a circular economy model.

BIOnTOP is a **high impact** project for the improvement of the bioeconomy in Europe. Thus, it is hinged on delivering **cost competitive packaging solutions that can be mechanically recycled or industrially/domestically composted** combining different approaches to tailor the bioplastic biodegradation. In this way, **the sustainability from cradle to cradle of several new bio-based high-volume applications will be maximised.** This impact will be achieved by performing **research work** with the objective of making this concept **commercially viable** in practice, analysing the potentially **successful exploitation pathways** and **assessing and managing any related risks**. As shown in fig. 10, BIOnTOP significantly contributes to the KPIs set out in the overall BBI-JU roadmap and more specifically to the expected impacts in the topic BBI.2018.SO3.R10- *Develop bio-based packaging products that are biodegradable/ compostable and/or recyclable*, as described below:

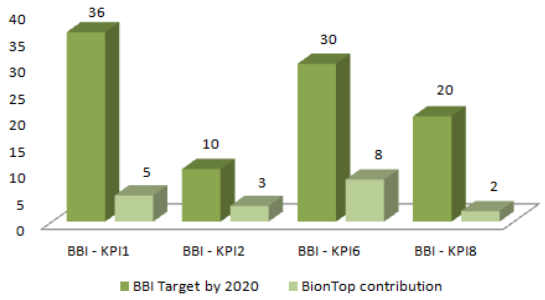

Fig. 10: BIOnTOP contributions to BBI JU KPIs

- Creation of at least 1 new cross-sectorial interconnection in bio-based economy clusters (KPI 1)

The BIOnTOP proposal builds upon the BBI goal of **achieving the full potential of the bio-based industries**, which requires swift and concerted action of today's dispersed stakeholders across various sectors, disciplines, and Member States. Connecting different partners, such as researchers, technology developers and producers, through clustering and networking is key to developing new value chains and supporting the emergence of new bio-based processes and products. To that end, BIOnTOP

106 COM(2012) 60 final. The Commission Communication of 13 February 2012 entitled "Innovating for Sustainable Growth: A Bioeconomy for Europe".
107 COM/2015/0614 final. The Commission Communication of 2 December 2015 entitled "Closing the loop - An EU action plan for the Circular Economy".

will **define the relevant sustainable circular value chains & cross-sectorial connections** in a dedicated task in WP7. The **active involvement of all key players along the value chains** (including additional sectoral associations in the Advisory Board) will allow us to create **long-lasting links** with related clusters, necessary to fill a clear gap in the marketplace for sustainable plastics especially at EoL, safe for human health and the environment but also, and above all, effective in terms of performance.

With BIONTOP we will establish 1 new bio-cluster connection and further intensify 4 connections, among the sectors of bioplastics, packaging, agri-food, personal care and textile sectors as explained below.

BIOnTOP will create an unprecedented systematic, cross-border and cross-sectorial, connection all actors from 'cradle' to 'grave' for the considered bioplastics-based end use applications. Indeed, **our consortium covers the cradle-side, i.e. biosourcing**, represented through the bio-based feedstock (e.g. food processors and monomers providers in our AB) **and plastics producers** associated with EUBP**. But, the BIONTOP consortium also covers the grave-side, i.e. waste management aspects**, represented by ROMEI, OWS(terminated)/OWS RF and ISWA (in our AB, in particular its working Group on Biological Treatment of Waste will contribute). Interconnecting these key organizations means bringing together the missions and commitments of their respective industrial members. Their presence in the consortium favours the generation of positive spill over effects among their affiliates leading to push-pull effects for the new BIOnTOP technologies.

While the current recycling of bioplastics is almost inexistent industrially, this **new connection will allow launching concerted advocating for an improved framework for circular bio-based plastics**. Potentially including incentive schemes both at EU and national levels (especially while the waste management landscape varies very significantly between the EU Member States) and providing valuable input for on-going standardisation efforts at EU level. This will contribute to several legislative initiatives in Europe aimed at reducing the use of conventional petrol-derived plastics by promoting the use of bio-based plastics, which are required to be biodegradable and compostable. For example, a new packaging law in Germany encourages the use of materials from renewable resources or recycled materials for the production of packaging, supports the use of certified bio-based and compostable plastic bags for the mandatory separate collection of biowaste. Likewise, in 2016, France adopted its 'Green Growth and Energy Transition law' and stipulated that bio-based, home-compostable packaging was mandatory for certain applications (e.g. thin F&V bags in the supermarket, mail order wrappings, etc.). A Decree specified percentages of bio-based content to be reached by defined dates: 30% bio-based content by 2017, 40% by 2018, 50% in 2020, and 60% in 2025. In BIOnTOP, by bringing together the producers of bioplastics and key players advocating efficient and sustainable waste management, we expect EUBP and ISWA´s (AB member) lobbying towards policy and decision makers to support concrete measures for the transition towards a more efficient circular bio-based economy.

BIOnTOP will further intensify 4 emerging cross-sectorial interconnections in bio-based economy cluster:

- We will reinforce the **bond that the bioplastics sector has with the packaging sector** (which is by far the largest consumer of biodegradable and bio-based plastics) by providing new demonstrators in food and personal care applications. That will be made possible thanks to BIOnTOP solutions' unique EoL tailored to their needs. This new link is for example illustrated at the level of our Large packaging converter innovation centre, EMSUR/INNOTECH, which so far has only made very preliminary tests with commercial biopolymers but does not have yet any bio-based packaging in its associated companies' product portfolio (due to insufficient properties for the main application served). The participation in BIOnTOP will ensure EMSUR/INNOTECH to drive the development towards actual market needs.
- **Relying on the transition from linear to circular value chains, we close the loop connecting bioplastics producers with biomass generators, which are also end users of compost**. This way we will create **new upstream links with producers of unvalorised and unavoidable processing waste** opening up unprecedented business opportunities for actors that conventionally are positioned at the end of the packaging value chain. BIOnTOP aims to deliver close to 100% bio-based formulations using highly sustainable biomass feedstocks. Whenever possible we will use by-products, in particular, agro-wastes (e.g. monomers, NF, waxes from oil byproducts as compatibilizers for natural fillers and citrates plasticisers from citrus wastes, proteins from cheese byproducts), as well as sustainably sourced FA from oil. In particular, third generation malic acid coming from enological sector thanks to the collaboration with OENO, and natural fillers from wheat middling by WEAREBIO, will be valorized. These are particularly relevant by-products considering that the EU is the world's largest producer of wheat (156 million MT) and fruits (40 MT[108]) leading to waste per year of respectively roughly 30% and 20%[109]. The use of these in BIOnTOP high-value applications will create new incomes for agri-food sectors. These biomass fractions will reduce raw material cost vs. currently available pure PLA, increase disintegration rate, and allow extending the accessible uses for our bioplastics. As such, the current waste challenge turns into an enabler for innovation allowing a significant decrease in the environmental impact of the agri-food sectors and of the value chains of the BIOnTOP solutions. We will **have the agri-food processing industry playing a strategic polyvalent end user role within the packaging value chain**. While, as seen, they act as material supplier along the new value chains, they need to be supplied with compost to nurture F&V and plastic packaging to package food. Incontrovertibly, this magnifies **the strength of the new links and the resonance of our solutions in the new circular bio-based value chains.** In our consortium, for example, this cross-over role is assumed by WEAREBIO, who validates the new packaging solution in organic food application especially F&V, but also supplies NF. This strategic role is reinforced by its organic food specialization and related vision that

108 areflh.org/index.php?option=com_content&view=article&id=123%3Aleurope-des-fruits-et-legumes&catid=3%3Aterritoires-dintervention&Itemid=74&lang=en
109 FAO. 2011. Global food losses and food waste – Extent, causes and prevention.

all steps of the value chain should be environmentally sound. Likewise, ENTREPINARES can supply whey from cheese making which in turn is used as oxygen barrier coating for the packaging used to pack the dairy products.

- Besides using textile as primary packaging, the link **between the packaging (as supplier of EoL material) and the textile sector** will be reinforced. Indeed, while the latter is already using a fair amount of the recycled plastic from e.g. PET bottles, with BIOnTOP, new bio-based SRM but also coatings, will allow extending the applications covered (e.g. secondary packaging like bags with improved functionalities).
- **Cross-sectorial interdependencies will be created through the synergies among the needs of the sectors considered in the demonstrators** (e.g. similar materials serve the needs for MAP and personal care). We will build on such synergies to reach out to sectors beyond the consortium, like pharma packaging. This sector equally needs barrier properties, especially to humidity for thermoformed blisters as well as sachets. Further links with the agriculture sector can also be established though the discussion of the post project potential of the new materials with enhanced biodegradability (e.g. mulching films, pots and nets). All these links, considered when building our AB, can contribute to the outmost exploitation of our results.

- Creation of at least 1 new bio-based value chain based on the innovative products developed (KPI 2)

To contribute to the achievement of the KPI 2 aimed at delivering "10 new bio-based value chains by 2020" and maximise the potential to deliver new bio-based products, BIOnTOP will successfully prepare for the future creation of operative new value chains based on extensively tested and validated novel bioplastic solutions. As such, BIOnTOP consortium leverages partners at every stage of the supply and value chain, as well as additional stakeholders in the AB. The development of a suite of novel bioplastic packaging solutions that respond to unmet demands in terms of functionality and EoL requires connecting a critical mass of actors that are able to generate value working in a synergistic collaboration. We aim to create **3 new stable and effective value chains not only by new players interconnections at the upstream of the chain** (to obtain the new bio-based raw materials and process polymers) but also **remodelling existing value chains** (from compounders, plastics converters onwards) that will substitute their petrol-derived products with our bioplastics solutions. As illustrated in Fig. 11, upstream players crossover the three targeted packaging sub-sectors (food, personal care and textile).

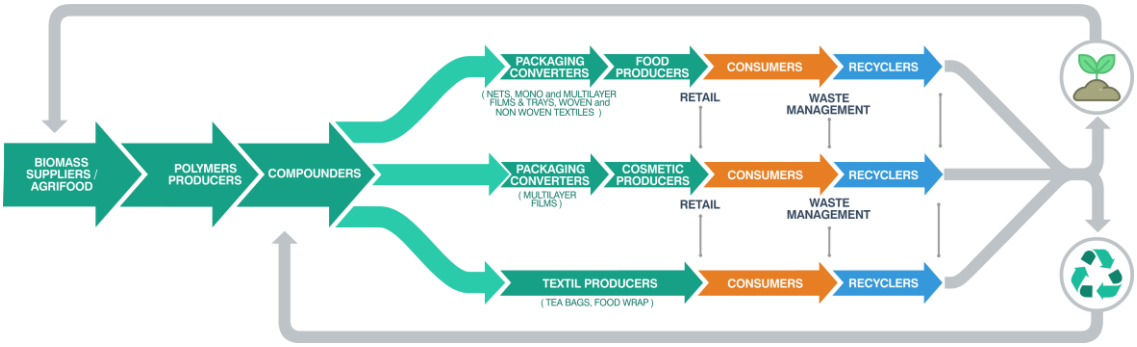

Fig. 11: BIOnTOP value chains (New links in KPI1 across the value chains shown by grey arrows)

BIOnTOP is uniting expertise in food technology, materials science, plastic processing technology, compostability, recyclability, biodegradability, LCA, engineering, as well as **industry actors at different levels of the value chain**, capable of steering the research in tune with their market knowledge. To reach the project impact of the delivery of **bio-based packaging which plastics match specific EoL requirements** that are currently not satisfied by their conventional counterparts, joint actions will be needed at different levels of the value chain, e.g. between material producer, final item designer and waste managing body so the recycling is actual and not a mere recyclability label. The biodegradation predictive model delivered in WP4 will be a common tool used at these different levels facilitating this collaboration.

Currently almost 40% of all plastic consumed in Europe is used for packaging[1], BIOnTOP is therefore **embracing the most significant applications for bioplastics in its new value chains**. The value chain approach will also guarantee that access to selected knowledge occurs among the partners to reach a seamless exploitation. The BIOnTOP actors will work towards similar goals and a proportion of Large end users, who will test and validate the new bio-based solutions in the selected target demonstrator applications, are included as early adopters at the end of the value chain to stimulate the demand. This will particularly benefit SMEs (IRIS, CMSA, PLANET, OWS(terminated)/OWS RF, WEAREBIO, BIOMI, ROMEI, ARCHA) as it will provide them with new up- and down-stream partners. Cooperation among BIOnTOP members will be particularly important in the post-project phase, as results of the project are pre-competitive and additional investment will be required before BIOnTOP is ready for commercialisation. As addressed through specific tasks in WP7, setting the basis for a robust value chain will be key to generate real value on the market. Building on the experience of strong players in the packaging and biopolymers sectors will play a massive role in designing promising products and shaping their successful market entry phase. As such, we are confident that our developments, meeting true market demand, can convert into a sustainable supply & value chain of bio-based material with functionalities that outperform existing plastic materials at a competitive market price.

- Creation of at least two new demonstrated 'consumer' products based on bio-based chemicals and materials that meet market requirements (KPI 6)

The consumer is given an important position in our bio-packaging value chains, as direct user of all BIOnTOP products in single or multiple uses. His perception and aspirations will be studied in WP7 to maximise our ability to satisfy his demand for environmentally-friendly products.

BIOnTOP will deliver **two main types of biopolymers derived innovations** resulting respectively for WP2 and 3:

1) **Copolymers and compounds** with controlled biodegradation profile even in mild conditions;
2) Whey, PLA and FA **coatings** with unique functionalities (barrier to oxygen, aroma, humidity, easy emptying, waterproofing, low tackiness) still compatible with optimal EoL of derived coated materials. Those can enhance the consumer's experience, facilitating the product release from the packaging, reducing waste and maximising products organoleptic characteristics.

Those 2 new families will be broken down in a series of tailored compounds and coatings grades that can be combined in the material structures shown in figure 1. From those materials, the following packaging application with tailored EoL will be validated with selected packed products as **demonstrators** (EoL compatibility and preferences in table 7):

1) Monomaterial **trays & films for F&V**
2) Multilayer **trays & films compatible with MAP** for e.g. **dairy product**, and multilayer **sachet for single serve or flow pack for personal care** products such as wet wipes
3) **Nets for F&V**
4) Monofilament **woven fabric for tea bags**
5) Reusable **textile food wraps**
6) **Multiple use bags** as a secondary packaging solution from SRM
7) Reusable **coated woven carrier bag** from SRM

In terms of market requirements, beyond the **suitability of the properties** of the proposed bio-plastics, also **economic competitiveness** has been given great concern since the proposal set up (see benchmarking table 5 at end of section 1.4 for more details on the expected performances of the BIOnTOP solutions for each application). The biopolymers, raw materials and additives have been carefully selected to ensure that the BIOnTOP packaging solutions can be cost effective vs. their conventional counterparts currently used in the target application sectors but also more affordable than emerging bio-based solutions. Likewise, required equipment and processing costs will be given attention to facilitate successful introduction and uptake in the market. In WP1, the market costs and performances in comparison with benchmark materials to replace will be considered when establishing the specifications of the materials developed while the overall products eco-sustainability will be verified in WP4 by LCA.

The choice to use of PLA and related copolymers in our packaging innovation is justified by its competitive price vs. other biodegradable polymers (fig. 11[110], still representative of the overall comparative situation according to EUBP), high maturity and volume available. In 2016, the price level of the PLA was around €2/Kg (depending on the grade, which is 2.5 to 7-fold less than PHAs) and the global production capacity for PLA was 213,847 tonnes, which equals a share of 5.1% of global bioplastics production capacity (which stood at 4.12 million tonnes) and **22.2% of biodegradable polymers**. Europe was the largest market for PLA demand and is expected to grow at CAGR of 19.2% from 2017 to 2022 in light of regulatory norms on the implementation of bioplastics in various end-use sectors including food & beverages[111].

Packaging accounted for 59.6% of the overall PLA market in 2013 and is expected to remain the dominant application segment over the next period[112]. Textiles are expected to be the fastest growing segment for PLA, at a CAGR of 17.5% in terms of revenue from 2014 to 2020. In 2021, PLA production capacity is expected to rise to 323,147 tonnes, which is 5.3% of the estimated 2021 global bioplastics production of 6.11 million tonnes. The growth of PLA greatly outperforms that of other biodegradable materials and keeps up with the impressive general bioplastic growth. Nevertheless, in the next 5 years, Europe's share in PLA production is forecast to stagnate at a low level (2016: 4%; 2021: 3%). In BIOnTOP, tapping upon broad EU demands for products with sound EoL, we aim to revert the current deficit in EU production of PLA compounds and related copolymers from local resources. To fill such scarcity, TOTAL-CORBION, partner of the project, which has a tenfold production capacity comparing to the first EU PLA producer and is fueling the project with a unique PLA know-how and over 25 years of expertise in the field of biodegradable biopolymers. TOTAL-CORBION continues evaluating the options of powering up a new industrial plant in EU by himself or through future partnership with EU companies for local PLA production as could be triggered by innovations like BIOnTOP.

Introduction of low cost natural fillers and additives will be a key contributor to allow cost-efficiency vs. commodity plastics and indeed significant decreases vs. currently available pure PLA. That is why we expect to boost the market for the lactic polymers family. Indeed, the initial comparative cost estimation of the BIOnTOP solutions provided in section 1 (table 5) indicates that our formulations are expected to cost a moderate premium vs. the non-bio-based counterparts (difference <40%) and significantly less than pure PLA. However, the calculated price is positively balanced by the extended shelf-life in case of barrier packaging, the reduction of relative weight but also in all cases the very high bio-based content of our plastics (close to 100%) and currently unmet sustainable EoL options. It is worth specifying that the price of fossil-based plastics is depending on oil prices and fluctuates

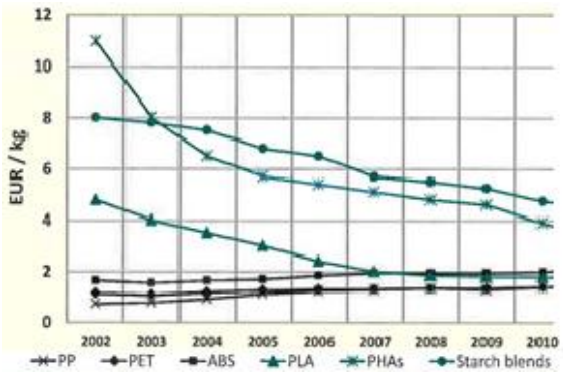

Fig. 12: PLA cost /kg vs other biopolymers between 2002 - 2010

110 ENDRES, H.-J. & SIEBERT-RATHS, A. 2011. Engineering Biopolymers - Markets, Manufacturing, Properties and Applications, Munich, Hanser-Verlag
111 https://www.gminsights.com/industry-analysis/lactic-acid-and-polylactic-acid-market
112http://www.prnewswire.com/news-releases/lactic-acid-and-poly-lactic-acid-pla-market-worth-43122-million-and-21696-million-respectively-by-2020-grand-view-research-inc-512033561.html

while the prices of PLA, and in general bio-based plastics, depend on biomass prices that exhibit a decreasing trend as further feedstocks are grown and refined for diversified purposes. In fact, although greater than that of polyolefins, PLA price is very close to market prices of general purpose PS and PET[113], so improving its barrier properties and other functionalities to meet markets requirements can unlock new applications.

Validation of at least one new and improved processing technology reflecting the TRL gain since the start of the project (KPI 8)

As discussed with TRL table 3, we will follow a scale up approach of different technologies where the manufacturing readiness level MRL gain will follow that of the TRL (at least 5 at project end). To maximise the possible transfer to new applications with tailored EoL, we selected broadly used conversion processes such as extrusion (in its different variants like compounding, flat film/sheet extrusion, blown extrusion, welding/sealing filament melt spinning) and further steps like gravure coating, thermoforming (e.g. nets and thermoformed pots are also used for agriculture, textiles are also used for carrier bags, etc.). Although we will work on optimising any step, those are not expected to be significantly different for the new materials.

In addition, we will develop 2 novel processes in terms of **copolymerisation and FA grafting** (considered as our main contribution to KPI8). As opposed to the prior art, we will mainly focus on the development of novel bio-copolymers from LA/LD and malic or succinic acid in combination with a diol with tailored properties by polycondensation and ROP routes that will be performed in batch or by a continuous REX process in order to obtain the PLA copolymers. As it was described in the state of the art, REX has emerged as a powerful tool with potential to replace chemical batch reactor in which a twin-screw co-rotating extruder is used implying several advantages: solvent free melt (no downstream processing and drying needed, no by-products), low cost continuous process, starting from monomers/ prepolymers and control of the chemistry throughout the polymerization. REX can enhance the commercial viability and cost-competitiveness of these materials as already demonstrated about biodegradable polymers like PLA. In view of the critical mass of very promising lab and pilot scale results using REX, it seems now extremely timely to deal with its industrialisation for the production of new high added values biopolymers. The current TRL of these technologies is 3 and will be increased until 5. Trials will be subsequently made at AIMPLAS and at the SME partner PLANET.

As described in the state of the art, within the BIOnTOP project, an innovative approach will improve the water vapour barrier properties of bio-based films and coatings by **green chemical grafting with fatty acids from renewable resources**. This will result also in waterproofing and easy emptying behavior[114]. Previously developed transfer methods will be adapted for the FA application on the materials from the project and then upscaled on a gravure printing/coating line at INNOTECH (pilot in industrial environment trials on a machine in fig. 13 at TRL5 to 6 will be made). Since it is solvent free and very low amount of coating (nm range) are deposited, the process is very resource efficient both in terms of energy and materials.

Furthermore, **IRIS will bring industry 4.0 paradigm/factory of the future concepts expertise**. Wherever applicable, advanced processing approaches will be followed to allow speeding up the upscaling and facilitate the future industrialisation of our results: that includes the use of tailored monitoring systems for automatic identification of materials to be recycled but also of predictive models developed through advanced artificial intelligence methodologies for supporting decision making e.g. for eco-designing the formulations with biodegradation behaviour suited to the specifications of the application. Further process modeling will be used for predicting the process conditions to dry protein coatings so they lead to optimal properties according to Wheylayer patent.

Fig 13: left to right: REX pilot plant at TRL5, printing line at TRL 6 and Nordmeccanica Labo Combi 400 pilot coating line at TRL5

Reduction of the environmental footprint associated with the end-of-life phase of developed packaging products by at least 30 % compared with existing products for similar applications

Using our novel BIOnTOP compounds and coatings, the target **EoL will be the starting point to develop our sustainable-by-design bio-based packaging solutions**, which will suit biodegradability in different conditions and materials recycling requirements. BIOnTOP will ensure that **the impact of plastics on the environment** is minimised by increasing the share treated in waste management facilities or domestically composted to avoid the collection challenges for certain types of packaging. Indeed flexible/light packaging, nets and small format items are waste-intensive and, even if they represent 10% of the packaging market by weight, such items are very prone to leakage into the environment and are disproportionately present as marine litter. In our project, we will also **deliver biodegradation predictive models** taking into account the EoL scenario required. This will equip plastic manufacturers with a validated tool that permits them to accelerate the selection of the right formulations for eco-designing new packaging based on our materials.

EoL considerations are particularly important for brand owners adopting bio-based plastic solutions[115]. The project focuses on packaging that, as opposed to their synthetic counterparts which are often landfilled or incinerated, can be organically (i.e. industrially or in home-composting conditions) recycled or even suitable for AD. Table 7 shows how BIOnTOP can **instil a new**

113 Bio-based and biodegradable plastics - Facts and Figures. 2017. Wageningen Food & Bio-based Research number 1722, ISBN- 978-94-6343-121-7
114 Schmid et al. (2014) Water Repellence and Oxygen and Water Vapor Barrier of PVOH-Coated Substrates before and after Surface Esterification. Polymers 6(11) 2764-2783.
115 Open-BIO H2020 project, Deliverable 9.2 / A II: Acceptance of Bio-Based Products in the Business-to-Business Market

bio and circular-by-design ethos in the packaging sector (reinforced by the fact that some virgin materials are in fact by-products from agri-food processing). Recyclability of post-production wastes will be possible for all our materials. Recycling of post-consumer packaging would be technically feasible for all our materials but could be limited by the application (i.e. in presence of significant amounts of food leftover in the waste plastic). In such case, organic recycling is more suited, but it of course depends on the actual availability of a selective collection system for the organic waste fraction, which still lack in many EU countries. In such case home composting gains further relevance, AD is also seen as a valuable organic recycling procedure as also biogas is produced. BIOnTOP, as home compostable, will also degrade in mesophilic AD-plants. Due to the lag phase, it will probably be only partly degraded and then it will need to degrade further in compost or soil. Finally, despite incineration is surely not our target EoL, as opposed to the fossil-based materials, incineration of bio-based plastics would provide renewable energy with a lower greenhouse impact compared to e.g. polyolefins. Large scale recycling and biodegradation tests under different environment as well as compostability and recyclability trials will be performed in WP4 to confirm we meet the target EoL of biodegradation under industrial, home composting and anaerobic conditions and to show the harmlessness of the bio-sourced material on soil. Results from the tests on final packaging and textile prototypes will confirm that BIOnTOP solutions meet the conditions of Vinçotte's "OK compost home" among others.

Table 7: End of life options (x: compatible, X: preferred) of developed BIOnTOP materials vs. market available ones

Application	Solution	Material	Material recycling	Industrial composting	AD	Home composting	Incineration with energy recovery	Landfill
Monolayer trays & films for F&V	BIOnTOP	copolymers or PLA compounds based tray and lid	X for tray in absence of spoiled F&V	X	X	x	x	
	Benchmark	PP tray and PE-lid	Only for tray in absence of spoiled F&V				X	X
Multilayer tray & film (dairy, personal care)	BIOnTOP	Copolymers or PLA compounds based nets, whey and FA coating	X for tray in absence of spoiled F&V	X	X	x	x	
	Benchmark	PS or PE/EVOH tray and PE/EVOH/PET film					X	X
Nets for F&V	BIOnTOP	Copolymers or PLA compounds based textiles	No (too light & contaminated)	X	x	X	x	
	Benchmark	PET or PP or PE					X	X
Monofilament fabric for tea bag	BIOnTOP	Copolymers or PLA compounds based textiles, aroma barrier	No (too light & contaminated)	x	x	X	x	
	Benchmark	PET, PA, PP					X	X
Textile food wrap	BIOnTOP	Copolymers or PLA compounds based nonwoven coated textiles	X	X	x	x	x	
	Benchmark	Benchmark: PET					X	x
Secondary packaging textile bags	BIOnTOP	r- copolymers or PLA compounds based nonwoven coated textiles	X	X	x		x	
	Benchmark	Benchmark: PP, PE (coated)					X	x
Multiple use shopping bags	BIOnTOP	r- copolymers or PLA compounds based films	X (if > 50 microns due to collecting challenges below)	X	x	X	x	
	Benchmark	Benchmark: PP, PE					X	x

As mentioned before, a very limited share of plastic waste is currently recycled. In fact, several conventional petrol-derived light weight packaging (trays, nets, lids) are very difficult to handle by recycling plants and even create troubles for the recycling of the main stream rigid packaging and are therefore better considered for organic recycling as in BIOnTOP. Furthermore, as opposed to standard PLA, our new materials aim to meet industrial compostability criteria even at high thickness. That will open further avenues for the use of BIOnTOP materials in thick thermoformed and injection moulded items used in different sectors including agriculture and packaging.

In a **holistic approach to mitigate current EoL challenges**, besides delivering new materials, **BIOnTOP will also develop a specific monitoring device that will ensure the right material sorting for either organic or mechanical recycling**. Those will prevent the landfilling of recoverable post-consumer waste. Although we bring novelty in terms of biodegradation in mild conditions, keeping with the waste management hierarchy, we will also show that the recycling of BIOnTOP packaging, post-industrial wastes and rigid fractions is viable contributing to increase the mechanical recycling rates of plastic in the future CE (target <10% landfill in municipal solid wastes by 2030). To this aim reprocessing trials will be performed. The need to explore the establishment of industrial recycling facilities which are dedicated to non "drop-in" bio-based plastics such as PLA, currently very limited in spite of the high value of these materials, will be also discussed.

Although whenever possible controlled waste management should be favoured, in BIOnTOP we will develop **modular formulations for home composting** which, besides meeting previously discussed national regulations, will be an asset for countries with limited composting plants and for domestic valorisation since a fair share of F&V are spoiled while still unpacked at consumer level. Likewise, some food packaging textiles like tea bags require alternative EoL options since the contamination of plastic with other (organic) materials hinders the quantitative removal for mechanical recycling or industrial composting. As such, combined home composting will favour the market penetration of bioplastics for specific applications. Furthermore, apart from biodegradation, our solutions will be also suitable for AD. Organic recycling by treatment in industrial biogasification plants will be evaluated, by the option of subsequent aerobic composting to represent industrial biogasification plants with additional

composting step. Investigating AD as EoL option, **BIOnTOP will contribute to the definition of standards that currently do not exist**.

In terms of **environmental footprint** of EoL phase, according to the EC DG ENVI "Plastic waste in the environment"[116] study from 2011, plastic film recycling could potentially save **0.74 kg CO_2 eq./kg recycled plastic in comparison to landfill (>50%):**

In case of organic recycling, a drastic reduction of the carbon footprint and GHG emissions is also observed. The captured CO_2 during biomass growth stays in the material over its whole life-cycle and is set free at the end of its life (by biodegradation processes) closing the carbon loop. If controlled home composting is achieved there will be additional environmental benefit due to saving of organic waste transport to compost facilities, absence of energy and man work consumption for compost production.

Table 8 below considers PLA (as the closest material to BIOnTOP) versus others most commonly used packaging materials (e.g. PE films and PP trays) and compares their typical EoL scenario. This shows advantages of managing waste packaging by composting vs. incineration greater than 90% in greenhouse gases (GHG) emission. All in all, as mentioned the waste management landscape is very variable in different countries so any comparison will have a specific applicability but we are confident that all our target EoL scenario meet the topic target of a 30% reduction in environmental foot print.

Table 8: Comparative EoL CO_2 foot print for relevant packaging materials [117]

EoL	CO2 foot print
PLA collection and industrial composting	0.0612 kg CO2-eq./kg PLA
PLA home composting	Ca. 0 kg CO2-eq./kg PLA (if effective, aerobic biodegradation; assume no transport and manual composting).
PE collection and incineration as Municipal Solid Waste (MSW)	3.03 kg CO2-eq./kg PE
PP collection and incineration as PE collection and incineration as MSW	2.55 kg CO2-eq./kg PP

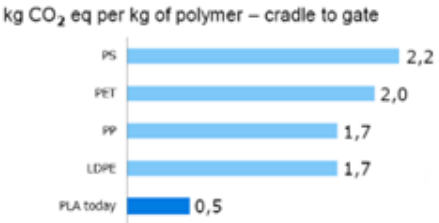

Fig. 14: Emission from polymers production (further PLA aims to become carbon neutral)

The improved environmental footprint using bio-packaging with sustainable EoL is not limited to their waste management phase but can resonate throughout their life cycle. In terms of sourcing, among the LCA impact criteria, the main benefits of using bioplastics vs. conventional plastics are generally in terms of saving non-renewable resources and low toxicity connected to human cancer. According to the BBI JU's SIRA[5], bio-plastic has been identified as a major sector that determine the decarbonisation of our economies. In terms of **resources**, bio-based plastics have the unique potential over conventional plastics for reducing GHG emissions or even being carbon neutral (1 metric ton of **bioplastic** generates between 0.8 and **3.2 fewer metric tonnes of CO_2** as compared to 1 metric ton of **petroleum-based plastics**)[118],[119], as shown on Fig. 14[120]. When possible, the project will consider the use of raw materials from waste (e.g. MA, NF, etc.). Indeed, by-products are assigned a null impact in LCA, which will help minimising our materials' CO_2 footprint. Also, the use of natural fillers in trays and textiles, will reduce the use of biopolymer considerably and will result in a reduction of the environmental impact in accordance to the biofiller content. Biofillers have a very low impact on fossil CO_2 production: for example, 1 Kg of wheat has just 0.177 Kg fossil CO_2 on Ecoinvent databases. Furthermore, the overall **density** of the biocomposite (trays and textiles) packaging will be **reduced thanks to the NF by an estimated 10%.** This will have an impact on the overall use phase of the products and also on its **EoL phase that will benefit from 10% CO_2 footprint** reduction. Savings in transport would be even more meaningful for reusable products like carrier bag with long use time.

Finally, in terms of use, **the barrier packaging ensuring a longer shelf life will have a hugely positive impact on the food chain.** Likewise, whereas packed food typically binds much more resources than the packaging itself, the reduction of products waste due to easy emptying effects provided will have a large effect on the LCA results expected in the project. All in all, in BIOnTOP, the **carbon footprint** of the considered bio-based products will be decreased along the new value chains as will be shown by LCA during the project in WP4.

Reduce the costs associated with the end-of-life phase of the developed packaging products by at least 30 % compared with the disposal costs for existing solutions for similar applications

In different OECD countries, packaging waste represents a significant, yet increasing, share of total household waste. This represents a significant economic burden too. The applicable EoL options and related fees vary per type of packaging material and mainly depends on how easy it is to separate it in the waste management stage. Gate fees for EoL management for similar treatment options can vary substantially at national level but also at regional and local level. Due to the specificities of plastic packaging, the costs linked to plastics EoL treatment tend to be significantly higher than the costs related to the treatment of other materials. Despite the recycling rate is increasing following a shift towards a CE, currently only 31% of plastic is recycled and in excess of 41% incinerated and 27% landfilled[1]. These latter two EoL options lead to significant resource depletion. Indeed, even the energy recovery achieved by incineration plants only has an average efficiency of 35%. Therefore, gate fees at material

116 http://ec.europa.eu/environment/waste/studies/pdf/plastics.pdf
117 Data source: Ecoinvent 3.4 APOS, specifically the following datasets (APOS, U (of project Ecoinvent 3 - allocation at point of substitution - unit): - 1 kg Biowaste {RoW}| treatment of biowaste, industrial composting - 1 kg Waste polyethylene {Europe without Switzerland}| treatment of waste polyethylene, municipal incineration - 1 kg Waste polypropylene {RoW}| treatment of waste polypropylene, municipal incineration)
118 Fondazione Istituto Italiano di Tecnologia, Bioplastics and Bioelastomers
119 www.mckinsey.com/business-functions/sustainability-and-resource-productivity/our-insights/the-new-plastics-economy-rethinking-the-future-of-plastics
120 www.lca.plasticseurope.org www.lca.plasticseurope.org

recovery facilities are substantially lower than those for incineration and landfilling. Usually, factors influencing gate fees are mainly related to the landfill tax (which enables competing treatments to raise their gate fees as the tax increases); the quality and quantity of material being presented; energy costs, etc. Currently, industrial composting is so rarely used for plastics packaging that it is assumed to remains at 0% in the baseline scenario[121].

Based on a EU study showing the example of gate fees in different countries, table 8 reports those for selected representative consortium countries[122]. Comparing composting to more common incineration and landfill used for e.g. multilayer packaging clearly shows that waste treatment average fees would reduce by above 30% in case of applying BIOntTOP new potential EoL.

Table 8: Comparative EoL costs in €/T (including treatment costs minus revenues from products/energy obtained)

Gate fees (€/T)	Composting	AD	Incineration	Landfill
Germany	62 (40,000 tpa), 56 (60,000 tpa)	109	250 (50 ktpa and below) 105 (200ktpa), 65 (600ktpa)	35-220
Italy	25-43 (20,000 tpa)	--	41-93	70-75
Netherlands	80 (10,000 tpa), 30-60 (50,000 tpa)	50-84	70-134	107-164

According to the EBB, more than 90% of materials currently disposed off in incinerators and landfills could be reused, recycled and composted. According to EPA, a well-run curbside recycling program can cost anywhere from €40 to more than €122 per ton whereas trash collection and disposal programs, cost from €57 to more than €160 per ton. Recyclers pay variable amount for post consumers and post industrial plastic wastes depending on their cleanliness and purity (e.g. from 0 to 70 € per ton). Then they have to separate, clean, grind and reprocess the compounds and subsequent materials (e.g. fibres for SILON or bags for BioMi). Vs. operative costs involved, a positive balance can be obtained only if such materials can be sold (processed) above 1.5€ per kg which is totally feasible for materials with the performances planned for BIOnTOP.

For organic waste (food or yarn), composting activity leads to revenues 2-30€ per T¡Error! Marcador no definido.. Furthermore 1 kg of PLA can lead to ca. 0.33 kg of compost[123]. Finally, while transporting materials to waste treatment and valorisation facilities has a significant cost that can range from €16 to €57 per T[124], promoting an EoL based on **home composting allows reducing waste management costs for municipalities**, as home composting can reduce a significant weight fraction of domestic waste (more than 20%). This is even more relevant in case of **absence of separate biowaste collection system** from households, or for countries with very **limited amount of AD or composting plants**.

All in all, the cost competitiveness of the proposed packaging and textiles solutions is to be considered in a (multiple use) cradle to cradle perspective linked to their mechanical recyclability, industrial compostability or biogasification, which will provide a new source of revenues rather than constitute an EoL burden as too often nowadays.

Other social and environmental impact

As mentioned, besides addressing unmet demands for specific EoL in controlled waste management context, BIOnTOP will also contribute to other environmental and socially important impacts especially in terms of pollution and consumer health.

Unquestionably, our **quality of life** is linked to the quality of our environment and measures are needed for lowering the environmental burden of our activities and consumption patterns. According to a report of the Ellen MacArthur foundation[7], only a very low amount of plastic materials is recycled (14%) while high amounts end up in landfills (40%). It is important to highlight the most alarming figure in this report: **the amount of plastic wastes leaking from our waste management streams shows as much as 32% of plastic ending up in our environment** (figure 15). This is especially critical for single use ultra lighweight bags (e.g. for F&V). If handled at the source by responsible consumers in controlled home composting process, there could also be less chances of leak in the environment along the many collection, transport, distribution steps of waste management.

Long term plastic persistence in the environment is aggravated by a **cumulative effect,** due to both accidental and intentional littering. The slow decomposition of plastic into small pieces leads to damages to the fauna via ingestion and entanglement[125],[126]. Recognized as one of the major international environmental challenges by the European Environment Agency (EEA)[127], **plastics account for 89-95% of marine litter** (roughly 8 million tons every year), in particular packaging (bags, bottles, lids, etc.) leading to irreversible consequences for marine ecosystems. Besides, **25 MT of plastic waste is accumulated annually in soil worldwide**, being detrimental to the environment (irreversible soil contamination, degradation of soil quality, pollution of rivers and groundwater) and human health (heavy metals, dioxins and furans released during plastic burning are associated with endocrine disruption, heart disease, cognitive and motor disabilities, and human carcinogen)[128]. At the EU level, the recent BOHEMIA study[129] recognises ocean and marine pollution as key priorities to be tackled by the EU post-H2020 R&I framework programme (FP9). Concerted actions are required to sustainably solve the huge plastic waste accumulation issue.

121 Mudgal et al. "Options to improve the biodegradability requirements in the packaging Directive " DG Environment-EC, February 2012, Biointelligence Service.
122 Costs for Municipal Waste Management in the EU , Final Report to Directorate General Environment, European Commission, Ecotec, 2017
123 Cosate de Andrade et al. LCA of PLA: Comparison Between Chemical Recycling, Mechanical Recycling and Composting, J Polym Environ (2016) 24:372–384
124 https://www.transparencymarketresearch.com/plastic-recycling-market.html
125 http://www.kimointernational.org/MarineLitter.aspx
126 http://www.onegreenplanet.org/environment/ways-plastic-pollution-impacts-animals-on-land/
127 State of Europe's seas, EEA Report No 2/2015, ISSN 1977-8449
128 D. Briassoulis et al, Review, mapping and analysis of the agricultural plastic waste generation and consolidation in Europe, Waste Management & Research, 2013, Vol 31, Issue 12, pp. 1262 – 1278
129 Beyond the Horizon: foresight in support of future EU Research and Innovation Policy- Doi: 10.2777/906131

Promoting the use of recycled plastics in value added application, BIOnTOP solutions will influence the increase of the quantity of plastic waste recycled mechanically. Furthermore, BIOnTOP biodegradability in mild conditions is a step towards the existence of plastic packaging that can biodegrade also when accidentally littered in the environment. As such, biodegradation in marine water and soil will be tested in WP4. Likewise, while supporting greater amount of packaging can actually be recycled in controlled waste management conditions, BIOnTOP will contribute to consumer access to healthier soils and possibly seas and therefore food that contains less harmful substances leaking from plastics or microplastic fragments. Health concerns are also relevant to food packaging and since in the past, synthetic substances widely used in plastics approved for food contact were finally restricted due to the leaching of harmful compounds, e.g. plasticizers like phthalates, BPA, or PVC /DEHA /aluminium used in food wraps, all leading to high health concerns, as discussed section 1.4. In contrastthe project will use harmless/innocuous substances which often derive from food (by-products like wheat fibres which are certified to be safely edible, protein, oil derivatives, fruits by products). Also, barrier packaging will provide consumers with healthy food along longer shelf life.

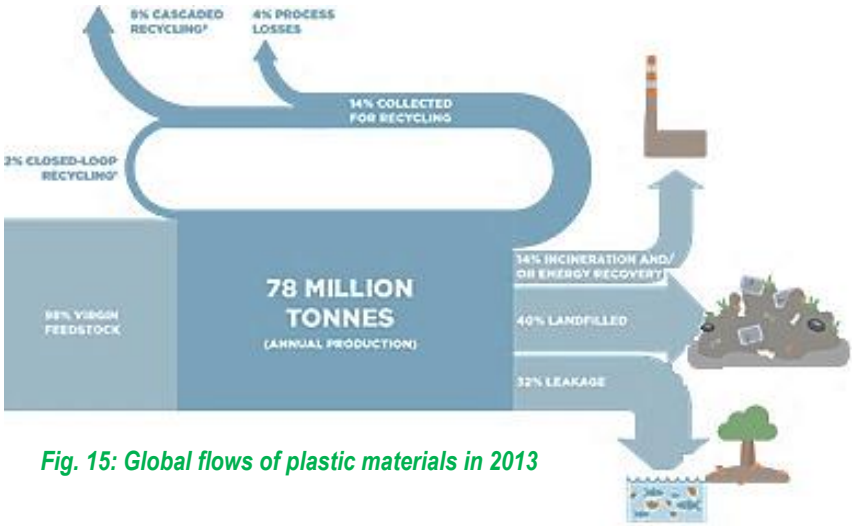

Fig. 15: Global flows of plastic materials in 2013

Furthermore, in terms of coated textiles, our new PLA and /or FA coatings will impart durable hydrophobicity enabling industry **to replace fluorocarbons which are currently used in hydrophobic coatings but are toxic, persistent and bioaccumulative.**

The demand for our consumer products, BIOnTOP packaging, is expected be driven by green- and health- conscious consumers concerned about the environment[130]. Furthermore, societal benefits will be assessed by the active involvement of MC, a consumer organisation which will give the alignment of BIOnTOP to the consumers' needs and perceptions. In particular, BIOnTOP will ensure a strong coverage at European level, on consumers attitudes on the issue and their requirements, mainly thanks to the huge network of MC, which will contribute to encourage the knowledge of the positive benefits of bio-based packaging as well as raising awareness about economic, societal and environmental challenges.

Contribution to improving the innovation capacity and the integration of new knowledge

While the generation of new knowledge is positive, it is in its industrial use that innovation capacity is increased, and impact is realised. As such, to allow generating truly market replicable innovation, the BIOnTOP consortium unites the industry value chain necessary for market success, and the project has integrated several transversal activities such as environmental, regulatory, safety and economic assessments, and finally business planning activities to maximise the post-project exploitation of the results. The project gives trans-sectorial industry and transdisciplinary research and technology players an opportunity to collaborate and share all required technical knowledge and market expertise. Effective knowledge transfer from the RTDs (AIMPLAS, ASU, INSTM, CTB) to the participating enterprises, especially SMEs often lacking own R&D capacities, laying down the basis for supporting the industry partners to carry forward the results post-project, is fundamental to increasing the likelihood of market uptake. To this end, a dedicated task, T7.5.4 has been included in the work plan to ensure seamless knowledge transfer occurs.

As discussed in the innovation potential section, unique and potentially patentable bio-packaging solutions relying on novel **PLA copolymers and coatings** formulations will be developed to achieve optimal functionalities and biodegradation in milder conditions than industrial composting. The predictive modeling of the biodegradation is a novelty. Mechanistic models, following the approach most researchers would attempt, where the physico-chemical-biological laws behind the biodegradation behavior would have to be modelled, are too complex in view of the number of stochastic and deterministic variables. As such, the route we propose is a data driven modeling. This knowledge will not only provide new insight for the scientific community who lacks systematic studies on biodegradation in mild conditions, as well as also help in the shaping of new related standards, but also be used with 2 clear industrial targets: i) support the eco-design of the products meeting target EoL scenario (as will be preliminarily tested for WP5-6 demonstrators) and ii) support organic recyclers, together with the composition monitoring system, to ensure they sort suitably the new post consumer waste e.g. checking if they meet the conditions for industrial compostability.

Addressing our societal and environmental challenges, e.g. the depletion of non-renewable resources and sub-optimal EoL compatibility of current materials with existing waste management scenarios, is recognized to be a major catalyst for increasing our innovation capacity, and for the integration of new knowledge generated as a result of research into our practical societal and industrial contexts. The global paradigm shift from oil-based towards bio-based materials represents a unique opportunity for Europe to kick-start a world leading competitive bioeconomy with significant economic, environmental and societal benefits. The CE also unlocks tremendous potential for the development of innovative new materials and products derived from bio-based

130 About 80% of European consumers want to buy products with a minimal impact on the environment (Eurobarometer survey, EC, 2013).

sources, including from agri-food waste. In such context, the project will provide new knowledge on potential applications for fruit waste derived monomers, wheat fillers and cheese by-product valorisation.

Today, the European bioeconomy sectors are worth 2 trillion euros in annual turnover and account for 22 million jobs in the EU, which is 9% of the EU's workforce[131]. Bioplastics have been designated a lead market by the EC. Bioplastics are experiencing a huge growth, but the vast majority of currently produced bioplastics come from Asia (75%) or Latin America (12%) and Europe only represents a 7% share (sources EUBP). Furthermore, bio-sourcing and biodegradability have recently emerged as major market drivers especially in the packaging sector. In tune with the current bioplastics growth rate of about 20% per year, our new solutions will meet the growing number of brand owners that are seeking solutions with sustainable sourcing and EoL to satisfy market demand, as well as to gain product differentiation.
To increase the innovation capacity of the industry, our approach is to capitalise on a valuable opportunity for the EU bioplastic, packaging and the end-user industries to substitute currently used synthetic counterparts with bio-based compounds and coatings. BIOnTOP represents a **very timely development of new recyclable, cost-efficient and high-performance bio-based materials with controlled biodegradation,** into growing 'green' and 'bio-based' industries.

This will result in **great business opportunities reinforced by current** bioplastics **market trends,** the growth in our target end use sectors, **increased** visibility, and policy awareness and support for such solutions**.** Indeed, we tap into the massive **EU plastic business** (€350 billion **turnover** per annum). Plastics demand in Europe is ca. 50 million tonnes, almost 40%, or **ca. 20 million tonnes, of this is used for packaging**, of which 17.3% is for food trays and films[1]. Worldwide, over 90 million tons of textile are produced, ca. one quarter goes to technical textiles[132]. Within the technical textiles where EU has a leading position[133], so-called packtech, represents ca 8%, which means that **worldwide around 2 million tonnes of textiles are used for packaging** (assumed at 30% in EU). BIOnTOP is estimated to reach **a market penetration of 0.5% of these plastic segments** (totalling a potential of 4 million T per year) **by 2030 in EU**, generating close to **€40 million turnover and ~170 new jobs**[134] **at the biopolymer production upstream** level of the value chain. **Such figures would be reinforced at the downstream (end users) level** who benefit from new unique products**.** Besides direct partners' exploitation, these figures, equivalent for current limited European PLA production to increase by a factor over 2 by 2030, correspond to an overall deployment through further licensing, business alliances with additional converters. Estimates about how partners respectively expect to grow their business and create jobs with the new BIOnTOP solutions are given in section 2.2. Should Europe get a strong foothold in the market here considered by being first to market with a strong brand that is scientifically backed (as a results of this RIA), and derived packaging producers gain an invaluable head-start in the global competition.

Barriers/obstacles/framework conditions that may determine if and to what extent expected impacts will be achieved

In a green economy, growth in income and employment are driven by public and private investments to enhance resource efficiency, reduce carbon emissions and pollution, and prevent the loss of biodiversity and ecosystem services. These investments need to be catalysed and ideally supported by targeted public expenditure, policy reforms, tax breaks and incentives as well as regulation changes. Exploitation activities in the project will include addressing standard-making bodies at European and international levels related with the project technologies and fields of application (in subtask T7.5.2). On-going standard developments[135] especially in the areas of bio-based content determination and biodegradation in specific environments will be monitored and contemplated under the lead of EUBP (participating in CEN TC 249 - WG 17- and TC 411) and OWS(terminated)/OWS RF (participating is tests definition for such standards).
Regarding REACH regulation (EC No 1907/2006), whereas all polymers are exempt from registration and evaluation any considered monomers (already market available) and other additives used in the project will be checked in terms of their REACH status (WP7) and compliance will be ensured.
Plastic materials for food contact, e.g. packaging trays and films (lids, pouches) and also nets, fall within the scope of the Framework Regulation (EC) No 1935/2004, which provides the harmonized legal framework by setting out general principles and safety requirements for all kinds of food contact materials (FCM). More specifically, food contact materials and articles made of *plastics* are covered by the Regulation (EU) No 10/2011 (last amendment by Regulation (EU) No 2017/752). It sets out rules on the composition of the plastic FCMs by establishing a positive list of substances approved for use in the manufacture of plastic FCMs ('Union list') and lays down migration limits (a generic overall migration limit as well as specific migration limits for individual substances). Compliance with regulatory demands is essential for putting new packaging materials on the market. Therefore, we have decreased the possible impact of the legal requirements on FCMs by selecting materials and starting substances that have been previously approved for such purpose (e.g. LA: FCM No 99; SA: FCM No 247; MA: FCM No 499, FA: FCM No 12). As food ingredient, whey protein is recognized safe for food contact according to scientific principles and/or can also be used provided that migration is below ppb levels as used behind a functional barrier that is approved[136].
In addition, BIOnTOP will be a key enabler for implementing the Circular Economy Package (CEP). Specifically, BIOnTOP contribute to delivering reusable packaging (bags, wraps), to minimize packaging needs (e.g. tea bags with aroma barrier avoiding

131 European Bioplastics 2012; EC-Communication "Innovating for Sustainable Growth: A Bioeconomy for Europe", 13/02/2012
132 ECP4 Strategic Research Agenda 2016
133 https://ec.europa.eu/growth/sectors/fashion/textiles-clothing/eu_en
134 Based on a new job being created for every €230k of growth, taking into account an estimated sales price of ca. 2€/kg
135 http://ec.europa.eu/growth/single-market/european-standards/policy/benefits/
136 http://www.wheylayer.eu/files/2012/08/Safety-compliance-letter.pdf

the need of secondary sachets, as well as of industrially and organically recyclable solutions. This will contribute to the implementation **directive on the packaging and packaging waste,** which sets new legally binding targets and fixed deadlines for waste recycling and the reduction of landfilling. Indeed, the package includes a common EU target for recycling 65% of packaging waste by 2025, and 70% by 2030; in this context, the plastic sector has a separate target 50% by 2025 and 55% by 2030. This represents a significant increase from the current target imposed by the Packaging Directive 94/62/EC, which set a 22.5% recycling target. Awareness about the right sorting of the new materials and EoL treatment will be required to foster the best practices among waste management bodies, recyclers, and composters to make it happen also in practice. This will be greatly facilitated by educative workshops which will be continued in post-project efforts to create adequate policies, demonstrate economic viability, and encourage intra and inter value chain cooperations.

Possible barriers may finally arise from the need for specific knowledge of how to reach the desired and required biodegradation profile depending on the formulation to enable its correct selection. We have therefore included in the workplan extensive training and knowledge sharing actions, and the easy to use biodegradation modeling tool will support new users of the materials. The required investments to produce the new copolymers will also be considered as a potential barrier and therefore we will favour materials that can be processed with existing infrastructures, which will be adopted by the potential exploiters of the oligomers and copolymers by batch or REX processing, i.e. respectively TOTAL-CORBION and PLANET.

Additional investments

In terms of further contributions to maximizing the impact of this BBI action, significant **in-kind contributions** are planned to reflect the industrial context of the action and will definitely contribute to increasing the impact and reaching the project objectives. T**he consortium covers 31% (1.3k€) of the overall project budget**. This co-funding, comprising the full budget of the large companies as well as a 3% flat rate for all others partners, is justified by the genuine conviction of the partners that they will on the long term obtain a return on their investment when the BIOnTOP solutions reach the market. Our success securing in the consortium world leaders in the bioplastics, packaging, textile, food and personal care fields (e.g. TOTAL-CORBION, EMSUR/INNOTECH, SILON SIOEN, ENTREPINARES, UBESOL) who will fully cover their own costs by themselves, and members of the AB who will make also significant in-kind contribution (in providing samples, assessment, advice, etc.) shows that we have a winning concept. In addition, some partners also provide **additional investments towards the BBI-JU's objectives** (table 9).

Table 9: Summary and cost of the additional investments (note: no financial transfer between partners are made)

Entity	Activity	Contribution to the BBI JU objectives	Estimated private investment (M€)	Estimated value public investment (M€)	Other EU funding received
1.AIMPLAS	Improve control of recycling steps, adaptation of facilities for bioplastics. Synthesis of biopolymers (2019-2021)	KPI1 (polymer processing), KPI2 (polymer processing), KPI6 (RTD for food packaging processing)	0.55M€	Investments are supported by regional authorities via own resources and European regional funds (50%).	
3 CTB	Extension of lab facilities and pilot lines to help the reconversion towards a circular bio-based economy (2018-2020)	The equipment being installed will allow to perform the filament extrusion foreseen in BIOnTOP.	1.5 M €	1.0M€ Belgian Province W-Flanders invests in the building & infrastructure	Investments supported via EU regional funds (50%)
4.ASU	Innovation Campus – Model Factory and Laboratory for sustainable food production and food packaging	KPI1 (polymer processing), KPI2 (polymer processing), KPI6 (RTD for food packaging processing)	2 M€	1.2M€	0.8M€
8. BIOMI	Investment in pilot and demo scale lines for obtaining new materials and products which are bio-based. Co-funding in another project activities	Creation and development of new bio-materials and sustainable packaging products	0.5 M€	0.7 M €	No
9 / 22 EMSUR / INNOTECH	Investment in extrusion line to work with biopolymers for films and trays, coatings and pouches	Moving forward to obtain sustainable packaging products	€2.5 M€	0	No
16.OWS (terminated) 23. OWS RF	Expansion of capacity for disintegration and biodegradation testing.	Disintegration and biodegradation testing in biopolymers	0.3 M€	-	-
17 ROMEI	Investment between 2019-2022 in: pet/pla recycling capacity upgrade, industry 4.0 feeding plants for 3 extruders, start-up new logistic warehouse, FV production upgrade.	New recycling and compounding capacity for PLA, actually processed with pet plants	2.2 M €	-	-

In short, the foundation for achieving the expected impact of the project has been laid: an innovative research agenda, a resourceful action plan for the contextualisation and exploitation of the expected results, and an excellent conglomerate of actors in the consortium who are in pursuit of a shared common goal.

2.2 Measures to maximise impacts

Our vision is to lead the transition towards a sustainable **post-petroleum society** in the plastic sector while decoupling economic growth from resource depletion and environmental impact by serving industry with the provision of **cost-effective high**

performance sustainable multifunctional bioplastic packaging solutions with tailored materials and organic recyclability and biogas potential at EoL. In order to assist Europe in the transition towards a bio-economy new sustainable cross-sectorial value chains will be established providing competitive bio-based alternatives in lightweight packaging, such as nets and trays and textile applications. In line with a number of EU and national regulations the use of non-biodegradable shopping bags and lightweight multipurpose bags as well as those banning potentially polluting single use items, e.g. disposable tableware (e.g. in France), but also other enforcement measures for the broader EU SET plan[137], combined increasing incentives for significant bio-sourcing (e.g. in Germany), there is a clear need for new materials with high bio-based content and for meeting the demand for plastics with innovative EoL options. In order to realise this vision, post-project exploitation pathways have been carefully planned already during BIOnTOP proposal preparation. We have devised our **initial strategy for the dissemination and the exploitation of the project results**, with a view to increasing the projects' innovation & business outcome as well as social & environmental impacts.

Outline of the dissemination plan

The dissemination plan for BIOnTOP comprises a set of actions to **spread the non-confidential results generated during the project** to ensure that as many relevant audiences as possible have access to them**.** The aim underpinning the dissemination strategy is to heighten awareness about the tangible and intangible outputs of BIOnTOP, to contribute to **knowledge building** and **advancement of innovation** in Europe as well as to avoid duplication of research work. In order to safeguard the commercial opportunity, the dissemination activities will be subject to a confidentiality policy. By default, and building on the consortium experience in collaborative projects, any result of the project is to be treated as confidential and can be disseminated only after agreement of the project partners following the relevant clauses of the consortium agreement (signed during the grant agreement preparation phase in keeping with initial agreements on ownership reflected herein and respecting the general terms of the grant agreement). This approach allows us to retain complete control and make sure that no "exploitable" information is disclosed before being protected. The protection of the IPR is of invaluable importance for the participating partners. Therefore, the non-confidential results and information on the project to be publicly disseminated will be carefully identified. The RTD participants in the consortium will publish results in peer-reviewed open access journals, which, will be an indicator of overall project success (in terms of progress vs. the scientific state of the art) and an important element of dissemination towards industry, standardisation bodies, and policy makers. The industrial partners and market actors will focus on showcasing and disseminating results at conferences and trade fairs and through other industrial channels such as technical/sectorial magazines, industry network meetings, and the like (to demonstrate the industrial novelty and commercial potential of our project outcomes).

The IPR Manager (Dr. Marco de la Feld from ENCO) will work closely with the project innovators to manage the identification of non-confidential disclosable information that can be published/disseminated. He will lead the project partners in strategically planning and effectively implementing the dissemination during a dedicated task in the Work Plan, *Task 7.5 Exploitation and Dissemination Planning*. All the measures (outlined in table 10) will be tailored taking into account the project needs, the target groups' background and the expertise of the project partners. Our project dissemination activities will intersect dissemination strategies of the BBI-JU, via full members BioMi and PLANET (in process) and associated BIC members AIMPLAS, INSTM, CTB and EUBP, as well as other EU PPP as SPIRE through IRIS and ENCO, which will contribute to the distribution of project outputs across the relevant European and international communities and amplify the project dissemination effort. All the dissemination activities will be developed with an eye towards the exploitation of the project results by other parties, in order to lay down the basis for subsequent innovation actions as well as other private initiatives which can accelerate the route towards the take-up of new technologies within the bio-based plastic sector, also beyond the BIOnTOP selected applications.

Beyond above discussed use of traditional dissemination channels (presentations & posters at conferences, publications, industry magazines, newsletters and on-line platforms), we will also tap into more vanguard channels, such as social media platforms (ResearchGate, Academia.edu, SlideShare, Twitter, YouTube, Vision2020 open innovation platform), etc. Also, the project website will serve as a lasting virtual hub for disseminating downloadable public deliverables and outputs of BIOnTOP, as it will house also the 'BIOnTOP Green Channel' for open access to the peer-reviewed publications of the project.

In particular, the network of EUBP across the European bioplastics value chain should be highly instrumental in identifying and approaching prospective business partners and potential licensees. With around 70 member companies from across the bioplastics value chain that are active in the European market, EUBP works closely together with all major stakeholders in this industry. The association also has access to several unparalleled channels of communication within the bioplastics industry, ranging from the largest European bioplastics conference, to the bioplastics-focussed newsletter with the widest circulation, to one of the most widely-followed (4000+ followers) Twitter accounts focussing on bioplastics topics. To amplify our impacts, we also capitalise on ISWA's (in our AB) communication channels to reach the waste management sector.

Table 10: Outline of the BIOnTOP Dissemination Plan

Target Audience	Tools & Channels	How contributing in achieving Impacts
Consumers, Farmers, Retailers, General Public	Project Leaflet, Video Clips, Website, Blogs and Magazines e.g. Food Production Daily, Bio-Based World News, PlastEurope, Farmers Weekly, ESOF (EuroScience Open Forum)	Market pull. Public acceptance of the proposed innovative bio-based plastic applications that impact daily life stimulating the feeling they are getting value for money from public funded research.

137 http://ec.europa.eu/energy/en/topics/technology-and-innovation/strategic-energy-technology-plan

Monomers producers, Copolymers producers, Compounders, Agro feedstock suppliers	***Trade fairs/exhibitions:*** EFIB, Compounding World Expo ***Open Access Journals (OAJ):*** *Polymers*	Widespread of the use of the project innovation. Multiply success of the project by inspiring new initiatives based on the project results.
Packaging converters Textiles producers Food processors Organic Food producers Cosmetics & personal care companies Other packaging users	***Trade exhibitions and conferences***: Interpack (GE). International Conference and Exhibition on Food Processing & Technology (UK); Alimentaria (ES); Cosmetics Business (DE), Anuga FoodTec&Anuga (DE); Bioplastics Business Breakfast (GE), ICTF: Annual International Conference on Textiles and Fashion; Aachen-Dresden International Textile Conference; European Textile Platform conference, Textile ETP, Dornbirn Conference, Techtextil ***Magazines***: Fibers and Filamant, Plastics News Europe Project's training events.	Foster new supply chains and new cross-sectorial value chains (which will be reinforced by the presence of our expert Association EUBP) Raise awareness about BIOnTOP materials and suitable end of life options tailored on a vase by case basis. Multi-actor approach to support the transition towards a post-petroleum economy
Industry Multipliers- Sectorial Associations (PlasticsEurope, CEFIC, Plastics Europe, PAFA - Packing and Film Association, Copa Cocega, European Crop Protection Association, European Association of Agricultural Economists) and other associations.	***Conferences, seminars and workshops*** (participation in organising, scientific committees, with poster, papers, presentations, or chairing): Annual World Food and Innovation Forum; Bio-based World, Biopolpack, Biopol, Sustainable Bioplastics, Bioplastics, Biorefining, Bio-based Materials, etc. ***Features in Newsletters and e-bulletins*** *(plastics today, food production daily, plastic news, agronotizie,* CosmeticsMag, *etc.)*	Dissemination of new knowledge and technologies for transitioning to a bio-based economy. Communication to retailers and consumers about the bio-based materials used in BIOnTOP. Increase target industry sustainability for consumer and environmental protection. Building new value chains and cross sectorial interconnections among clusters.
Standardisation bodies (CEN, ISO, DIN CERTCO, AfOR, COBRO, Vinçotte), and policy stakeholders, national food safety authorities, EFSA, EC, Expert Group for Bio-based Products (E02886), National governmental bodies, FAO, EUROPEN, INCPEN.	***Policy dialogue***: joining roundtables fostering multilevel and horizontal discussion on BIOnTOP results. Collaboration with public administration will drive better informed decision-making processes across Europe also with a view in improving waste management legislation and standards.	To increase the influence of the project and its results in policymaking towards the creation of a positive framework for the take up of BIOnTOP (and in general bio-based plastics with tailored EoL).
Public/Private waste management, Recyclers (industrial organic and mechanical recycling)	***Conferences:*** ISWA World Congress, Waste Management, Ecomondo, Waste & Recycle Conference, Plastics Recycling Conference ***OAJ Articles:*** Waste Management & Research (ISWA-run Academic Journal) ***Magazines***: Waste Management World (ISWA Magazine)	To support the transition towards a post-petroleum economy improving circularity Market & public organisation push
Research and academic stakeholders	***Magazines***: Pack World, Bio-based News, Plastics films, plasticseurope.it plasticsnewseurope.com, Bioplastics Magazine, Sustainable Plastics, Labiotech. Plasticeurope, Plastiker, Bio-based World News, Farmers Journal (IE) ***OA Journal Articles***, Advances in Materials Science and Engineering ***Specific events*** as Open day at Universities, Researchers' Nights, Knowledge Open Festivals. Project training events.	Effective knowledge sharing among the research and scientific communities in order to accelerate the paradigm shift from oil-based towards bioeconomy.
Early stage researchers		Attract more young people to scientific and research careers, enriching their curricula with knowledge from the project. Early stage researchers inspired by the results, will be motivated to remain in EU- research and possibly work in areas that are beneficial for society, BBI, sustainability and innovation.
Other H2020 BBI projects and international/ national projects	REFUCOAT, SuperBIO, Open-Bio, BioSTEP, InnoNet, BIOCOMPLACK, CommeBeBiz, AGRIMAX, PolyBioSkin, Hyperbiocoat, MultiCycle	Cooperation, knowledge interchange and reuse between previous and future projects.

Initial Exploitation and Business Plans for BIOnTOP Key Exploitable Results (KERs)

The BIOnTOP consortium mobilises the necessary partners not only for the pilot scale production of novel bio-packaging based on bio-coatings and copolymers with multiple tailored EoL (WP2-3) but also to validate the new bioplastics applications (WP5-6) including environmental and economic sustainability, testing and modelling biodegradation in specific conditions (WP4), a set of activities which are all highly relevant for future market exploitation (WP7).

A market-oriented approach will drive BIOnTOP R&D activities with Task 1.3 capitalising on the participating lightweight packaging, such as trays and nets as well as textiles, end users' and the AB members' expertise, and Task 7.4 ensuring that sufficient knowledge is collated to steer the research towards delivering a consumer-oriented solution with good market potential. Our intention is to realize all exploitation pathways for the results:

a) **Commercial exploitation:** a key exploitation route will be the industrial use of the project knowledge for the **creation of new innovative bio-based and biodegradable packaging materials, within and beyond the validated applications, new processes and advisory services to deliver high competitiveness and growth impacts**.
b) **Education and training:** the use of the new knowledge generated for upskilling and for enriching curricula in European education will have a high impact for raising the European knowledge base to prepare lightweight textiles and films-based packaging, and end user industry sectors for the transition to the bio-based economy; 3 training events will be organised.
c) **Further research:** planning future spin-off research and development projects especially in areas that will contribute to the transferability of the project results to new applications;
d) **Contribution to standards:** while making available new bio-based materials with innovative EoL options, the project results could on the long term contribute towards new regulations in the field of disposable plastic waste favouring the choice of materials that are home compostable, wherever a better option like industrial composting or material recycling is not available or viable due to food scraps e.g. from F&V and tea.

Detailed business plans will be developed in *Subtask 7.5 Business and exploitation planning,* and annexed to the Plan for the

Exploitation and Dissemination of results (PEDR), in order to provide the details on how the project's results will be used in commercial exploitation activities, for training purposes, further research as well as standardisation activities. Further measures will be planned in the framework of this task to increase the likelihood of continuity of the established ecosystem and value chains, as well as market uptake of BIOnTOP KERs for European industrial sustainability, competitiveness, growth, and jobs. During the proposal stage, each partner has communicated clear exploitation expectations (Table 11), as well as expected growth derived from the project in line with the share of the respective previously discussed overall growth of the sector he represents and that he could tap into. This will be planned in greater detail during WP7 to maximize the innovation impact but while growing large companies by few percent has already a significant impact, it can be a game changer for very small companies like the young start ups PLANET, BioMi or WEAREBIO.

Table 11: Summary of exploitation intentions of the BIOnTOP partners and expected growth year 7 post project (% vs. current)

no	Short Name	Intention for use of project results	Turnover increase	New Jobs
1	AIMPLAS	Further developments to design and produce at lab/pilot plant scale of specific copolymers with tailor made properties using a broad range of different bio-based monomers. Possible patents on these formulations will be developed for food packaging and personal care products among others. Use of the knowledge generated in further projects related to development of full biodegradable plastic packages for a wide range of food and non-food products. Further knowledge about the most suitable copolymers formulations with tailored EoLs will be applied to future projects as well as their behavior once recycled for different applications with demanding requirements.	NA, non profit	
2	INSTM	Further knowledge in processing of bio-based polymers and co-polymers for packaging and personal care applications used in services and projects. Possible patent (to license) on formulation for production of PLA based materials based on PLA and bio-based reactive plasticizers, or additives to promote biodegradability.	NA, non profit	
3	CTB	Offering consultancy, pilot facilities and testing services in the domain of biopolymers for textile applications. Extending future portfolio in research and development projects for biopolymers in textile applications.	NA, non profit	
4	ASU	Further knowledge in FA grafting of proteins for packaging applications used in services and projects. Possible patent licensing on grafted protein materials.	NA, non profit	
5	EUBP	Promoting the increasing use of innovative bio-based and biodegradable materials and supporting research into the further development of bioplastics with the goal of creating performance-improved & more sustainable applications.	NA, non profit	
6	TOTAL-CORBION	sales of LD and PLA prepolymers for REX, of copolymers by batch process, as well as of PLA for the coatings. Grown presence in EU.	Not possible to estimate	
7	PLANET	Industrial production and commercialization of masterbatch based on BIOnTOP formulations.	50%	3
8	BIO-MI	Industrial production of materials, films and final packaging applications.	15%	5
9	EMSUR	Commercialisation of home compostable trays and lids with differential end of life trough associated companies.	2%	15
10	CMSA	Commercialisation of home compostable nets with differential end of life.	10%	5
11	SILON	Extending market of bio-based compounds (possibly from SRM) and staple fibres with tuned biodegradability.	3%	2
12	SIOEN	Extending market of bio-based textile with tuned biodegradable products.	10%	15
13	WEAREBIO	Commercialisation of F&V packed in sustainable materials. Extension to further organic food products.	15%	2
14	ENTREPI.	Commercialisation of cheese packed in sustainable materials. Possible future supply of whey for barrier coatings.	3%	2
15	UBESOL	Commercialisation of personal care products packed in sustainable materials. Extension to cosmetic products.	4%	1
16 23	OWS (terminated) OWS RF	Position as a leader in environmental assessment services for bioplastic sectors, especially through proposed improved test standards.	5%	7
17	ROMEI	Commercialisation of BIOnTOP SRM and BIOnTOP based compounds.	10%	3
18	IRIS	Further development (customization service) of the biodegradation predictive models and licensing of the derived app. Commercialisation of full monitoring systems or the specific chemiometrics package to be installed in existing recycling plant to enable the sorting of the new bioplastic. Use of the knowledge generated in further projects related with bioplastics.	10%	3
19	ARCHA	Exploitation of new coatings as water repellent finishing agent for clothing textiles. Strengthen its position as analytical and consulting service provider across EU.	5%	3
20	ENCO	Use of the knowledge in bio-based plastics market in further related projects and in consultancy services.	10%	1
21	MC	Serve the interest of consumers for better products	NA, non profit	
22	INNOTECH	Commercialisation of home compostable trays and lids with differential end of life trough associated companies.	2%	15

The following table gives a summary description of a selection of the expected BIOnTOP KERs:

Table 12: BIOnTOP Key Exploitable Results (Knowledge, products or technologies)

WP	KER	Owner	Envisaged IPR Tool	Type of exploitation	Target end-users	Timetable
2	Copolymers synthesis route	AIMPLAS	Patents	Licensing to REVERDIA and others	Polymer producers	2/3 years from the end of the project
	Compounds formulations	INSTM	Patent or trade secret	Licensing to PLANET	Polymer converters	Directly post project for PLA blends and 3 years for copolymers
3	FA grafting	ASU	Patent or trade secret	Licensing to ARCHA to supply INNOTECH associated companies	Polymer converters	Directly post project for in terms of licensing, 2-3 years for upscaling

	Process to allow FA combination with whey coating	ASU, IRIS	Whey coating processing patented, so protection already granted, but patenting combination is considered	Licensing to INNOTECH associated companies	Polymer converters	Directly post project for in terms of licensing, 2-3 years for upscaling
	Fibres coating	CTB, ASU	Patent or trade secret	Licensing to ARCHA, SILON and SIOEN	Textiles converters	Directly post project for in terms of licensing, 2-3 years for upscaling
4	Predictive models and derived app for biodegradation kinetics of BIOnTOP	IRIS	Confidential algorithms that will be used in software development secured by code	Licensing	Polymer producers and converters as well as waste sorters	2/3 years post project for integration of the predictive model in a software product
	Biodegradation testing and LCA	OWS(terminated)/OWS RF	NA, specific standard test	Services	Polymer producers and converters	Directly post project
	Monitoring system for BIOnTOP materials	IRIS	Confidential know-how for hardware calibration + design, software secured by a code	Sales of turn key solutions	Recyclers	5 years after the project when EoL BIOnTOP amounts are sufficient to recycle
5	Coated laminates	INNOTECH	Confidential know-how	Sales of products - associated companies	Packaging converter and food packers	Approx. 4 years post project
	Secondary packaging	BioMi	Confidential know-how	Sales of products	Packaging and converting sector	1-3 years post project
	Films & trays packaging	INNOTECH	Confidential know-how	Sales of products - associated companies	Food & personal care products packers	Approx. 4 years post project
6	Processability of novel biopolymer formulations into textiles	CTB	Confidential know-how on processing parameters/ properties relationship	Consultancy	Textile producers	Services and support offered after project finalisation
	Process to obtain nets	CMSA	Confidential know-how	Sales of products	Food packers	Approx. 4 years post project
	Specific biodegradable formulations for textiles	Sioen	Confidential know-how / Patents for specific end products	Own production / licensing	Packaging end-users of specific textiles (food, retail)	Product development within 1 to 3 year timeframe.

A Business Model Canvas has been prepared for the BIOnTOP materials, featuring core KERs along the value chain (Fig. 16). Similar tools will be completed for each KER in WP7. This will be instrumental to develop and pivot the related business models during the project. The final business plans (D7.5) will provide the details on how the project's results will be used in commercial exploitation activities, for training purposes, for further research activities as well as for standardisation activities. It will also specify in detail for each result both joint and individual exploitation strategies in terms of technology roadmaps, finance requirements, commercial opportunities and related business roadmap, taking into account IPR.

Key Partners
- Bio-based monomers producers
- Biopolymers producers (esp. PLA)
- Wheat, citric and wax converters (for by products)
- Packaging convertors
- Waste management bodies (composters and recyclers)
- Distributors
- Licensees

Key Activities
- Application consulting
- Integration assessment/know how transfer
- Manufacturing of the different products along the value chain
- Sales and distribution

Key Resources
- Financial capacity to cover fixed costs until breakeven (possible further public funding, private investments to be studied in WP7)
- Technical &Sales force human capital
- Operational managing structure for business development
- Production infrastructures among partners

Value Proposition
Formulating novel coatings as well as **PLA copolymers and compounds,** Biontop will deliver recyclable-by-design **cost competitive** packaging solutions that can be mechanically **recycled** or industrially/ **domestically** composted or even suitable for AD. Each solution will have a tailored EoL and performance validated for trays, films, nets and textiles packaging for different products. We also provide eco-design biodegradation predictive models, the know how needed for converting, using and recycling them.

Customer Relationships
- Samples based trials for client engagement
- Free initial application assessment for client capture
- Direct marketing and sales of the different products by the different partners

Promotion Channels
- Specialised tradeshows
- Specialised events
- Professional press
- DEM (Direct Email Marketing)
- SEM (Search Engine Marketing)
- Lobbying on influencers
- Direct targeting to current costumers and new prospects

Customer Segments
- Plastic Converters (packaging, textiles as early adopters of SRM; in a second phase, agricultural, tableware but also medical and sanitary sectors)
- Packaging end users, especially for food in the organic sector, but also cosmetics and pharmaceutical sector
- Consumers of packed products.

Cost Structure
- Cost of raw material + production of the different final products
- Sales structure costs (technical assessment)
- Human capital costs

Revenue Streams
- Sales of PLA copolymers and compounds, novel coatings, packaging solutions (trays, pouches, nets and textiles)
- Sales of services/equipments to support Biontop implementation including the monitoring systems for the EoL for the new materials and biodegration modelling derived services
- Royalties on licensing

Fig. 16: Business model canvas to be applied to each KER in WP7

Fig 17 represents the expected full scale coordinated deployment of the different semi-finished materials and finished products along the supply and value chain to reach our target markets within 3 to 5 years post-project, as well as the adoption to new applications in the year after that. In terms of the **measures that will be in place in order to effectively plan for the exploitation**

of the results of the project, the robust industry involvement in the consortium (SMEs and Large companies will unique market expertise in bioplastics & packaging), backed by strong knowledge management, knowledge transfer and market exploitation plans, and their key role in the project activities will boost an industry-driven approach to the project respecting market, consumer and regulatory requirements. This will ensure that BIOnTOP is successful in transitioning knowledge towards the market overcoming potential barriers that may hinder the swift uptake of the results of BIOnTOP research in the post-project phase.

Effective knowledge transfer, under *Subtask 7.5.4.,* from technical partners to SMEs (internally),), third-party enterprises and end-users (externally) will also be fundamental for paving the way towards commercialization, guaranteeing that sufficient knowledge and competencies to exploit results are acquired. To this end, the involvement of RTDs of international renown in the fields of plastic formulations (INSTM), plastic processing & synthesis (AIMPLAS), packaging technologies (ASU), PLA compounding and textile extrusion & processing (CTB), and who have a track record in working closely with and in transferring research results to industry, as well as providing training and educational programmes, is a valuable asset, which will ensure that we bridge the innovation gap that frequently prevents industry adoption of R&D outputs. In particular, AIMPLAS performs very frequent broad scope trainings for its over 600 associated companies in the plastic sector. The non-industrial partners will exploit the results of BIOnTOP to enhance their educational offer with the novel concepts and system design principles that will be identified and developed during the project. Some of the topics tackled in the project shall become part of Master degree courses, as well as research topics for Master & PhD theses, offering students a taste of research problems that have practical impact and interest, and the opportunity to perform research in a high quality international environment. The inclusion of ENCO with extensive experience in translating R&D output into actual commercialisation and market penetration, is another important added value for the consortium and will contribute to ensuring the post-project stage. Importantly, as opposed to greenwashing claims sometimes found for materials having minor bio-based content or low overall sustainability, we will work towards very high bio-based content of our formulations (close to 100%) and we will follow a cautious and transparent communication policy both in terms of sourcing and EoL, backed up by consequential LCA results, which will contribute to raising industry confidence delivering a positive **breakthrough** in the context of **environmental-friendly food packaging**, as well as carrier bags.

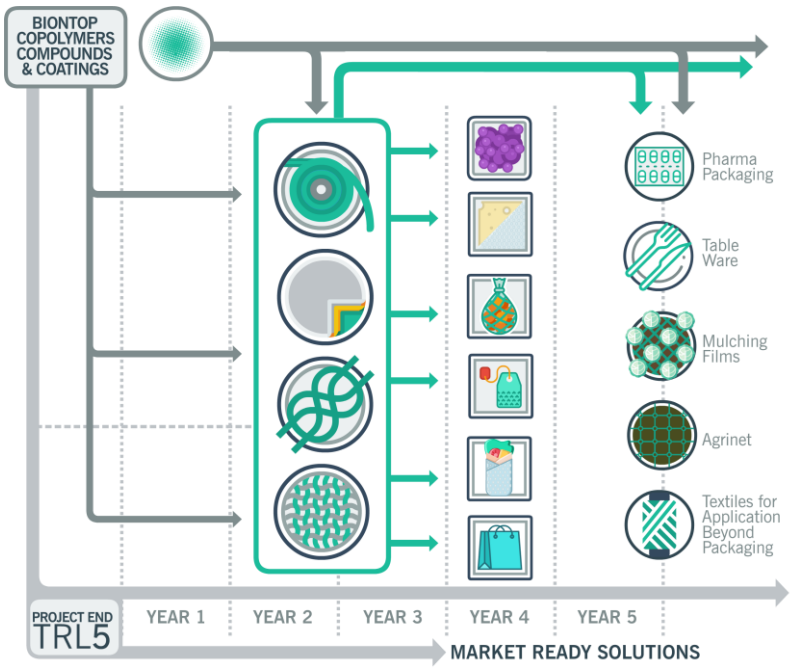

Fig. 17: Market deployment plan of main final BIOnTOP KERs

Sound management of the knowledge generated during the project, active planning in relation to post-project exploitation activities, further research work for escalating along the TRLs, as well as scouting for possible viable financing routes to commercialisation that will allow us to quickly scale up, bridge the innovation gap, navigate the 'valley of death', and swiftly overcome the critical stage between pre-competitive results and the market according to previously outlined market deployment timeline (fig 17).

How the participants will manage the research data generated and/or collected during the project Although the exploitation interests will be favoured by default, BIOnTOP will participate in the Pilot Open Access Research Data scheme for a selected amount of research data generated by the project. A data management plan (DMP) has been included among WP8's deliverables. Non-confidential results will be widely disseminated in open access media and repository (after anonymising critical information e.g. on composition). All publications and selected results will be considered for this so as to provide the scientific community with a better understanding of e.g. the influence of the structure of the copolymers on biodegradation after an agreed embargo time allowing excluding the commercially sensitive details in case of selected successful formulations. All publicly available data will be protected by copyright statements.

Strategy for knowledge management and protection

The Knowledge Management Strategy (Fig. 18) which will be followed throughout the duration of BIOnTOP is heavily interconnected with the exploitation and dissemination activities to pave the way for post-project exploitation. An integrated approach will be followed for the identification, capture, retrieval, distribution, sharing, use and reuse of BIOnTOP information and knowledge assets. This will be essential to drive the value and benefits of these information and knowledge assets and contribute to maximising the innovation impact of the action. Knowledge management sessions will take place regularly during the project meetings. These sessions will be chaired by the Project Coordinator (Dr. Belen Monje, AIMPLAS) and the IPR Manager (Dr. Marco de la Feld, ENCO), attended by all project beneficiaries, and will facilitate the disclosure of the IP and the novelty of the project results. The aim of these sessions is to review: i) expected results per WP, ii) evaluation of the IP (technology watch/prior art review), iii) evaluation of protection options (IPR tools evaluation), iv) third party components included in the project activities including tools and datasets, v) confidentiality of project results, vi) expected publications per partner and vii) exploitation intentions per partner.

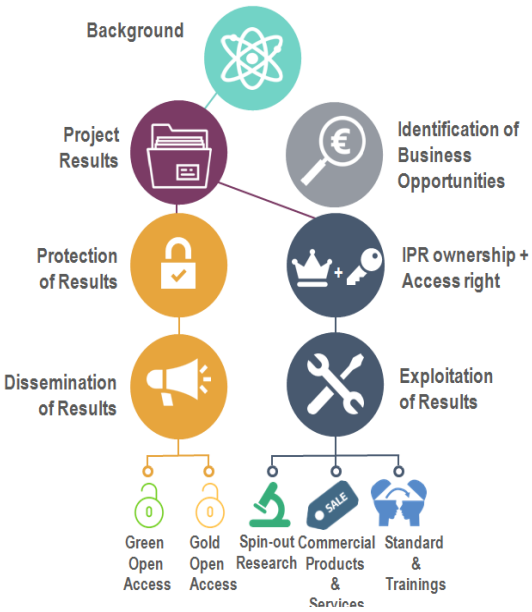

Fig. 18: Knowledge Management Strategy

Background IPR and access rights

Background, that is, information and knowledge (including inventions, databases, etc.) held by the partners prior to their accession to the EC grant agreement, as well as any intellectual property rights which are needed for carrying out the project or for using results held by the consortium, or indeed by third parties, is defined in the proposal preparation phase (pre-Grant Agreement). The current status of the background being brought to the project by the consortium is listed in Table 13. Access to Background necessary for the implementation of the project will be granted royalty free and background needed to use the results of the project will be provided either royalty free or under fair and reasonable conditions.

Table 13: Background per partner

No	Name	Background
1	AIMPLAS	Know-how in copolymerisation and reactive extrusion synthesis of biopolymers. Know-how in biopolymers for food packaging applications. Know-how in extrusion and thermoforming processes. Know-how in recycling technologies. Several patents in the field of bio-based and biodegradable polymer like EP 2784114 A1, ES 2364831 B1.
2	INSTM	Know-how on biodegradable polyester based materials extrusion. Knowledge on polyester recycling through reactive extrusion methods. Patent related to Copolymers based on reactive polyesters and plasticisers for the manufacture of transparent, biodegradable packaging film, WO 2013164743 A1. Patent related to blends of polyesters and aromatic polycarbonate, WO2012025907A1. Patent on toughened post-consumer condensation polymers (including PET and PET/PE from multilayers based formulations), Italian patent, P12011A000090.
3	CTB	Know-how in formulation (including functionalisation) and processing of biopolymers for textile, composite and packaging applications.
4	ASU	Know-how in formulation (including functionalisation) and processing of biopolymers for flexible packaging applications.
5	EUBP	Leading bio-based and biodegradable plastics industry association in EU with extensive insights into the global biopolymer and bioplastics market; decade-long experience with communication on and dissemination of information relating to these materials.
6	TOTAL - Corbion	Knowledge in bio-based biodegradable plastics as well as monomers, converting processes, application, end of life, markets
7	PLANET	Knowledge on compounding and REX. Licence of IMSTM's patents WO 2013164743 A1. WO2012025907A1. P12011A000090
8	BIO-MI	Know-how in flexible packaging production.
9	EMSUR	Know-how in flexible and rigid packaging for a broad range of applications.
10	CMSA	Know-how in the field of nets for a broad range of applications.
11	SILON	Know-how on compounding and staple fibre extrusion of recycled PET and PLA.
12	SIOEN	Know-how on textile converting including PLA.
13	WEAREBIO	Knowledge in food sector specialised in organic food and use of eco-friendly packaging for bulk and retail uses.
14	ENTREPI.	knowhow on dairy food production and packaging requirements.
15	UBESOL	knowhow on personal & home care and cosmetics products production and their packaging requirements.
16 23	OWS (terminated) OWS RF	Knowledge about analysis of (organic) recyclability, evaluation of compostability (e.g. industrial composting: the complete EN 13432 testing program; home composting NF T 51-800), biodegradation and disintegration under anaerobic conditions and Life Cycle Assessment.
17	ROMEI	knowhow on packaging recycling including PLA.
18	IRIS	Know-how in monitoring systems especially including NIR, HIS -Knowledge on data mining for predictive modeling and bioplastics. License of the Wheylayer patent "Whey coated films" PCT / IB 2011 / 05327 and the Wheylayer trademark.
19	ARCHA	Expertise as analytical and consulting service provider related with regulatory compliance on the environmental aspects, H&S on products

		and in workplaces, implementation of Quality, Environment and Safety systems.
20	ENCO	Knowledge in communication, business & market analysis. Knowledge in innovation, exploitation and IPR management.
21	MC	knowhow on consumer perception studies, and communication towards them.
22	INNOTECH	Know-how in flexible and rigid packaging for a broad range of applications.

Results definition, protection, ownership and access rights

Selected results from the project were previously listed in Table 12 along with the expected scenarios of results ownership. This table will apply for all project results, and it will be periodically revised and updated during the project in line with developments. The project results will be continually assessed in terms of novelty, patentability and protection through Task 7.4, as part of an effective knowledge management strategy. The choice of the most suitable form of IP protection, as well as the duration and geographical coverage depends on the results at stake, but also the business plans for their exploitation and legitimate interests of consortium partners. By default, results will be owned by the partner generating them. To avoid doubt or confusion over ownership, a protocol of procedures will be defined, which will be followed by each partner to clearly show the development of the generation of knowledge or results. In case a result is jointly generated (i.e. where the separate parts of some result are attributed to different participants), it will be jointly owned. Consequently, Joint Ownership Agreements will regulate the allocation and the terms of exercising this joint ownership(s). However, an overriding principle will be that the participating industry partners are always provided with the access that they need to the results to exploit them in keeping with the needs and strategy of their business. As such, access rights to results for implementing the project will be granted on a royalty-free basis and in case of access rights for use purposes will be granted either royalty-free, or on fair and reasonable conditions (these conditions will be established in Exploitation Agreements that will be negotiated between the requester and granter of the access rights). A first draft of the Exploitation Agreement will be available by M12 and will be continually revised and edited throughout the project, and a final version will be signed by all partners at M48 (annexed to D7.6).

Open access to peer-reviewed scientific publications

BIOnTOP partners will provide peers and end-users with free of charge online access to the project peer-reviewed scientific research articles. Partners can follow two routes towards open access: (1) by self-archiving their journal articles in an open access repository, also known as 'green' open access, or (2) by publishing in an open access journal, known as 'gold' open access. Whereas fees to cover publishing in gold open access journals have been budgeted, BIOnTOP will privilege green open access, whereby the partners will publish in peer-reviewed scientific publications and then self-archive a version in repositories (using only those indexed in specialised search engines like Scopus). The project website will complementarily house the 'BIOnTOP Green Channel' for open access to the peer-reviewed publications of the project. In relation to expected publications from the project, these include but are not limited to: results from food bio-packaging and textile value chain analyses; results from demonstration of recyclability, biodegradability and home compostability of BIOnTOP compounds; LCA; novel results and applications of PLA and copolymer blends.

Fig. 19: BIONTOP Communication Toolbox

Communication activities

Communication activities are of the outmost importance for effectively promoting our action and outputs towards the stakeholders that we want to reach to fully exploit this BBI action. To successfully connect and cooperate with, and acquire positive support from relevant stakeholders and the public at large, a strategic plan will be implemented in Task 7.5.5, on top of the preliminary outline presented in Fig. 18. The overall communication campaign, which covers the whole project duration, will be led by the Innovation, Communication & Dissemination Manager entity EUBP, which holds an established extensive network and communication capacity to ensure that benefits and impacts of the project are genuinely conveyed to key stakeholders and the general public (in particular through MC). Our aim is to i) generate understanding of what BIOnTOP seeks to achieve, ii) stimulate interest and support, and iii) motivate the collaboration of actors along the value chain as well as of the general public in order to boost the transition to the bio-based economy in EU. The BIOnTOP communication campaign leverages one-way, bilateral, and multilateral strategic channels.Following a preliminary stakeholder analysis (which will be developed in Task 7.3), the target audiences of BIOnTOP communication activities comprise monomers & biopolymer suppliers, agro-food processors, plastic packaging manufacturers and converters, personal care and food manufacturing industry and retailers, other packaging end users, consumers & consumer organizations, agricultural associations, public authorities, standardisation & safety agencies/organisations, scientific communities, and recycling & industrial composting companies. Particular efforts will be made to engage with and raise **awareness of the general public**. It is important to address consumers' concerns by informing the public about our **strong dependence on** fossil-based **conventional plastics**

to which **renewably sourced, bio-based, high-performance, and cost competitive bioplastics can be the more sustainable alternative.** The awareness raising actions carried out will aim to generate acceptance and open the path to building trust towards innovative biopackaging solutions, through optimal communication and transparency using appropriate language enabling EU consumers to make informed purchasing and consumption decisions. A set of key messages will be developed specifically to popularise complex subjects and to bring the issue closer to people's lives by making it more understandable. The campaign will be rolled out through various channels and tools (figured in Fig. 19), with the following approach: 1. 'Push' communication (graphic identity, newsletters, press releases, promotional materials, gadgets, etc.) and 2. 'Pull' communication and interaction (website, social media, liaison with other initiatives and projects, joint communication, etc). The fact that our demonstrators are broadly focused on consumers applications like packaging (which in turn was a prominent reason for their selection) will allow maximising the awareness of the impact of the bioeconomy, and indeed BBI programmes, on their everyday life.

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

Brief presentation of the overall structure of the work plan

A competitive work plan has been devised in order to deliver on the objectives of the BIOnTOP project providing bioplastic solutions with tailored end of life for a number of packaging and textile applications. This will preserve our environment and resources but also meet the expectations of green-minded consumers. The project objectives have been used as the starting point for planning the work that needs to be implemented within the project (Section 1.1). All the actions that are required to meet the objectives of this Research and Innovation project have been collectively analysed by the interdisciplinary consortium. It was agreed how best to distribute the workload among the various experts in keeping with their relevant fields of knowledge, capacity, infrastructure, resources and in the context of bioplastics supply, packaging textile industry value chains, as well as waste management companies. This bottom-up planning process has greatly facilitated the task of arriving at a **clear, comprehensive and realistic work plan** (see Gantt Chart and Pert Diagram below), whereby all partners their ownership.

To fulfilling the project objectives, our work plan is split into 8 WPs logically distributed along the value chain, each with clear tasks that will enable the project's objectives to be fulfilled through a coherent structure that will also facilitate project. To ensure the clear focus towards the target application of this Research and Innovation Action building upon past successful research, most WPs, except WP2-3 dedicated to initial material development, are led by industrial partners. The **first phase** (WP1) deals with industry-led **specification definition** to guide the following innovation work to match the market needs. The **second phase** (WP2) deals with the **lab scale development and upscaling** of the new copolymers and compounds to meet the goals of tailored properties and biodegradability for each packaging application. In a third phase (WP3) the **development of multifunctional barrier coatings** will be performed through a first stage of formulation that will be later applied on different substrates. WP4 is a transversal stage that will be focus on the **environmental aspect including recycling, biodegradation** trials with a toll gate at midterm to provide inputs to transition from phase 2 to phase 3 and also **environmental impact** evaluations. Regarding the next two phases (WP5 and 6) deals with the pilot production and validation of the novel materials designed in WP2 and 3 for the different target applications: mono and multilayer films and trays in WP5 and nets and other textile solutions in WP6, whereby synergies will be created between those using similar processes (e.g. trays converted by sheet extrusion followed by thermoforming as opposed to nets and textiles from extrusion melt spinning) for different food and personal care end uses. WP7 addresses the **economic viability performance** of all the developments as well as the **innovation impacts evaluation, exploitation and disseminations** activities to prepare the market uptake and maximise the project impact. To complement the first 7 WPs linked with scientific & technical (S&T) development and validation activities, in the **last phase** (WP8) will address the full management of the project in order to make all the proposed developments in line with defined BIOnTOP objectives.

The work plan is clearly linked to the objectives set out in Section 1.1 and the consortium is confident that this work plan is robust to ensure that the project's S&T and innovation goals are fulfilled. The research and development work are encapsulated in a comprehensive risk assessment and management strategy, sound communication and training to heighten the impact of the knowledge generated, as well as knowledge management to pave the way for planning and implementing the project's exploitation and dissemination pathways, and robust project coordination and ISO certified project management procedures. The work packages and tasks form a complex inter-weaved matrix of actions, many interdependent, with very close interaction between the WP and task leaders, which is essential for the delivery of the full suite of outputs. During the work planning stage, consideration has been given to logistical matters, such as the time needed for ordering parts/components, the time needed for shipping of prototypes from development to test sites, installation times, the 'what if' scenarios of parts not arriving, installation work throwing up challenges, delays in samples, delays in necessary inputs/results from feed in tasks or subtasks, etc. The link between WPs in terms of needed output material/sample quantity required by next activities were also accounted for. As this is a highly experienced development team, and in particular with many members of the consortium experienced in applied R&D under tight timeframes, as well as experience in integration work at industrial sites, the work planning has benefited from this 'on the ground' pragmatic experience, especially in terms of consideration of interdependencies and timings. The potential risks have been identified and contingency plans proposed, which are discussed in Section 3.1 under "Critical risks and mitigation measures". Importantly, the collective planning of the work plan has achieved the 'buy-in' of all the project developers and participants and an early understanding of the importance of a team effort, team planning and active communication among all WP and task